Data Engineering with Cloud Pipelines

Subtopics:

- Data pipeline architecture
- ETL vs ELT frameworks
- Data lakes vs warehouses
- Apache Spark basics
- Real-time data streaming
- Cloud storage optimization
- Data version control
- Data governance standards
- BigQuery fundamentals
- Workflow orchestration
- Data observability
- Scalable pipeline deployment

Subtopic 1: Data Pipeline Architecture (Full 150)

Phase 1: Easy Level (1–50)

1. What is the primary purpose of a data pipeline?

A) To create data from scratch

B) To transport data from a source to a destination

C) To delete old hardware

D) To replace a data scientist

Answer: B

2. In a pipeline, what is the "Source"?

A) The final database where data is stored

B) The system or location where the data originates

C) The person reading the final report

D) The cloud service billing department

Answer: B

3. What does the "Ingestion" phase involve?

A) Deleting data from the source system

B) Bringing data into the pipeline from an external system

C) Visualizing data in a bar chart

D) Encrypting a developer's laptop Answer: B

4. Which of the following is a common "Sink" in a data pipeline?

A) A web scraper

B) A Data Warehouse

C) A user's keyboard

D) A temperature sensor

Answer: B

5. What is "Batch Processing"?

A) Processing data immediately as it is created

B) Processing groups of data at specific, scheduled times

C) Manually typing data into a system

D) Deleting data in very small pieces

Answer: B

6. What is "Streaming" or "Real-time" processing?

A) Processing data once every few months

B) Processing data continuously as it arrives

C) Printing data on physical paper

D) Storing data in a cold, offline archive

Answer: B

7. What does "Latency" measure in a pipeline?

A) The total physical size of the data

B) The time delay for data to move through the system

C) The monthly cost of the servers

D) The number of columns in a database table

Answer: B

8. What is "Throughput"?

A) The total delay in the system

B) The amount of data processed in a given time period

C) The security level of the pipeline encryption

D) The physical weight of the server hardware

Answer: B

9. Which format is most commonly used for semi-structured data?

A) MP3

B) JSON

C) PNG

B) Checking if the data follows rules before it is loaded

D) EXE

Answer: B

10. What is a "CSV" file?

A) Compressed System Video

B) Comma-Separated Values

C) Cloud Storage Variable

D) Central Security View

Answer: B

11. What is the role of an "Orchestrator" (like Apache Airflow)?

A) To write the data directly to a disk

B) To manage the timing and order of pipeline tasks

C) To design the database schema

D) To encrypt user passwords

Answer: B

12. What is "Data Validation"?

A) Selling data to a third-party company

C) Moving data from a warehouse to a lake

D) Creating a manual backup of the data

Answer: B

13. What is "Metadata"?

A) Encrypted secret data

B) Data that provides information about other data

C) Any dataset larger than 1 Terabyte

D) Records that have been deleted

Answer: B

14. Which cloud service is a popular "Object Storage" used in pipelines?

A) Amazon S3

B) Google Search

C) Microsoft Word

D) Adobe Acrobat

Answer: A

15. What is a "Pipeline Trigger"?

A) An error that crashes the entire system

B) An event or schedule that starts the pipeline

C) A specialized data engineer

D) A type of database index

Answer: B

16. What is "Data Transformation"?

A) Moving data from one city to another

B) Changing the format, structure, or values of data

C) Deleting the source data permanently

D) Buying more cloud storage space

Answer: B

17. What is "Data Cleaning"?

A) Dusting the physical server racks

B) Fixing or removing incorrect, corrupted, or duplicate data

C) Deleting the entire database

D) Changing the owner of the data file

Answer: B

18. What does "API" stand for?

A) Automated Private Internet

B) Application Programming Interface

C) Advanced Processing Integrated

D) Archive Point Index

Answer: B

19. What is a "Database Schema"?

A) The price of the database software

B) The formal structure and organization of data in a database

C) The speed of the database queries

D) The physical location of the data center

Answer: B

20. What is "Reliability" in a pipeline?

A) How fast the pipeline can move data

B) The ability of the pipeline to function correctly without failing

C) The total cost of building the pipeline

D) The number of people allowed to see the data

Answer: B

21. Which tool is often used for simple, automated data movement?

A) Photoshop

B) Fivetran

C) Slack

D) Zoom

Answer: B

22. What is "Scalability"?

A) The ability to handle more data by adding resources

B) The ability to delete data quickly

C) The weight of a server rack

D) The number of colors in a dashboard

Answer: A

23. What is "Data Redundancy"?

A) Deleting unnecessary data

B) Storing the same data in multiple places for safety

C) Making data run faster

D) Encrypting data twice

Answer: B

24. What is a "Job" in a data pipeline?

A) A person's employment

B) A specific unit of work or task to be executed

C) A type of programming language

D) A physical server

Answer: B

25. What is "Automation" in a pipeline?

A) Hiring more people to move data

B) Using software to run the pipeline without human intervention

C) Manually checking every row of data

D) Turning off the servers at night

Answer: B

26. Which of these is a "Structured" data source?

A) A collection of random photos

B) A SQL Server database table

C) A long audio recording

D) A handwritten letter

Answer: B

27. What is "Data Inconsistency"?

A) Data that is always exactly the same

B) When the same data has different values in different systems

C) Data that is heavily encrypted

D) Data that is very large in size

Answer: B

28. What is a "Data Connector"?

A) A physical power cable

B) A software component that links a source to the pipeline

C) A person who types in data

D) A specialized computer monitor

Answer: B

29. What is "Workflow"?

A) The hours a person works in a week

B) The sequence of processes through which data passes

C) A type of office software

D) The physical path to the server room

Answer: B

30. What is "Downstream"?

A) The source systems at the start

B) The processes or systems that receive data later in the pipeline

C) The physical floor of the building

D) The start of the pipeline

Answer: B

31. What is "Upstream"?

A) The systems that provide the data at the start of the pipeline

B) The final dashboard or report

C) The cloud provider's main website

D) The very end of the pipeline

Answer: A

32. What is a "Flat File"?

A) A 3D digital model

B) A simple text file where each record is on a single line (like CSV)

C) A broken hard drive

D) A secure encrypted folder

Answer: B

33. What is "Encryption"?

A) Compressing a file to make it smaller

B) Converting data into a secret code to prevent unauthorized access

C) Deleting old data from the system

D) Translating data into another language

Answer: B

34. What is "Load" in the context of ETL?

A) The physical weight of the data

B) The process of writing data into the destination system

C) The number of users visiting a website

D) The total cost of the pipeline project

Answer: B

35. What is "Extraction" in the context of ETL?

A) Deleting data from a source

B) The process of gathering data from various sources

C) Visualizing data in a chart

D) Sending data via an email attachment

Answer: B

36. What is "Data Availability"?

A) The total cost of the data

B) The assurance that data is ready for use when needed

C) The number of people who can read the data

D) The total size of the database

Answer: B

37. What is a "Log" file?

A) A piece of wood used for cooling

B) A file that records events and errors that happen in the pipeline

C) A type of data transformation

D) A data storage bucket in the cloud

Answer: B

38. What is "Parallel Processing"?

A) Doing only one task at a time

B) Dividing a task into parts and running them at the exact same time

C) Deleting two files at once

D) Running a pipeline once a year

Answer: B

39. What is a "Schema Registry"?

A) A list of all database users

B) A tool that stores and manages versions of data structures

C) A billing system for the cloud

D) A network security firewall

Answer: B

40. What is "Data Integrity"?

A) The speed of the data transfer

B) The accuracy and consistency of data over its entire lifecycle

C) The price of the data on the market

D) The color of the data icons in the UI

Answer: B

41. Which of these is an "Unstructured" data source?

A) A SQL Table

B) A video file

C) An Excel sheet

D) A CSV file

Answer: B

42. What is a "Dashboard"?

A) The front panel of a car

B) A visual display of data insights and key metrics

C) A place to store raw, unedited data

D) A type of programming code

Answer: B

43. What is "Cold Storage"?

A) A refrigerator for the servers

B) Storage for data that is rarely accessed and needs to be kept cheaply

C) A high-speed RAM cache

D) A server located in a cold country

Answer: B

44. What is "Hot Storage"?

A) A server that is currently overheating

B) Storage for data that needs to be accessed frequently and quickly

C) A broken or non-functional hard drive

D) Data stored in a desert environment

Answer: B

45. What is a "Data Warehouse"?

A) A physical building used for storing servers

B) A system used for reporting and data analysis of structured data

C) A place to store raw, unedited photos

D) A type of web browser software

Answer: B

46. What is a "Data Lake"?

A) A large body of water

B) A repository that stores vast amounts of raw data in its native format

C) A highly organized SQL table

D) A tool used for deleting data

Answer: B

47. What is "Uptime"?

A) The time it takes to upload a file

B) The period of time a system is functional and available

C) The time spent writing code

D) The time the sun is visible

Answer: B

48. What is "Downtime"?

A) The time spent sleeping

B) The period of time a system is unavailable or not functioning

C) The time it takes to download a file

D) The total cost of a server failure

Answer: B

49. What is "Data Profiling"?

A) Creating a social media profile

B) The process of examining data to understand its content and quality

C) Selling data to advertisers

D) Encrypting user profiles

Answer: B

50. What is a "Point-to-Point" connection?

A) A complex network of many interconnected systems

B) A direct link between one source and one destination

C) A wireless internet connection

D) A type of data visualization chart

Answer: B

Phase 2: Medium Level (51–100)

[Image comparing Lambda Architecture vs Kappa Architecture]

51. What is "Idempotency" in a data pipeline?

A) The speed at which data is processed.

B) The property where running a job multiple times produces the same result.

C) The ability to compress data automatically.

D) The security level of the data source.

Answer: B

52. Which architectural pattern uses a "Batch Layer" and a "Speed Layer"?

A) Kappa Architecture

B) Monolithic Architecture

C) Lambda Architecture

D) Microservices Architecture

Answer: C

53. What is the main purpose of "Change Data Capture" (CDC)?

A) To delete all data every 24 hours.

B) To capture only the data that has changed in a source database.

C) To change the data format from JSON to CSV.

D) To capture user passwords for security logs.

Answer: B

54. What is "Backpressure" in a streaming pipeline?

A) Increasing the speed of the data source.

B) A signal to the producer to slow down when the consumer is overwhelmed.

C) A physical pressure build-up in server cooling.

D) The process of data pushing back against a firewall.

Answer: B

55. A "Directed Acyclic Graph" (DAG) ensures:

A) Tasks are performed in a random order.

B) There are no circular loops in task dependencies.

C) Data is stored in a graph database only.

D) Every task is run 10 times.

Answer: B

56. What is the benefit of a "Decoupled Architecture"?

A) It makes the system cheaper to run on a single laptop.

B) It allows components to scale and fail independently.

C) It reduces the number of data engineers needed.

D) It eliminates the need for cloud storage.

Answer: B

57. "Schema-on-Read" is most commonly associated with:

A) Relational Databases (SQL)

B) Data Lakes (Hadoop/S3)

C) Excel Spreadsheets

D) Physical Ledger Books

Answer: B

58. What is "Vertical Scaling"?

A) Adding more servers to a cluster.

B) Increasing the power (CPU, RAM) of a single existing server.

C) Moving data from a low floor to a high floor.

D) Reducing the number of users.

Answer: B

59. What is "Horizontal Scaling"?

A) Upgrading the RAM of a single machine.

B) Adding more machines or nodes to a system.

C) Making the database rows wider.

D) Using a wider monitor.

Answer: B

60. What is "Data Lineage"?

A) The cost history of a data project.

B) The visual map showing data origin and journey.

C) The family tree of the data team.

D) The physical location of fiber cables.

Answer: B

61. What is a "Message Broker" (like Kafka) used for?

A) To store data for 50 years.

B) To act as a buffer between producers and consumers.

C) To calculate math formulas.

D) To design website interfaces.

Answer: B

62. In a "Push-based" ingestion model:

A) The pipeline asks the source for data.

B) The source sends data to the pipeline as it becomes available.

C) Users must manually click "Upload".

D) The database is physically moved to the cloud.

Answer: B

63. In a "Pull-based" ingestion model:

A) The source forces data into the pipeline.

B) The pipeline requests data from the source when ready.

C) Data is deleted after it is read.

D) No data is moved at all.

Answer: B

64. What is "Schema Evolution"?

A) Deleting a database schema.

B) The ability to handle structure changes without breaking the pipeline.

C) Moving from on-premise to cloud.

D) Data losing value over time.

Answer: B

65. Which file format is "Columnar" and optimized for analytics?

A) CSV

B) Parquet

C) JSON

D) XML

Answer: B

66. What is "Exactly-Once" processing?

A) Data is processed at least once.

B) Data is guaranteed to be processed exactly one time.

C) Data is only processed if the user is online.

D) Data is processed once a year.

Answer: B

67. What is "At-Least-Once" delivery?

A) Data might be lost.

B) Data is guaranteed to arrive but might have duplicates.

C) Data is only delivered if health is 100%.

D) Data is never delivered.

Answer: B

68. What is the role of a "Staging Area"?

A) Final place where users see data.

B) Intermediate storage for processing/cleaning before the final load.

C) A backup of the source.

D) A training room.

Answer: B

69. "Micro-batching" is used to:

A) Process massive data once a year.

B) Process data in small groups at high frequency.

C) Delete data in small increments.

D) Reduce the salary of the team.

Answer: B

70. What is "Data Partitioning"?

A) Deleting a database.

B) Dividing a dataset into smaller parts based on a key (like Date).

C) Combining all data into one giant file.

D) Giving users different passwords.

Answer: B

71. What is "Stateful" transformation?

A) Depends only on the current row.

B) Requires info from previous rows (like a running total).

C) Changes the state of the server physically.

D) Done in a specific country.

Answer: B

72. What is "Stateless" transformation?

A) Remembers every row seen.

B) Each record is processed independently.

C) Happens when the server is off.

D) Does not use any code.

Answer: B

73. What is "Windowing" in stream processing?

A) Cleaning office windows.

B) Dividing a continuous stream into finite chunks based on time.

C) Closing the program after 10 mins.

D) Looking at data on a small screen.

Answer: B

74. What is a "Tumbling Window"?

A) Follows a user.

B) Fixed-size, non-overlapping time intervals.

C) A window that never closes.

D) Deletes data.

Answer: B

75. What is a "Sliding Window"?

A) Moves in fixed increments and can overlap.

B) Only opens once a day.

C) Used for glass cleaning.

D) Only stores the first record.

Answer: A

76. What is "Data Shuffling"?

A) Deleting data randomly.

B) Redistributing data across different partitions or nodes.

C) Alphabetical sorting only.

D) Encryption before sending.

Answer: B

77. What is "Checkpointing"?

A) Security guard check.

B) Periodically saving pipeline state for recovery.

C) Checking the cloud bill.

D) Deleting logs.

Answer: B

78. Which architecture uses only one processing engine (no batch layer)?

A) Monolithic

B) Kappa Architecture

C) Three-Tier

D) Client-Server

Answer: B

79. What is "Fan-out"?

A) Cooling the broker.

B) Sending one message to multiple destination systems.

C) Reducing message count.

D) A system crash.

Answer: B

80. What is "Fan-in"?

A) Multiple sources to a single destination.

B) Single source to multiple destinations.

C) Turning on a fan.

D) Deleting data.

Answer: A

81. What is "Service Discovery"?

A) Searching for cloud providers.

B) Automatic detection of devices/services on a network.

C) Finding IT support.

D) Finding expensive data.

Answer: B

82. What is "Observability"?

A) Watching the pipeline on screen.

B) Understanding internal state based on outputs (logs/metrics).

C) Installing cameras.

D) Measuring server room size.

Answer: B

83. In an "Event-Sourcing" pattern:

A) Store only final state.

B) Store every change as immutable events.

C) Delete events after processing.

D) Only capture weekend events.

Answer: B

84. What is "Data Skew"?

A) Perfect distribution.

B) Concentration in one partition causing a bottleneck.

C) Incorrect data types.

D) Encryption key.

Answer: B

85. What is "Load Balancing"?

A) Carrying a server.

B) Distributing tasks across servers to prevent overload.

C) Making the database lighter.

D) Balancing the budget.

Answer: B

86. What is "Heartbeating"?

A) Sound of a server.

B) Signal sent to indicate a service is alive.

C) Encryption.

D) Deleting logs.

Answer: B

87. What is a "Webhook"?

A) Physical cable hook.

B) Automated HTTP callback when an event happens.

C) Internet browser.

D) Fishing for data.

Answer: B

88. What is "Cold Start"?

A) Snowy climate start.

B) Delay when a function executes for the first time after being idle.

C) Low temp failure.

D) Nightly shutdown.

Answer: B

89. What is "Schema on Write"?

A) Check structure on read.

B) Enforce structure at entry to the database.

C) Paper schema.

D) No schema.

Answer: B

90. "Exactly-Once" vs "At-Most-Once":

A) Exactly-once guarantees delivery; At-most-once prevents duplicates but may lose data.

B) Same thing.

C) Slower vs expensive.

D) Batch vs streaming.

Answer: A

91. What is "Data Governance"?

A) Government control.

B) Set of rules for quality and security.

C) Deleting 5-year-old data.

D) Political management.

Answer: B

92. What is "Data Sovereignty"?

A) Data speed.

B) Data subject to laws of its physical location.

C) Owned by a king.

D) Non-movable data.

Answer: B

93. What is "Metadata Management"?

A) Deleting metadata.

B) Maintaining info about data assets.

C) Hiding metadata.

D) Selling metadata.

Answer: B

94. What is "Data Compaction"?

A) Deleting data.

B) Merging small files into large ones for performance.

C) Small font.

D) Zip file.

Answer: B

95. "Zero-Copy Data Transfer" means:

A) No backups.

B) Moving data without intermediate software buffers.

C) Delete after move.

D) Copy zero times.

Answer: B

96. What is "Data Freshness"?

A) Cleanliness.

B) Time since last update.

C) Age of hard drive.

D) Unused data.

Answer: B

97. What is "Rate Limiting"?

A) Charging per row.

B) Controlling requests per time period.

C) Increasing speed.

D) Limiting engineers.

Answer: B

98. What is a "Side-Car" pattern?

A) Small server next to big one.

B) Helper service attached to a main app.

C) Motorcycle delivery.

D) Power supply.

Answer: B

99. What is "Data Virtualization"?

A) VR headset data.

B) Querying without knowing physical location/format.

C) Fake database.

D) Deleting physical servers.

Answer: B

100. What is a "Circuit Breaker" pattern?

A) Room fuse.

B) Mechanism to stop requests to a failing service.

C) Internet shutdown.

D) SQL speedup.

Answer: B

Phase 3: Master Level (101–150)

101. In Kappa Architecture, how is historical data re-processed?

A) Using a separate MapReduce batch job.

B) Replaying the data log from the stream broker.

C) Manual SQL inserts.

D) It cannot be re-processed.

Answer: B

102. What is the primary difficulty of "Exactly-Once" in distributed systems?

A) Storage space.

B) The Two Generals' Problem (coordinating state across nodes during failure).

C) Coding language limitations.

D) High cloud costs.

Answer: B

103. What is a "Dead Letter Queue" (DLQ) used for?

A) Storing spam emails.

B) Isolating problematic messages for debugging without stopping the flow.

C) Deleting messages automatically.

D) Speeding up the message broker.

Answer: B

104. Which core principle defines "Data Mesh"?

A) Centralized data management.

B) Decentralized, domain-driven ownership.

C) Using only one vendor.

D) No data engineering required.

Answer: B

105. What does "Write-Ahead Logging" (WAL) ensure?

A) Faster UI updates.

B) Atomicity and Durability by logging changes before applying them.

C) Logs are written in English.

D) Data is compressed before logging.

Answer: B

106. What is "Watermarking" in stream processing?

A) Adding a digital logo.

B) A timestamp threshold to handle late-arriving data in time-windows.

C) Liquid cooling for servers.

D) Measuring data depth.

Answer: B

107. What is "Eventual Consistency"?

A) Data is never consistent.

B) Data will be consistent across all nodes given enough time.

C) Data is consistent immediately upon write.

D) Data is only consistent on weekends.

Answer: B

108. What is "Strong Consistency"?

A) Consistent eventually.

B) All reads return the most recent write immediately.

C) Physically strong servers.

D) High compression rates.

Answer: B

109. What is "Backfill"?

A) Filling the server room with air.

B) Running historical data through a new or updated pipeline.

C) Deleting the last 10 rows.

D) Reverting to an old version of code.

Answer: B

110. The "CAP Theorem" states you can only have two of these three:

A) Cost, Accuracy, Power.

B) Consistency, Availability, Partition Tolerance.

C) Code, Architecture, Performance.

D) Cloud, Automation, Privacy.

Answer: B

111. What is "Sharding"?

A) Deleting a table.

B) Horizontal partitioning of data across multiple database instances.

C) Vertical partitioning of columns.

D) Encryption of small data bits.

Answer: B

112. What is "Data Gravity"?

A) Speed of falling data.

B) The idea that data and apps are drawn to where the data is stored.

C) Weight of the server.

D) Physical law of storage.

Answer: B

113. What is "Log-Structured Merge-Tree" (LSM)?

A) A tree in a forest.

B) A data structure optimized for high-write throughput.

C) A way to organize folders.

D) A security protocol.

Answer: B

114. What is a "Bloom Filter"?

A) A type of air filter.

B) A space-efficient probabilistic data structure to check set membership.

C) A color filter for charts.

D) A data cleaning tool.

Answer: B

115. What is "Two-Phase Commit" (2PC)?

A) Committing code twice.

B) A distributed algorithm to coordinate a transaction across multiple nodes.

C) A fast SQL query.

D) A two-person approval process.

Answer: B

116. What is "Quorum" in a distributed system?

A) A type of database.

B) The minimum number of nodes that must agree for a transaction to succeed.

C) A meeting of data engineers.

D) A security key.

Answer: B

117. What is "Gossip Protocol"?

A) Employees talking about data.

B) A method for nodes to spread state information in a decentralized network.

C) A type of social media.

D) A leaked data report.

Answer: B

118. What is "Write Amplification"?

A) Making text larger.

B) When the amount of data written to storage is greater than the data actually sent.

C) Increasing the frequency of writes.

D) Using a louder server.

Answer: B

119. What is "Read Amplification"?

A) Larger font for reading.

B) When one logical read requires multiple physical storage reads.

C) Reading data twice.

D) High-speed reading.

Answer: B

120. What is "Service Mesh"?

A) A physical net for servers.

B) An infrastructure layer to manage service-to-service communication.

C) A group of shared databases.

D) A type of cloud billing.

Answer: B

121. What is "Immutability" in Big Data?

A) Data changes constantly.

B) Data that cannot be modified after it is created.

C) Data that is very small.

D) Data that is free to use.

Answer: B

122. What is "Functional Data Engineering"?

A) Coding while standing.

B) Using stateless, idempotent transformations on immutable data.

C) Using only functional languages like Lisp.

D) Only building functions that work.

Answer: B

123. What is "Data Mesh" vs "Data Fabric"?

A) Mesh is organizational; Fabric is a technology layer/tooling.

B) They are identical.

C) Mesh is old; Fabric is new.

D) Mesh is for SQL; Fabric is for NoSQL.

Answer: A

124. What is "Serverless Orchestration"?

A) No servers used at all.

B) Managing workflows where the underlying compute is handled by the provider (e.g., AWS Step Functions).

C) Manual task management.

D) Orchestrating a physical band.

Answer: B

125. What is "Hot-Cold Architecture"?

A) Mixing temperatures in the data center.

B) Separating frequently accessed data from rarely accessed data.

C) A broken cooling system.

D) Switching between two cloud providers.

Answer: B

126. What is "Polyglot Persistence"?

A) Speaking many languages.

B) Using different databases for different data needs within one app.

C) Using only one SQL database.

D) Storing data in many countries.

Answer: B

127. What is "Distributed Tracing"?

A) Drawing a map of the network.

B) Monitoring requests as they move through various microservices.

C) Finding where a wire goes.

D) Tracking the cost of a single node.

Answer: B

128. What is "Chaos Engineering" in pipelines?

A) Working in a messy office.

B) Intentionally injecting failures to test system resilience.

C) Writing bad code on purpose.

D) Having no project manager.

Answer: B

129. What is "Throughput-Latency Tradeoff"?

A) Faster speed always means more volume.

B) Optimizing for more data often increases the delay for single items.

C) There is no tradeoff.

D) Higher cost fixes both.

Answer: B

130. What is "Semantic Layer"?

A) A layer of code comments.

B) An abstraction that presents data using business terms rather than technical table names.

C) A translation service.

D) The top layer of a data lake.

Answer: B

131. What is "Data Observability" vs "Monitoring"?

A) Monitoring is what is broken; Observability is why it is broken.

B) They are the same.

C) Monitoring is for hardware; Observability is for software.

D) Monitoring is manual; Observability is automatic.

Answer: A

132. What is "Data Governance by Design"?

A) Designing a pretty logo for data.

B) Integrating security and quality controls directly into the pipeline code.

C) Hiring a designer to write code.

D) Fixing data after it is loaded.

Answer: B

133. What is "Push-Down Optimization"?

A) Pushing a server down the stairs.

B) Moving the transformation logic to the source or destination database to save move-time.

C) Deleting code to make it faster.

D) Lowering the price of the cloud.

Answer: B

134. What is "Columnar Compression"?

A) Making columns shorter.

B) Exploiting similarities in column data to reduce storage size.

C) Deleting columns.

D) Using a zip file on a table.

Answer: B

135. What is "Predicate Pushdown"?

A) Pushing a button.

B) Filtering data at the source before it is moved into the pipeline.

C) Filtering data after it is loaded.

D) Not using filters.

Answer: B

136. What is "Vectorized Execution"?

A) Using a drawing tool for data.

B) Operating on a batch of data values at once using CPU SIMD instructions.

C) Moving data in a straight line.

D) A type of security encryption.

Answer: B

137. What is "Just-In-Time" (JIT) Compilation in data engines?

A) Compiling code just before the deadline.

B) Converting query plans into machine code during execution for speed.

C) Writing code very quickly.

D) Compiling code only when it breaks.

Answer: B

138. What is "Partition Pruning"?

A) Cutting branches off a tree.

B) The optimizer skipping irrelevant partitions during a query.

C) Deleting old partitions.

D) Merging partitions together.

Answer: B

139. What is "Broadcast Join"?

A) Joining a TV show.

B) Sending a small table to all nodes to join with a large distributed table.

C) Joining two massive tables.

D) Joining data over the radio.

Answer: B

140. What is "Shuffle Join"?

A) Randomly joining data.

B) Redistributing both tables by the join key across all nodes.

C) Joining data without moving it.

D) A very slow join.

Answer: B

141. What is "Data Compaction" in LSM Trees?

A) Deleting logs.

B) Background merging of sorted files to remove duplicates and deleted entries.

C) Making the tree smaller.

D) Encrypting the tree.

Answer: B

142. What is "Write-Behind Caching"?

A) Writing data to disk first.

B) Writing to cache immediately and to disk later in the background.

C) Not using a cache.

D) Writing logs after the crash.

Answer: B

143. What is "Write-Through Caching"?

A) Writing to cache and disk simultaneously.

B) Writing only to disk.

C) Deleting the cache.

D) Writing logs during the crash.

Answer: A

144. What is "Multi-Tenancy"?

A) Having many house guests.

B) A single system serving multiple distinct customers or groups securely.

C) Using multiple cloud providers.

D) Having many passwords.

Answer: B

145. What is "Shared-Nothing Architecture"?

A) Not sharing any secrets.

B) Each node has its own private memory and disk; they communicate via network.

C) All nodes share one giant disk.

D) A system with zero users.

Answer: B

146. What is "Shared-Disk Architecture"?

A) All nodes share the same central storage.

B) No nodes share storage.

C) Using a CD-ROM.

D) Deleting the disk.

Answer: A

147. What is "Cloud-Native" Architecture?

A) Living in the cloud.

B) Applications designed specifically to exploit cloud features like elasticity and microservices.

C) Using a computer with an internet connection.

D) Using a laptop in an airplane.

Answer: B

148. What is "Infrastructure as Code" (IaC)?

A) Writing software for a website.

B) Managing infrastructure using configuration files (like Terraform).

C) Coding on a physical server.

D) Buying a server with a credit card.

Answer: B

149. What is "Zero-Trust" Security?

A) Not trusting your coworkers.

B) Requiring strict verification for every person and device trying to access resources.

C) Having no security at all.

D) A password-free system.

Answer: B

150. What is "Data Sovereignty"?

A) Data speed.

B) Legal requirement that data remains within the borders of the country where it was collected.

C) Data that is free from taxes.

D) Data that is owned by the government.

Answer: B

Subtopic 2: ETL vs. ELT Frameworks (Full 150)

Phase 1: Easy Level (1–50)

1. What does the "L" stand for in ETL and ELT?

A) Link

B) Load

C) Log

D) List

Answer: B

2. In the ETL process, where does the transformation usually happen?

A) In the destination database

B) On a separate staging/transformation server

C) Inside the source system

D) In the user's web browser

Answer: B

3. In the ELT process, where does the transformation happen?

A) Before the data is loaded

B) Within the target data warehouse

C) On a physical hard drive

D) In a third-party API

Answer: B

4. Which framework is considered the "traditional" approach for on-premise systems?

A) ELT

B) ETL

C) Manual Entry

D) Paper records

Answer: B

5. Which framework is optimized for modern Cloud Data Warehouses (Snowflake, BigQuery)?

A) ETL

B) ELT

C) API-only

D) Batch-only

Answer: B

6. ETL is primarily a _____ process.

A) Real-time

B) Batch

C) Random

D) Continuous

Answer: B

7. Which process is better for data privacy (masking PII before it hits the warehouse)?

A) ELT

B) ETL

C) Both

D) Neither

Answer: B

8. ELT allows for "Schema-on-Read." What does this mean?

A) You must define the table structure before loading.

B) You define the structure only when you are ready to query the data.

C) You never need a schema.

D) You read data from a book.

Answer: B

9. What is the primary goal of the "Transform" step?

A) To move data from A to B

B) To clean, format, and aggregate data for analysis

C) To delete the source data

D) To encrypt the hardware

Answer: B

10. In a standard ETL pipeline, is raw data stored in the final warehouse?

A) Yes, always

B) No, only the transformed/cleaned data is stored

C) Only if the warehouse is full

D) Only if it is requested

Answer: B

11. Which approach is typically faster for the initial "Ingestion" of large datasets?

A) ETL

B) ELT

C) Manual CSV upload

D) SQL Insert statements

Answer: B

12. In ELT, what provides the power for the "Transform" step?

A) The source system CPU

B) The destination warehouse's compute resources

C) A mid-tier processing server

D) A local laptop

Answer: B

13. Which framework is most common when loading data into a "Data Lake"?

A) ETL

B) ELT

C) Batch Processing

D) Scripting

Answer: B

14. What is "Extraction"?

A) Deleting data

B) Reading/Gathering data from a source system

C) Writing data to a disk

D) Sorting data alphabetically

Answer: B

15. What is a major disadvantage of traditional ETL for "Big Data"?

A) It is too fast.

B) The transformation server can become a bottleneck.

C) It uses too much storage.

D) It is too cheap to maintain.

Answer: B

16. Which process is more flexible if you don't know the final use case for the data?

A) ETL

B) ELT

C) Both are equal

D) Neither

Answer: B

17. What is a "Staging Area"?

A) A place for public speakers

B) A temporary storage space used during the ETL process

C) The final report dashboard

D) The source system

Answer: B

18. Which framework supports "Raw Data Persistence"?

A) ETL

B) ELT

C) Manual loading

D) Excel

Answer: B

19. What is "Data Mapping"?

A) Drawing a physical map

B) Defining how source fields relate to destination fields

C) Calculating distances between servers

D) Locating lost files

Answer: B

20. Is Python a common language for transformations in ETL?

A) Yes

B) No

Answer: A

21. Is SQL the primary language for transformations in ELT?

A) Yes

B) No

Answer: A

22. Which is chronologically older?

A) ETL

B) ELT

Answer: A

23. Why did ELT become popular recently?

A) Cloud storage and compute became cheap and scalable.

B) Internet speeds became slower.

C) Data volumes decreased globally.

D) SQL was removed from databases.

Answer: A

24. In ETL, "Transform" happens ____ "Load".

A) After

B) Before

C) At the same time as

D) Never

Answer: B

25. In ELT, "Transform" happens ____ "Load".

A) After

B) Before

C) Never

D) Only on weekends

Answer: A

26. Which is better for structured, small-scale datasets?

A) ETL

B) ELT

Answer: A

27. Which is better for unstructured, massive-scale datasets?

A) ETL

B) ELT

Answer: B

28. Does ETL require a dedicated processing engine separate from the DB?

A) Yes

B) No

Answer: A

29. What is "Data Cleansing"?

A) Part of Transform (removing errors/duplicates)

B) Part of Extract

C) Part of Load

D) Deleting the server

Answer: A

30. What is "Data Aggregation"?

A) Splitting data apart

B) Summarizing data (e.g., calculating totals)

C) Moving data between clouds

D) Encrypting data

Answer: B

31. What is a "Dimension Table"?

A) A table with descriptive data (e.g., Product Name)

B) A table with only numbers

C) A temporary table

D) A deleted table

Answer: A

32. What is a "Fact Table"?

A) A table containing quantitative metrics or events

B) A table for metadata

C) A table with names only

D) An empty table

Answer: A

33. Does ELT reduce "Time-to-Insight" for raw data exploration?

A) Yes

B) No

Answer: A

34. Which framework often requires Spark or Hadoop for the "T" step?

A) ETL

B) ELT

Answer: A

35. Which framework is "Schema-on-Write"?

A) ETL

B) ELT

Answer: A

36. What is "In-database transformation"?

A) ETL

B) ELT

Answer: B

37. Can ELT handle nested JSON data?

A) Yes

B) No

Answer: A

38. What is a "Data Pipeline"?

A) A physical pipe

B) A series of steps for moving and transforming data

C) A type of code

D) A cloud provider

Answer: B

39. Is "Informatica PowerCenter" a classic ETL tool?

A) Yes

B) No

Answer: A

40. Is "dbt" (data build tool) primarily used for ELT?

A) Yes

B) No

Answer: A

41. What is "Data Enrichment"?

A) Adding external data to your existing data

B) Deleting data

C) Saving money on data

D) Improving data speed

Answer: A

42. Does traditional ETL store a permanent copy of raw data?

A) Usually no

B) Always yes

Answer: A

43. Which framework is theoretically more scalable for compute?

A) ETL

B) ELT

Answer: B

44. What is "Data Integration"?

A) Combining data from different sources into a single view

B) Splitting data

C) Deleting data

D) Selling data

Answer: A

45. Which is usually cheaper to start with on very small data?

A) ETL

B) ELT

Answer: A

46. What is "Standardization"?

A) Making data follow a specific format (e.g., YYYY-MM-DD)

B) Deleting data

C) Moving data

D) Encrypting data

Answer: A

47. Is ELT more common in Data Lake architectures?

A) Yes

B) No

Answer: A

48. Is ETL more common in structured Data Marts?

A) Yes

B) No

Answer: A

49. What is "Load Frequency"?

A) How often data is moved (e.g., hourly, daily)

B) How much data weighs

C) The cost of data

D) The speed of the CPU

Answer: A

50. Which is easier to maintain if source columns change frequently?

A) ETL

B) ELT

Answer: B

Phase 2: Medium Level (51–100)

51. A company needs to mask customer credit card numbers before they enter the cloud. They should use:

A) ELT

B) ETL

C) Direct SQL

D) Manual upload

Answer: B

52. You are using Snowflake and want to transform data using its internal SQL engine. This is:

A) ETL

B) ELT

C) E-T-A

D) L-T-E

Answer: B

53. What is the risk of an unmanaged ELT process?

A) Data is never loaded

B) The warehouse becomes a "Data Swamp" of messy raw data

C) The source system crashes

D) SQL becomes obsolete

Answer: B

54. Which framework allows Data Scientists to access original, unformatted data?

A) ETL

B) ELT

C) Dashboarding

D) Reporting

Answer: B

55. In ETL, if the transformation server fails, what happens to the load?

A) It continues with raw data

B) The entire process stops

C) It loads empty tables

D) It deletes the source

Answer: B

56. In ELT, transformation logic is typically written in:

A) Java

B) SQL

C) Scala

D) C++

Answer: B

57. Which framework is better for "Change Data Capture" (CDC) into a warehouse?

A) ETL

B) ELT

Answer: B

58. You have 100 different API sources with varying schemas. Which is more flexible?

A) ETL

B) ELT

Answer: B

59. Which framework relies more on "Massively Parallel Processing" (MPP)?

A) ETL

B) ELT

Answer: B

60. What is "Push-down Optimization" in some ETL tools?

A) Moving the ETL process to the target DB (acting like ELT)

B) Deleting the data

C) Slowing down the pipeline

D) Encrypting the source

Answer: A

61. In an ETL pipeline, the transformation engine is often ____.

A) External to the database

B) Inside the database

C) At the user's desk

D) In a text file

Answer: A

62. ELT shifts the burden of work from the middle-ware to the ____.

A) Source system

B) Target Database

C) Data Engineer

D) User Interface

Answer: B

63. Which approach is better for real-time streaming analytics?

A) Batch ETL

B) Continuous ELT/Streaming

C) Manual CSV

D) Monthly Batch

Answer: B

64. What is "Data Deduplication" during the Transform phase?

A) Adding more data

B) Removing identical records

C) Renaming records

D) Moving records

Answer: B

65. Which framework is historically better for strictly regulated industries (Health/Finance)?

A) ETL (Cleaner control over data movement)

B) ELT

Answer: A

66. What happens to "Extraction" in ELT if the source is an API?

A) Data is fetched and stored raw in the destination

B) Data is transformed in the air

C) API is deleted

D) No extraction needed

Answer: A

67. Does dbt (Data Build Tool) perform the "E" or the "L"?

A) No, it only handles the "T" in ELT

B) It handles all three

C) It only handles "E"

D) It only handles "L"

Answer: A

68. Which framework is better for "Ad-hoc" queries on raw data?

A) ETL

B) ELT

Answer: B

69. Which pair is usually correct?

A) ETL is to Warehouse; ELT is to Lake

B) They are the same

C) ETL is to Lake; ELT is to Warehouse

D) None

Answer: A

70. Which costs more in terms of "Compute" inside the Cloud Warehouse?

A) ETL

B) ELT

Answer: B

71. What is "Self-service Analytics"?

A) Users doing their own data prep via ELT

B) Computers doing everything

C) Deleting the data team

D) Using a calculator

Answer: A

72. In ELT, what is the role of a "Landing Zone"?

A) To store raw files before they are loaded into tables

B) To land an airplane

C) To delete files

D) To host a website

Answer: A

73. Which allows for "Version Control" of data transformations more easily?

A) ELT (using tools like dbt + Git)

B) ETL

Answer: A

74. What is "Data Latency" in an ETL batch?

A) The delay between data generation and it appearing in the warehouse

B) The speed of light

C) The size of the file

D) The cost of the server

Answer: A

75. What is "Zero-copy cloning"?

A) Making a copy of data without using extra storage (common in ELT)

B) Not copying data at all

C) Deleting a copy

D) Manual data entry

Answer: A

76. In ETL, where is the 'Metadata' about transformations usually kept?

A) In the ETL tool's repository

B) In the destination table

C) In the source

D) In a text file

Answer: A

77. Which framework is better for "Legacy Mainframe Systems"?

A) ETL

B) ELT

Answer: A

78. What is "Business Logic"?

A) The rules used in the 'Transform' step

B) The price of a product

C) The office location

D) The CEO's name

Answer: A

79. Does ELT allow you to "replay" transformations on old raw data?

A) Yes

B) No

Answer: A

80. Does ETL allow you to "replay" transformations if you deleted the raw source?

A) No

B) Yes

Answer: A

81. Which framework supports "High-frequency" updates (every minute)?

A) ETL

B) ELT

Answer: B

82. What is "Scalable Compute"?

A) Adding more RAM

B) Cloud resources that grow as data grows

C) A very fast laptop

D) A large calculator

Answer: B

83. In ELT, what is the 'Staging Table'?

A) A table in the destination DB used for temporary processing

B) A table in the source

C) A CSV file

D) A physical table

Answer: A

84. Which framework usually requires more manual "Data Engineering" for initial setup?

A) ETL

B) ELT

Answer: A

85. What is "Orchestration" in ETL?

A) Coordinating the E, T, and L steps in order

B) Playing the violin

C) Deleting data

D) Buying servers

Answer: A

86. Is Apache Airflow used for ETL or ELT?

A) Both

B) Neither

Answer: A

87. What is "Columnar Storage"?

A) Storing data by columns (fast for ELT/Analytics)

B) Storing data by rows

C) Storing data in a circle

D) No storage

Answer: A

88. Which framework is "Target-agnostic"?

A) ETL (The engine is separate)

B) ELT

Answer: A

89. Which framework is "Target-dependent"?

A) ELT (Requires SQL/Compute from the target)

B) ETL

Answer: A

90. What is "Data Lineage" in ELT?

A) Tracking how raw data became the final report

B) Tracking the cost

C) Tracking the hardware

D) Tracking the employees

Answer: A

91. What is "Late-binding" schema?

A) ELT

B) ETL

Answer: A

92. What is "Early-binding" schema?

A) ETL

B) ELT

Answer: A

93. Which framework is more prone to "Resource Contention" in the warehouse?

A) ELT (Transformation and Querying share compute)

B) ETL

Answer: A

94. What is "ETLT"?

A) A mix (Transforming a little, loading, then transforming more)

B) A typo

C) A new programming language

D) A security protocol

Answer: A

95. What is "Data Discovery"?

A) Finding useful patterns in raw data (easier in ELT)

B) Deleting data

C) Selling data

D) Buying data

Answer: A

96. In ETL, if you need a new column, you usually must ____.

A) Re-run the entire pipeline from scratch

B) Just query the raw data

C) Change the source

D) Do nothing

Answer: A

97. In ELT, if you need a new column from raw data, you ____.

A) Just update your SQL transformation script

B) Must re-extract from the source

C) Cannot do it

D) Must wait for a month

Answer: A

98. Which framework is better for "Small, Structured Data"?

A) ETL

B) ELT

Answer: A

99. Which framework is better for "Unstructured JSON/Logs"?

A) ELT

B) ETL

Answer: A

100. What is the "T" in ELT usually doing?

A) Cleaning and Modeling

B) Moving and Deleting

C) Uploading

D) Printing

Answer: A

Phase 3: Master Level (101–150)

101. What is "Push-down Transformation"?

A) An ETL tool generating SQL to run on the target DB

B) Deleting data

C) Moving data to a lower server

D) Increasing pressure

Answer: A

102. How does "Schema Evolution" impact an ETL pipeline?

A) It usually breaks the transformation code.

B) It has no impact.

C) It speeds up the pipeline.

D) It makes it cheaper.

Answer: A

103. What is the "Data Mesh" perspective on ETL?

A) Centralized ETL is a bottleneck; domains should own their pipelines.

B) ETL should be banned.

C) Everyone should use one ETL server.

D) ELT is only for small companies.

Answer: A

104. In a Kappa architecture, do we traditionally use ELT?

A) No, it usually uses stream processing.

B) Yes, always.

C) Only for backups.

D) Only for SQL.

Answer: A

105. What is "Functional Data Engineering"?

A) Using immutable data and idempotent transformations (common in modern ELT).

B) Coding while standing.

C) Using only Excel.

D) Deleting bugs.

Answer: A

106. Which framework minimizes "Egress" costs between different cloud providers?

A) ETL (by filtering before moving)

B) ELT

Answer: A

107. What is "Idempotency" in a transformation?

A) Running the code again results in the same final state.

B) The code runs faster.

C) The code is written in Python.

D) The code is very long.

Answer: A

108. In ELT, how do you handle "Sensitive Data" (PII) without ETL-side masking?

A) Column-level encryption or hashing within the warehouse.

B) You don't.

C) By deleting the data.

D) By making it public.

Answer: A

109. What is "Micro-batching" in the context of ETL?

A) Breaking data into very small batches to simulate real-time.

B) Deleting small data.

C) Using a micro-computer.

D) Manual entry.

Answer: A

110. Which framework supports "Exploratory Data Analysis" (EDA) better?

A) ELT

B) ETL

Answer: A

111. What is "Computational Overhead"?

A) The extra resources used for transforming data.

B) The height of the server.

C) The price of a mouse.

D) The weight of a disk.

Answer: A

112. In modern ELT, what is the role of a "Semantic Layer"?

A) Defining business metrics once for all users.

B) Translating code to English.

C) Deleting tables.

D) Storing images.

Answer: A

113. What is "Data Decay"?

A) Data becoming less useful/accurate over time.

B) Data rotting like food.

C) Data being deleted.

D) Data moving to the cloud.

Answer: A

114. How does ELT support "Data Observability"?

A) Easier to check quality because raw and transformed data coexist.

B) It doesn't.

C) By using a camera.

D) By hiding data.

Answer: A

115. What is "Partition Pruning" in ELT transformations?

A) Querying only relevant parts of a table to save cost.

B) Deleting partitions.

C) Combining all partitions.

D) Renaming a partition.

Answer: A

116. Which framework is better for "Multi-structured" data?

A) ELT

B) ETL

Answer: A

117. What is "Data Democratization"?

A) Giving all users access to data for analysis.

B) Voting on data.

C) Making data free.

D) Deleting data.

Answer: A

118. In ETL, what is "Lookup"?

A) Checking a value in another table during transformation.

B) Looking at the ceiling.

C) Finding a file.

D) Searching Google.

Answer: A

119. What is "SCD Type 2" in a transformation?

A) Keeping full history by adding new rows for changes.

B) Overwriting old data.

C) Deleting old data.

D) Ignoring changes.

Answer: A

120. What is "SCD Type 1"?

A) Overwriting old data with new data.

B) Keeping history.

C) Deleting data.

D) Moving data.

Answer: A

121. Does ELT require a "Staging Database"?

A) Often, but it’s usually within the same Warehouse system.

B) Never.

C) Always a separate physical server.

D) Only for small data.

Answer: A

122. What is "Change Data Capture" (CDC) in ELT?

A) Stream of changes from a source DB into a warehouse table.

B) Deleting changes.

C) Capturing photos of data.

D) Manual updates.

Answer: A

123. Which framework handles "Schema Drift" more gracefully?

A) ELT (Ingest everything, fix SQL later).

B) ETL.

Answer: A

124. What is "Dark Data"?

A) Data that is collected but not used for analysis.

B) Encrypted data.

C) Data in a dark room.

D) Deleted data.

Answer: A

125. Is "AWS Glue" an ETL or ELT service?

A) Primarily ETL, but can do ELT.

B) Only ELT.

C) Neither.

D) A physical glue.

Answer: A

126. What is "Data Volatility"?

A) How often data changes.

B) How fast data moves.

C) The cost of data.

D) The size of data.

Answer: A

127. What is "Horizontal Partitioning"?

A) Sharding rows across different tables/nodes.

B) Splitting columns.

C) Deleting data.

D) Moving data.

Answer: A

128. What is "Vertical Partitioning"?

A) Splitting columns into different tables.

B) Splitting rows.

C) Deleting columns.

D) Moving tables.

Answer: A

129. Which framework is more "Cost-predictable"?

A) ETL (Usually fixed license/server cost).

B) ELT (Cloud compute can spike).

Answer: A

130. What is "Late-arriving facts"?

A) Data that arrives after the batch is processed.

B) Data that is wrong.

C) Data that is too early.

D) Data that is deleted.

Answer: A

131. In ELT, what is "Materialized View"?

A) A pre-computed table stored on disk to speed up queries.

B) A temporary view.

C) A hidden table.

D) A broken view.

Answer: A

132. What is "Computational Storage"?

A) Storage that can process data (helps ELT).

B) A hard drive.

C) A calculator.

D) Cloud storage.

Answer: A

133. Does ELT support "Unsupervised Learning" better?

A) Yes (access to raw features).

B) No.

Answer: A

134. What is "Data Orchestration" vs "Data Pipeline"?

A) Orchestration is the manager; Pipeline is the path.

B) They are the same.

C) Orchestration is manual; Pipeline is automatic.

D) None.

Answer: A

135. What is "Airbyte"?

A) An ELT/Extraction tool.

B) A type of airplane.

C) A data format.

D) A cloud provider.

Answer: A

136. What is "Fivetran"?

A) A managed ELT/Ingestion service.

B) A fast train.

C) A coding language.

D) A database.

Answer: A

137. What is "Data Bloat"?

A) Unnecessary storage of redundant data (common in ELT).

B) Data being deleted.

C) Data being fast.

D) Data being small.

Answer: A

138. Which framework is better for "Cold Data"?

A) ETL

B) ELT

Answer: B

139. What is "Serverless ETL"?

A) Running transformations without managing servers (e.g., Glue).

B) No computers used.

C) Manual transformation.

D) Desktop software.

Answer: A

140. What is "Data Refresh Rate"?

A) How often data is updated in the dashboard.

B) How fast the screen is.

C) The cost of update.

D) Number of users.

Answer: A

141. Is "Excel" technically an ETL tool?

A) Yes (Power Query).

B) No.

Answer: A

142. What is "Event Sourcing" in pipelines?

A) Storing every state change as an event.

B) Deleting events.

C) Finding events on a calendar.

D) Manual entry.

Answer: A

143. Which framework is more "Vendor-locked"?

A) ELT (Tied to Warehouse SQL).

B) ETL.

Answer: A

144. What is "SQL-on-Hadoop"?

A) Using SQL to query files in HDFS.

B) Deleting Hadoop.

C) A new database.

D) Moving data to SQL.

Answer: A

145. What is "Data Compression" in Load?

A) Shrinking data to save bandwidth.

B) Making data larger.

C) Deleting data.

D) Renaming data.

Answer: A

146. Is "Pentaho" an ETL or ELT tool?

A) ETL

B) ELT

Answer: A

147. What is "Data Portability"?

A) Ease of moving data between systems.

B) Carrying a server.

C) A small database.

D) Deleting data.

Answer: A

148. In ELT, what is "Flattening"?

A) Converting nested JSON/XML into rows and columns.

B) Compressing a file.

C) Deleting data.

D) Moving data.

Answer: A

149. What is "Parallel Loading"?

A) Loading multiple files at once.

B) Loading one by one.

C) Not loading.

D) Deleting loads.

Answer: A

150. Which framework is the "Modern" standard?

A) ELT (for cloud-scale analytics).

B) Manual ETL.

Answer: A

Subtopic 3: Data Lakes vs. Data Warehouses (Full 150)

Phase 1: Easy Level (1–50)

1. Which system is designed primarily for structured data?

A) Data Lake

B) Data Warehouse

C) NoSQL Database

D) Flat File

Answer: B

2. A Data Lake is best described as a repository for:

A) Only cleaned and processed data

B) Raw data in its native format

C) Deleted backup files

D) Website source code only

Answer: B

3. Which system typically follows a "Schema-on-Write" approach?

A) Data Lake

B) Data Warehouse

C) Text File

D) Public Cloud

Answer: B

4. Which system typically follows a "Schema-on-Read" approach?

A) Data Warehouse

B) Data Lake

C) Relational Database

D) ERP System

Answer: B

5. What is the primary storage format for a Data Warehouse?

A) Structured (Tables)

B) Unstructured (Images/Video)

C) Paper Logs

D) Audio Streams

Answer: A

6. Which is generally more expensive to scale for massive raw data?

A) Data Lake

B) Data Warehouse

C) Local Drive

D) USB Stick

Answer: B

7. Who is the primary user of a Data Warehouse for business reporting?

A) Data Scientists

B) Business Analysts

C) Graphic Designers

D) System Administrators

Answer: B

8. Who is the primary user of a Data Lake for data exploration and ML?

A) Sales Teams

B) Data Scientists and Engineers

C) Accountants

D) Human Resources

Answer: B

9. What does the term "Data Swamp" refer to?

A) A very large Data Warehouse

B) A poorly managed Data Lake that is hard to navigate

C) A cloud data center in a humid climate

D) A type of computer virus

Answer: B

10. Which system is better for storing unstructured data like PDFs and videos?

A) Data Warehouse

B) Data Lake

C) Excel Spreadsheet

D) SQL Server

Answer: B

11. Which system contains highly curated and "cleaned" data?

A) Data Lake

B) Data Warehouse

C) Scratchpad

D) Buffer

Answer: B

12. Amazon S3 and Azure Data Lake Storage (ADLS) are examples of:

A) Data Warehouses

B) Data Lakes

C) Text Editors

D) Compilers

Answer: B

13. Snowflake and Google BigQuery are primarily known as:

A) Data Lakes

B) Cloud Data Warehouses

C) Operating Systems

D) Web Browsers

Answer: B

14. What is the main benefit of a Data Lake?

A) Extreme organization

B) Low-cost storage and flexibility for all data types

C) Automatic report generation

D) Deletes data automatically

Answer: B

15. What is the main benefit of a Data Warehouse?

A) It stores raw video files

B) High performance for structured business queries

C) It has no fixed structure

D) It is always free

Answer: B

16. In which system is data "transformed" before it is saved?

A) Data Lake

B) Data Warehouse

C) Data Stream

D) Data Queue

Answer: B

17. True or False: A Data Lake can store data even if its future use is unknown.

A) True

B) False

Answer: A

18. Which architecture is considered more "Rigid"?

A) Data Lake

B) Data Warehouse

C) Neither

D) Both

Answer: B

19. Which system is optimized for "OLAP" (Online Analytical Processing)?

A) Data Lake

B) Data Warehouse

C) Cache

D) Buffer

Answer: B

20. What is the common cost model for a Data Lake?

A) High cost per query

B) Low-cost storage (pay per GB)

C) Fixed price per user

D) Free for everyone

Answer: B

21. Can a modern Data Warehouse store semi-structured data like JSON?

A) No, never

B) Yes, most modern warehouses support JSON

C) Only if printed on paper

D) Only if converted to CSV

Answer: B

22. Which system is better for "historical" point-in-time financial reporting?

A) Data Lake

B) Data Warehouse

C) Message Queue

D) Temp File

Answer: B

23. What is "Metadata" in a Data Lake?

A) The actual content of the files

B) Information about the files (labels, source, date)

C) The monthly cloud bill

D) The admin's username

Answer: B

24. Which is usually "Centralized" for an entire corporation?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: C

25. Which one is faster to "set up" for initial raw ingestion?

A) Data Warehouse

B) Data Lake

C) ERP System

D) CRM System

Answer: B

26. Data in a Warehouse is organized into:

A) Folders and Files

B) Tables (Rows and Columns)

C) Loose Documents

D) Audio Clips

Answer: B

27. Data in a Lake is organized into:

A) Strict SQL Tables

B) Folders and Files (Object Storage)

C) One single massive file

D) It cannot be organized

Answer: B

28. Which system is "Purpose-built" for specific questions?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: B

29. Which system is "General Purpose" for all data?

A) Data Lake

B) Data Warehouse

C) Neither

D) Both

Answer: A

30. What is the "Loading" process for a Warehouse called?

A) Streaming

B) ETL

C) Browsing

D) Surfing

Answer: B

31. What is the "Loading" process for a Data Lake usually called?

A) Careful Cleaning

B) Raw Ingestion

C) Manual Entry

D) Deletion

Answer: B

32. Which is better for AI and Machine Learning model training?

A) Data Warehouse

B) Data Lake

C) Calculator

D) Post-it notes

Answer: B

33. Which system typically has stricter "Data Governance"?

A) Data Lake

B) Data Warehouse

C) Public Folder

D) Desktop

Answer: B

34. Is a Data Lake a complete replacement for a Data Warehouse?

A) Yes, always

B) No, they are often used together (complementary)

C) Only for small companies

D) Only for tech startups

Answer: B

35. Which system is more likely to use "Hadoop" or "Spark"?

A) Data Warehouse

B) Data Lake

C) Microsoft Word

D) Powerpoint

Answer: B

36. Which system is more likely to use "SQL" as the only interface?

A) Data Lake

B) Data Warehouse

C) Image Editor

D) Video Player

Answer: B

37. What is "Structured Data"?

A) Data in tables with defined types

B) Photos and Videos

C) Audio recordings

D) Random text paragraphs

Answer: A

38. What is "Semi-structured Data"?

A) SQL Tables

B) JSON and XML

C) Hard drives

D) Printed books

Answer: B

39. What is "Unstructured Data"?

A) Excel sheets

B) Videos, Images, PDFs

C) Integers only

D) Boolean values

Answer: B

40. In a Warehouse, a change in source structure usually causes:

A) Faster queries

B) The pipeline to break/fail

C) No change at all

D) The data to delete itself

Answer: B

41. In a Lake, a change in source structure usually:

A) Breaks the whole system

B) Has minimal impact on ingestion

C) Costs extra money immediately

D) Changes the UI color

Answer: B

42. Which is more "Agile" for changing business requirements?

A) Data Lake

B) Data Warehouse

C) Static PDF

D) Paper Ledger

Answer: A

43. Which system provides a "Single Version of the Truth"?

A) Data Lake

B) Data Warehouse

C) Social Media

D) Email

Answer: B

44. What is the "Source of Truth" for raw, unedited data?

A) Data Warehouse

B) Data Lake

C) Dashboard

D) PowerPoint

Answer: B

45. Which one follows a "Bottom-up" architecture?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: B

46. Which one follows a "Top-down" architecture?

A) Data Warehouse

B) Data Lake

C) Neither

D) Both

Answer: A

47. "Delta Lake" adds warehouse-like features (ACID) to a:

A) Data Lake

B) Hard Drive

C) Memory Stick

D) Website

Answer: A

48. Which system focuses on query performance and efficiency?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: A

49. Which system focuses on storage durability and low cost?

A) Data Warehouse

B) Data Lake

C) RAM

D) Cache

Answer: B

50. Can you use both together in a "Lakehouse" architecture?

A) Yes

B) No

Answer: A

Phase 2: Medium Level (51–100)

51. What is the primary reason for a "Data Swamp"?

A) Lack of storage space

B) Lack of metadata and governance

C) High electricity costs

D) Using too much SQL

Answer: B

52. "ACID" compliance (Atomicity, Consistency, Isolation, Durability) is a core feature of:

A) Traditional Data Lakes

B) Data Warehouses

C) CSV files

D) Text editors

Answer: B

53. Which system is better for "Exploratory Data Analysis" (EDA)?

A) Data Warehouse

B) Data Lake

C) Static Report

D) Email

Answer: B

54. In a Data Lake, "Data Governance" is usually applied:

A) At the time of ingestion

B) At the time of consumption (Read)

C) Never

D) By the cloud provider only

Answer: B

55. Which system handles "Schema Drift" more easily?

A) Data Warehouse

B) Data Lake

C) Relational Database

D) Mainframe

Answer: B

56. A "Data Mart" is a subset of which system?

A) Data Lake

B) Data Warehouse

C) Data Stream

D) Data Queue

Answer: B

57. What is the main storage layer for a cloud Data Lake on AWS?

A) Redshift

B) S3

C) EC2

D) RDS

Answer: B

58. What is the main storage layer for a cloud Data Warehouse on AWS?

A) Redshift

B) S3

C) Glacier

D) EBS

Answer: A

59. Which system is better for real-time sensor data ingestion?

A) Data Warehouse

B) Data Lake

C) Excel

D) Word

Answer: B

60. Which system typically uses "Massively Parallel Processing" (MPP)?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: C

61. What is "Schema-on-Write"?

A) Defining the structure before loading the data

B) Defining the structure after loading the data

C) Not using a schema

D) Writing the schema on a piece of paper

Answer: A

62. What is "Schema-on-Read"?

A) Enforcing a structure during the load

B) Applying a structure to raw data when it is queried

C) Deleting the schema when reading

D) Reading without a schema

Answer: B

63. Which framework is often used to query a Data Lake using SQL?

A) Amazon Athena

B) Photoshop

C) Slack

D) Zoom

Answer: A

64. In a Data Lakehouse, what does the "Gold" layer contain?

A) Raw data

B) Cleaned, aggregated, business-ready data

C) Deleted data

D) Encrypted data

Answer: B

65. In a Data Lakehouse, what does the "Bronze" layer contain?

A) Raw, unedited data

B) Final reports

C) User passwords

D) Cloud bills

Answer: A

66. Which architecture is best for "Time Travel" (querying past versions of data)?

A) Basic Data Lake

B) Modern Data Warehouse / Lakehouse (e.g., Delta Lake)

C) Excel

D) CSV

Answer: B

67. Why is "Data Discovery" harder in a Data Lake?

A) There is no data

B) Data is raw and lacks a central catalog

C) Data is too small

D) Data is only in one language

Answer: B

68. Which system is more "Scalable" for compute and storage independently?

A) Data Warehouse (traditionally coupled)

B) Data Lake (natively decoupled)

C) Both are same

D) Neither

Answer: B

69. "Parquet" and "ORC" are columnar formats usually stored in:

A) Data Warehouses

B) Data Lakes

C) Both

D) Neither

Answer: B

70. What is the "Compute" in a Data Lake?

A) The storage disk

B) Engines like Spark, Presto, or Athena

C) The data itself

D) The cloud bill

Answer: B

71. Which system is optimized for "High Concurrency" (many users at once)?

A) Data Warehouse

B) Data Lake

C) USB Drive

D) Floppy Disk

Answer: A

72. What is "Data Lineage"?

A) The cost of data

B) The history and path of data from Lake to Warehouse

C) The name of the data

D) The size of the data

Answer: B

73. Which system is better for "Regulatory Compliance" (GDPR)?

A) Data Warehouse (Easier to track/delete specific rows)

B) Data Lake (Harder to find specific rows in raw files)

C) Neither

D) Both

Answer: A

74. What is "Object Storage"?

A) Storing data as tables

B) Storing data as discrete units (objects) with metadata (used in Lakes)

C) Storing data on a CD

D) Storing data in RAM

Answer: B

75. Which system is better for "Large-scale Joins" on billions of rows?

A) Data Warehouse

B) Data Lake

C) Both (depending on engine)

D) Neither

Answer: A

76. What is "Partitioning" in a Data Lake?

A) Deleting files

B) Organizing files into folders by date/category (e.g., year=2023/)

C) Combining files

D) Renaming files

Answer: B

77. What is "Small File Problem" in Data Lakes?

A) Files are too big

B) Too many tiny files causing slow query performance

C) Files are deleted

D) Files are empty

Answer: B

78. Which system is "Schema-Rigid"?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: B

79. Which system is "Schema-Flexible"?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: A

80. What is a "Data Catalog"?

A) A list of employees

B) A central directory that helps users find and understand data in a Lake

C) A price list

D) A type of database

Answer: B

81. In a Warehouse, "Transform" happens before "Load" (ETL). In a Lake, it is often:

A) ELT

B) ETL

C) Only E

D) Only L

Answer: A

82. Which system has higher "Ingestion Speed"?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: B

83. Which system has higher "Query Speed" for business dashboards?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: A

84. What is "Data Virtualization"?

A) Querying a Warehouse and Lake together without moving data

B) Using VR goggles

C) Deleting the cloud

D) Using a fake database

Answer: A

85. "HDFS" stands for:

A) High Data File System

B) Hadoop Distributed File System

C) Hard Drive File System

D) Hidden Data File System

Answer: B

86. Which system is "Decoupled"?

A) Modern Cloud Data Lake

B) Traditional On-prem Warehouse

C) Both

D) Neither

Answer: A

87. What is the "Silver" layer in a Lakehouse?

A) Raw data

B) Cleaned and filtered data

C) Business aggregates

D) Deleted data

Answer: B

88. Can a Data Lake store "Images"?

A) Yes

B) No

Answer: A

89. Can a Data Warehouse store "Images"?

A) Not efficiently (usually only links to images)

B) Yes, natively

Answer: A

90. Which is better for "Predictive Modeling"?

A) Data Warehouse

B) Data Lake

C) Calculator

D) Post-it

Answer: B

91. Which is better for "Financial Auditing"?

A) Data Warehouse

B) Data Lake

C) Neither

D) Both

Answer: A

92. What is "Data Granularity"?

A) The size of the grains

B) The level of detail in the data

C) The price of data

D) The speed of data

Answer: B

93. Which system usually has "Low Granularity" (summarized)?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: A

94. Which system usually has "High Granularity" (raw detail)?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: B

95. Is "Snowflake" a Data Lake?

A) No, but it has "Data Lake" capabilities

B) Yes, strictly

C) No, strictly

D) It is a web browser

Answer: A

96. Is "Databricks" a Data Warehouse?

A) No, it started as a Data Lake tool but created the "Lakehouse"

B) Yes, strictly

C) No, strictly

D) It is an email service

Answer: A

97. What is "Schema Enforcement"?

A) Checking data against a schema before allowing it in (Warehouse)

B) Deleting the schema

C) Writing a new schema

D) Ignoring the schema

Answer: A

98. What is "Schema Inference"?

A) Guessing the structure of raw data during a query (Lake)

B) Forcing a structure

C) Deleting the structure

D) Writing the structure

Answer: A

99. Which is more "Future-proof"?

A) Data Lake (Stores everything raw for future tools)

B) Data Warehouse

C) Neither

D) Both

Answer: A

100. In a Data Lake, "ETL" usually happens:

A) Before the Lake

B) Inside the Lake (T happening after L)

C) Never

D) In a text file

Answer: B

Phase 3: Master Level (101–150)

101. What is the "Medallion Architecture"?

A) A way to win a data competition

B) A framework for organizing a Data Lake into Bronze, Silver, and Gold layers

C) A type of database index

D) A cloud billing strategy

Answer: B

102. In a "Data Mesh", how is a Data Lake handled?

A) It is centralized for everyone

B) It is decentralized into domain-owned "Data Products"

C) It is deleted

D) It is replaced by Excel

Answer: B

103. What is "Data Compaction" in a Data Lake?

A) Deleting data

B) Merging small files into larger ones to optimize read performance

C) Compressing a single file

D) Making the disk smaller

Answer: B

104. Which consistency model do most Cloud Data Lakes (S3) follow?

A) Strong Consistency

B) Eventual Consistency (though S3 is now Strong for most operations)

C) No Consistency

D) Weekend Consistency

Answer: B

105. What is a "Z-Order" index in a Lakehouse?

A) Sorting data alphabetically

B) A technique to co-locate related information in the same files for faster multi-dimensional queries

C) A way to delete data

D) A type of encryption

Answer: B

106. How does "Delta Lake" provide ACID transactions?

A) Using a transaction log (JSON) to track all changes

B) Using a physical lock on the server

C) Using a secret password

D) It doesn't

Answer: A

107. What is "Pushdown Optimization" in a Data Lake query engine?

A) Pushing a button

B) Offloading filtering and aggregation to the storage layer to reduce data transfer

C) Deleting the query

D) Speeding up the CPU

Answer: B

108. What is the "CAP Theorem" implication for a distributed Data Warehouse?

A) You must choose between Consistency, Availability, and Partition Tolerance

B) It only applies to Lakes

C) It only applies to local drives

D) It is a marketing term

Answer: A

109. What is "Vectorized Execution" in modern Warehouse engines?

A) Using a drawing tool

B) Processing a block of rows (a vector) at once using CPU SIMD instructions

C) Moving data in a straight line

D) A type of security

Answer: B

110. Which is a "Shared-Nothing" architecture?

A) Every node has its own CPU, Memory, and Disk (e.g., Teradata)

B) All nodes share one disk

C) No one uses the system

D) Using a USB drive

Answer: A

111. What is "Multi-cluster Warehouse"?

A) Using many different cloud providers

B) Automatically spinning up extra compute clusters to handle high user concurrency

C) Using a very large monitor

D) Deleting clusters

Answer: B

112. What is "Iceberg" in the context of Data Lakes?

A) A piece of ice

B) An open table format for huge analytic datasets

C) A cold storage service

D) A type of computer virus

Answer: B

113. What is "Hudi"?

A) A brand of hoodie

B) An open-source framework to manage ingestion and storage in Data Lakes

C) A database engine

D) A programming language

Answer: B

114. How does a Data Lakehouse handle "Upserts" (Update + Insert)?

A) It can't

B) By rewriting files or using delta logs to merge changes

C) By deleting the table

D) By manual entry

Answer: B

115. What is "Data Shuffling" in Spark-based Data Lake queries?

A) Randomizing the data

B) Moving data between nodes across the network during a join or group-by

C) Deleting the data

D) Encrypting the data

Answer: B

116. What is "Broadcast Join"?

A) Joining a TV station

B) Sending a small table to all nodes to join with a large table locally

C) Joining two massive tables

D) Joining data via radio

Answer: B

117. What is "Data Governance" vs "Data Management"?

A) Governance is the rules; Management is the execution

B) They are the same

C) Governance is for Lakes; Management is for Warehouses

D) Governance is manual; Management is automatic

Answer: A

118. Which architecture minimizes "Data Silos"?

A) Centralized Data Lake

B) Decentralized Data Mesh

C) Both (if implemented correctly)

D) Neither

Answer: C

119. What is "Dark Data"?

A) Encrypted data

B) Data that is collected but not analyzed or used

C) Data in a dark room

D) Deleted data

Answer: B

120. What is "Write Amplification" in a Data Lake?

A) Making text larger

B) When a small update forces the rewriting of a very large file

C) Increasing the number of writes

D) Using a louder disk

Answer: B

121. What is "Predicate Pushdown"?

A) Pushing a button

B) Filtering data at the storage layer so only relevant data is read into memory

C) Filtering data after the query finishes

D) Not using filters

Answer: B

122. What is "Star Schema"?

A) A drawing of a star

B) A Warehouse design with a central Fact table and surrounding Dimension tables

C) A type of Data Lake

D) A security protocol

Answer: B

123. What is "Snowflake Schema"?

A) A schema that melts

B) A Star schema where dimensions are further normalized into sub-dimensions

C) A type of cloud provider

D) A database for snow data

Answer: B

124. What is "Normalizing" data?

A) Making it normal

B) Reducing redundancy by splitting tables (common in Warehouses)

C) Combining all tables

D) Deleting data

Answer: B

125. What is "Denormalizing" data?

A) Making it weird

B) Combining tables into one large table for faster reads (common in Lakes/Analytics)

C) Splitting tables

D) Encrypting tables

Answer: B

126. What is "Data Sovereignty"?

A) Data speed

B) Legal requirement that data stays within national borders

C) Data owned by a king

D) Free data

Answer: B

127. What is "Data Masking"?

A) Hiding data with a mask

B) Obfuscating sensitive data (PII) so it can't be seen by unauthorized users

C) Deleting sensitive data

D) Encrypting the whole disk

Answer: B

128. What is "RBAC" (Role-Based Access Control)?

A) Access based on user names

B) Access based on job roles (Admin, Analyst, etc.)

C) Access based on age

D) Access based on price

Answer: B

129. What is "ABAC" (Attribute-Based Access Control)?

A) Access based on alphabetical order

B) Access based on specific attributes (e.g., "Must be in US and HR department")

C) Access based on bank balance

D) Access based on computer speed

Answer: B

130. What is "Data Profiling"?

A) Creating a social media profile

B) Statistically analyzing data to find quality issues, nulls, and distributions

C) Selling data

D) Encrypting data

Answer: B

131. Which system is "Compute-Intensive"?

A) Data Warehouse

B) Data Lake

C) Both

D) Neither

Answer: C

132. What is "Storage-Intensive"?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: A

133. What is "Columnar Compression"?

A) Making columns shorter

B) Compressing similar data within a column to save massive space

C) Deleting columns

D) Renaming columns

Answer: B

134. What is "Bloom Filter" in a query engine?

A) A type of flower

B) A probabilistic structure that tells the engine if a value definitely isn't in a file

C) A color filter

D) A data cleaner

Answer: B

135. What is "Compute Isolation"?

A) Putting the server in a room alone

B) Giving different workloads separate compute resources so they don't slow each other down

C) Turning off the compute

D) Hiding the compute

Answer: B

136. What is "Tiered Storage"?

A) Moving data between Hot, Cool, and Archive tiers based on usage

B) Storing data in a wedding cake shape

C) Deleting data tiers

D) Using only one disk

Answer: A

137. Which system is better for "Semi-structured Web Logs"?

A) Data Warehouse

B) Data Lake

C) Neither

D) Both

Answer: B

138. What is "Idempotent Ingestion"?

A) Ingesting data once

B) Ensuring that ingesting the same file twice doesn't create duplicate data

C) Deleting data during ingestion

D) Fast ingestion

Answer: B

139. What is "Data Virtualization" vs "Data Federation"?

A) Federation is a type of virtualization across multiple databases

B) They are the same

C) Federation is manual; Virtualization is automatic

D) Federation is for SQL; Virtualization is for NoSQL

Answer: A

140. What is "Metadata Harvesting"?

A) Planting metadata

B) Automatically collecting metadata from various sources into a catalog

C) Deleting metadata

D) Selling metadata

Answer: B

141. What is "Semantic Layer"?

A) Translating code to English

B) An abstraction layer that uses business terms instead of complex table joins

C) A layer of comments

D) The top of the lake

Answer: B

142. Which system is better for "Exploratory AI"?

A) Data Lake

B) Data Warehouse

C) Both

D) Neither

Answer: A

143. What is "Data Freshness"?

A) How clean data is

B) The time difference between an event happening and the data being queryable

C) The age of the server

D) Data that is never used

Answer: B

144. What is "Cold Data"?

A) Data in the fridge

B) Data that is rarely accessed

C) Data from a cold country

D) Encrypted data

Answer: B

145. What is "Hot Data"?

A) Overheating servers

B) Data that is accessed very frequently (last 24 hours)

C) Data from a desert

D) Fast data

Answer: B

146. Which system is more "Immutable"?

A) Data Lake (Original files are rarely changed, only new ones added)

B) Data Warehouse

C) Both

D) Neither

Answer: A

147. What is "Data Obfuscation"?

A) Making data clear

B) Hiding data values to protect privacy (e.g., credit card ****)

C) Deleting data

D) Speeding up data

Answer: B

148. What is "Partition Overload"?

A) Having too many partitions in a table causing query planning to slow down

B) Deleting partitions

C) Not having enough partitions

D) Renaming partitions

Answer: A

149. Which is a "Cloud-Native" Data Warehouse?

A) Oracle on a server

B) Snowflake

C) Excel

D) CSV

Answer: B

150. What is the ultimate goal of the "Data Lakehouse"?

A) To replace the cloud

B) To combine the performance of a Warehouse with the flexibility/cost of a Lake

C) To delete all data

D) To make SQL obsolete

Answer: B

Subtopic 4: Apache Spark Basics (Full 150)

Phase 1: Easy Level (1–50)

1. What is Apache Spark?

A) A physical hardware server

B) A unified engine for large-scale data processing

C) A specialized video editing software

D) A type of network cable

Answer: B

2. Spark is primarily known for being _____ than Hadoop MapReduce.

A) Slower

B) Faster

C) Heavier

D) More expensive

Answer: B

3. What does "RDD" stand for in Spark?

A) Real-time Data Disk

B) Resilient Distributed Dataset

C) Random Data Distribution

D) Remote Database Driver

Answer: B

4. Spark stores intermediate data in _____ for faster processing.

A) Magnetic tapes

B) Memory (RAM)

C) Paper logs

D) External CDs

Answer: B

5. Which programming language is Spark natively written in?

A) Python

B) Scala

C) C++

D) PHP

Answer: B

6. Which Spark component is used for working with structured data using SQL?

A) Spark GraphX

B) Spark SQL

C) Spark MLlib

D) Spark Streaming

Answer: B

7. What is a "Spark Cluster"?

A) A single laptop running Spark

B) A group of connected computers working together

C) A specific type of folder

D) A collection of Spark books

Answer: B

8. Who is the "Master" in a Spark Cluster?

A) The person who wrote the code

B) The node that coordinates the work

C) The fastest hard drive

D) The internet provider

Answer: B

9. What are "Worker Nodes"?

A) Computers that do the actual data processing tasks

B) People who fix the servers

C) Cables connecting the machines

D) Backup batteries

Answer: A

10. What is "PySpark"?

A) A Spark version for Java

B) The Python API for Spark

C) A Spark tool for making fire

D) A database engine

Answer: B

11. What is a "DataFrame" in Spark?

A) A physical frame for a monitor

B) A distributed collection of data organized into named columns

C) A type of digital photo

D) A list of passwords

Answer: B

12. Which Spark component handles machine learning?

A) Spark MLlib

B) Spark SQL

C) Spark GraphX

D) Spark Core

Answer: A

13. Which Spark component is used for live data processing?

A) Spark Batch

B) Spark Streaming

C) Spark Offline

D) Spark Archive

Answer: B

14. What is a "Transformation" in Spark?

A) Deleting the entire cluster

B) An operation that creates a new dataset from an existing one

C) Changing the server's color

D) Renaming a file

Answer: B

15. Transformations in Spark are "Lazy." What does this mean?

A) They never finish

B) They are only executed when an "Action" is called

C) They use very little power

D) They are done by humans

Answer: B

16. Which of these is an "Action" in Spark?

A) map()

B) count()

C) filter()

D) flatMap()

Answer: B

17. Which of these is a "Transformation"?

A) collect()

B) show()

C) filter()

D) saveAsTextFile()

Answer: C

18. What is the "Driver Program"?

A) The program that runs the main() function and creates the SparkContext

B) A program that drives a car

C) A device driver for a mouse

D) A security firewall

Answer: A

19. What is an "Executor"?

A) A judge in a court

B) A process on worker nodes that runs tasks and stores data

C) A tool to delete code

D) A network router

Answer: B

20. What is the "SparkContext"?

A) The entry point to Spark functionality

B) A text file for notes

C) A hardware button

D) A type of cloud storage

Answer: A

21. What is "Partitioning" in Spark?

A) Splitting a dataset into smaller chunks for parallel processing

B) Deleting parts of the data

C) Combining all data into one row

D) Encrypting the data

Answer: A

22. How many partitions are usually needed for a Spark job?

A) Always 1

B) Depends on the number of CPU cores and data size

C) Exactly 100

D) Zero

Answer: B

23. What is "Fault Tolerance" in Spark?

A) The ability to ignore all errors

B) The ability to recover lost data automatically using RDD lineage

C) Paying for insurance

D) Preventing the user from typing

Answer: B

24. Spark can run on which of these cluster managers?

A) Hadoop YARN

B) Microsoft Word

C) Adobe Reader

D) Spotify

Answer: A

25. What is "Standalone Mode" in Spark?

A) Running Spark on a computer with no internet

B) Spark’s own simple cluster manager

C) Spark running inside a browser

D) A Spark version for phones

Answer: B

26. What does "Immutability" mean for RDDs?

A) They change every second

B) They cannot be modified after they are created

C) They are invisible

D) They are free to use

Answer: B

27. What is "Caching" in Spark?

A) Deleting data to save space

B) Keeping data in memory to reuse it later

C) Selling data for cash

D) Hiding data from the master

Answer: B

28. What is a "Job" in Spark?

A) A person’s career

B) A parallel computation consisting of multiple tasks

C) A type of variable

D) A server rack

Answer: B

29. What is a "Task" in Spark?

A) A long-term project

B) The smallest unit of work sent to an executor

C) A line of code

D) A user login

Answer: B

30. What is a "Stage" in Spark?

A) A place for a concert

B) A set of tasks that can run in parallel without data shuffling

C) A type of database

D) A Spark version

Answer: B

31. Which format is highly optimized for Spark?

A) TXT

B) Parquet

C) MP3

D) JPG

Answer: B

32. Can Spark read data from Amazon S3?

A) Yes

B) No

Answer: A

33. What is "GraphX" used for?

A) Drawing 3D models

B) Graph processing and social network analysis

C) Playing video games

D) Making pie charts

Answer: B

34. Spark SQL uses a component called "Catalyst." What is it?

A) A fuel for the server

B) A query optimizer

C) A type of keyboard

D) A backup generator

Answer: B

35. What is "Local Mode" in Spark?

A) Running Spark on a single machine for testing

B) Spark running in a specific city

C) Spark with no security

D) A manual way to process data

Answer: A

36. Is Spark a "Database"?

A) Yes

B) No, it is a processing engine

Answer: B

37. What is "Deployment Mode"?

A) How Spark is installed

B) Where the Spark driver runs (Client vs. Cluster)

C) The speed of the internet

D) The number of users

Answer: B

38. What is "Client Mode"?

A) The driver runs on the machine that submitted the job

B) Spark runs only for clients

C) A free version of Spark

D) Spark running on a phone

Answer: A

39. What is "Cluster Mode"?

A) The driver runs inside the cluster (on a worker node)

B) Spark runs on many clusters at once

C) A very expensive Spark version

D) Spark running on a hard drive

Answer: A

40. What is "Persistence" in Spark?

A) Never giving up

B) Storing an RDD/DataFrame on disk or memory for reuse

C) Deleting the RDD

D) Renaming the RDD

Answer: B

41. Which of these is a valid storage level?

A) MEMORY_ONLY

B) PAPER_ONLY

C) CD_ONLY

D) NO_STORAGE

Answer: A

42. What is "Narrow Transformation"?

A) Transformation that requires data from all partitions

B) Transformation where each input partition contributes to only one output partition

C) A transformation that makes data smaller

D) A very fast transformation

Answer: B

43. What is "Wide Transformation"?

A) Transformation that makes data wider

B) Transformation that requires data to be shuffled across the network

C) A transformation that uses a lot of RAM

D) A transformation for big screens

Answer: B

44. What is "Shuffling" in Spark?

A) Mixing data randomly

B) The process of redistributing data across partitions (expensive)

C) Playing music

D) Sorting data alphabetically

Answer: B

45. What is "Lineage" in Spark?

A) The family tree of Spark developers

B) The graph of all transformations used to build an RDD

C) The total size of the code

D) The physical path of the cables

Answer: B

46. Which tool can you use to monitor Spark jobs?

A) Spark UI (Web Browser)

B) Photoshop

C) Calculator

D) Task Manager

Answer: A

47. What is "Coalesce"?

A) Increasing the number of partitions

B) Decreasing the number of partitions efficiently

C) Deleting all partitions

D) Combining two different datasets

Answer: B

48. What is "Repartition"?

A) Deleting a partition

B) Reshuffling data to change the number of partitions (can increase or decrease)

C) Renaming a partition

D) Backing up a partition

Answer: B

49. Is Spark part of the Apache Software Foundation?

A) Yes

B) No

Answer: A

50. Can Spark process data in real-time?

A) Yes, via Spark Streaming/Structured Streaming

B) No, it is batch only

Answer: A

Phase 2: Medium Level (51–100)

51. What is the difference between a DataFrame and a Dataset?

A) DataFrames are untyped; Datasets provide type-safety (Scala/Java)

B) DataFrames are faster

C) Datasets are only for small data

D) They are the same thing

Answer: A

52. What is "Predicate Pushdown" in Spark?

A) Filtering data after it is loaded into memory

B) Moving filters to the data source level to reduce the data read

C) Pushing the "Run" button

D) Deleting the data filters

Answer: B

53. How does Spark handle "Data Skew"?

A) It ignores it

B) By using techniques like salting or repartitioning

C) By buying more servers

D) By deleting the skewed data

Answer: B

54. What is the "Catalyst Optimizer"?

A) A person who fixes Spark

B) A rule-based and cost-based engine that optimizes Spark SQL queries

C) A hardware acceleration card

D) A type of computer virus

Answer: B

55. What is "Broadcast Join" in Spark?

A) Joining a TV show

B) Sending a small table to all executors to avoid a large shuffle join

C) A join that uses radio waves

D) A join that deletes the small table

Answer: B

56. When should you use persist() instead of cache()?

A) When you want to specify a storage level (like DISK_ONLY)

B) When you have no RAM

C) Always

D) Never

Answer: A

57. What is the "DAG Scheduler"?

A) A tool for drawing graphs

B) A component that transforms a DAG of stages into a set of tasks

C) A tool to schedule meetings

D) A backup timer

Answer: B

58. What is a "Shuffle Map Stage"?

A) A stage that involves a shuffle operation

B) A stage for drawing maps

C) A stage that deletes data

D) A stage for testing

Answer: A

59. What is "Result Stage"?

A) The final stage that computes the result of an action

B) The first stage of a job

C) A stage with no data

D) A stage that fails

Answer: A

60. What is "Tungsten" in Spark?

A) A heavy metal in the server

B) An execution engine that optimizes memory and CPU efficiency

C) A type of battery

D) A coding language

Answer: B

61. What is the default number of partitions for Spark SQL shuffles?

A) 1

B) 200

C) 1000

D) 10

Answer: B

62. What is "Accumulator" in Spark?

A) A variable used for aggregating information across executors (like a counter)

B) A battery for the worker node

C) A specialized data scientist

D) A tool to delete rows

Answer: A

63. What is a "Broadcast Variable"?

A) A variable that changes every second

B) A read-only variable cached on each machine rather than sent with tasks

C) A variable used for music

D) A deleted variable

Answer: B

64. What is "Structured Streaming"?

A) Streaming video on Netflix

B) A stream processing engine built on the Spark SQL engine

C) A stream of plain text files

D) A manual way to move data

Answer: B

65. What is "Checkpointing" in Spark Streaming?

A) Checking the time

B) Saving the state of the stream to reliable storage for fault recovery

C) Stopping the stream

D) Deleting the stream logs

Answer: B

66. What is "UDF" in Spark?

A) Universal Data File

B) User Defined Function

C) Under Development Feature

D) Ultra Data Frame

Answer: B

67. Why are UDFs sometimes slower in PySpark?

A) Because they require data to move between the JVM and Python process

B) Because Python is a slow language

C) Because they use too much RAM

D) They are actually faster

Answer: A

68. What is "Vectorized UDF" (Pandas UDF)?

A) A UDF that uses drawings

B) A Python UDF that uses Apache Arrow to process data in batches efficiently

C) A UDF that only works on 1 row

D) A UDF for video files

Answer: B

69. What is "Dynamic Resource Allocation"?

A) Giving money to developers

B) Spark adjusting the number of executors based on the workload

C) Buying new servers every week

D) Manually setting the RAM

Answer: B

70. What is "Speculative Execution"?

A) Guessing what the user wants

B) Launching redundant tasks for slow-running tasks to finish faster

C) Betting on stock markets

D) Running code without testing

Answer: B

71. What is "Whole-Stage Code Generation"?

A) Writing code for a theater play

B) Collapsing a query into a single optimized function for the CPU

C) Generating code for every row

D) A manual coding process

Answer: B

72. What is "Adaptive Query Execution" (AQE)?

A) Queries that change based on user mood

B) Re-optimizing query plans during execution based on runtime statistics

C) A query that runs on phones

D) A slow query

Answer: B

73. In Spark memory management, what is "Storage Memory"?

A) Memory used for caching and internal data transfer

B) Memory used for the OS

C) Physical hard drive space

D) External USB space

Answer: A

74. In Spark memory management, what is "Execution Memory"?

A) Memory used for shuffles, joins, and aggregations

B) Memory used for writing code

C) Memory used for the master node only

D) Memory used for video streaming

Answer: A

75. What is the "Spilling" to disk?

A) Dropping liquid on the server

B) Writing data to disk when execution memory is full

C) Deleting data

D) Moving data to the cloud

Answer: B

76. What is "Py4J"?

A) A Python version for 4 years

B) A bridge that allows Python to call Java objects in Spark

C) A database

D) A network cable

Answer: B

77. What is "Apache Arrow"?

A) A weapon

B) A cross-language platform for in-memory data (used to speed up PySpark)

C) A type of cloud storage

D) A Spark component for graphs

Answer: B

78. What is the "Logical Plan"?

A) A plan that makes sense

B) An abstract representation of the computation steps

C) A physical map of the cluster

D) A list of project goals

Answer: B

79. What is the "Physical Plan"?

A) A plan for exercise

B) The actual executable strategy chosen by the Spark engine

C) The server room blueprint

D) The hardware specifications

Answer: B

80. What is "Explode" function in Spark SQL?

A) Crashing the database

B) Creating a new row for each element in an array or map

C) Deleting all columns

D) Encrypting the data

Answer: B

81. What is "Window Function" in Spark?

A) A function to clean windows

B) A function that calculates values over a specific range of rows

C) A function that opens a new tab

D) A function that works only on Windows OS

Answer: B

82. What is "Salting" in the context of Joins?

A) Adding seasoning to the server

B) Adding a random key to skewed data to distribute it more evenly

C) Deleting duplicate keys

D) Encrypting keys

Answer: B

83. What is "Spark-submit"?

A) A button on the Spark website

B) A utility to launch Spark applications on a cluster

C) A way to send emails to Spark developers

D) A manual code entry tool

Answer: B

84. What is spark.executor.memory?

A) The memory used by the driver

B) The amount of memory allocated to each executor process

C) The price of RAM

D) The speed of the CPU

Answer: B

85. What is spark.driver.memory?

A) The memory used for worker nodes

B) The amount of memory allocated to the driver process

C) The memory of the person using Spark

D) The cloud storage limit

Answer: B

86. What is "Data Locality" in Spark?

A) Processing data on the node where it is physically stored to reduce network traffic

B) Storing data in a local city

C) Using a local hard drive only

D) Writing data in a local language

Answer: A

87. What is "Serialization" in Spark?

A) Turning Spark into a TV series

B) Converting objects into a byte stream for network transfer or storage

C) Sorting data in a series

D) Deleting data one by one

Answer: B

88. Which is the more efficient serializer for Spark?

A) Java Serializer

B) Kryo Serializer

C) Text Serializer

D) PDF Serializer

Answer: B

89. What is "Micro-batching" in Spark Streaming?

A) Processing data in tiny discrete time intervals

B) Processing data once a month

C) Deleting small data

D) Manual data processing

Answer: A

90. What is "Watermarking" in Structured Streaming?

A) Adding a digital logo to the data

B) A way to handle late data by ignoring events older than a certain threshold

C) Measuring the depth of the data lake

D) A cooling system for the server

Answer: B

91. What is a "Trigger" in Spark Streaming?

A) An error that breaks the system

B) A policy that defines when the next batch of data should be processed

C) A person who starts the job

D) A specialized hardware button

Answer: B

92. What is "Exactly-Once" semantics?

A) Processing data at least once

B) Ensuring data is processed and reflected in the output exactly one time

C) Processing data only once a year

D) Deleting data after one use

Answer: B

93. What is "Bucketing" in Spark?

A) Putting data in a physical bucket

B) Storing data in pre-shuffled buckets to speed up future joins/aggregations

C) Deleting data in groups

D) Sorting data by size

Answer: B

94. What is "Spark Session"?

A) A meeting of Spark users

B) The modern unified entry point for Spark 2.0 and above

C) A single hour of Spark usage

D) A type of database connection

Answer: B

95. What is "Managed Table" in Spark SQL?

A) A table managed by a person

B) A table where Spark manages both metadata and data (deleting table deletes data)

C) A table with a manager

D) A very expensive table

Answer: B

96. What is "External Table" in Spark SQL?

A) A table stored in another city

B) A table where Spark only manages metadata (deleting table keeps the data)

C) A table on a USB drive

D) A table for external users only

Answer: B

97. What is "Z-Ordering"?

A) Alphabetical sorting

B) A data clustering technique to improve query performance in Lakehouses

C) A way to delete data

D) A Spark version

Answer: B

98. What is "Delta Lake"?

A) A lake with a delta

B) An open-source storage layer that brings ACID transactions to Spark

C) A cold storage service

D) A type of Spark graph

Answer: B

99. What is "Event Time" in streaming?

A) The time the data was processed

B) The time the event actually occurred at the source

C) The current time

D) The time the user logged in

Answer: B

100. What is "Processing Time" in streaming?

A) The time the event happened

B) The time the Spark engine actually processes the data

C) The time the server was built

D) The time it takes to write code

Answer: B

Phase 3: Master Level (101–150)

101. What is the "Two-Generals' Problem" in the context of Spark cluster coordination?

A) Two managers fighting over a project

B) The impossibility of guaranteeing agreement between two nodes over an unreliable link

C) A bug in the Scala compiler

D) A hardware failure in the master node

Answer: B

102. In "Tungsten," how does Spark avoid Java Garbage Collection (GC) overhead?

A) By not using Java

B) By managing memory explicitly "off-heap" using binary format

C) By restarting the server every hour

D) By using smaller variables

Answer: B

103. What is "Code Generation" (Janino)?

A) Using AI to write Spark apps

B) A library Spark uses to compile query plans into optimized Java bytecode at runtime

C) A way to generate random data

D) A manual coding tool

Answer: B

104. What is "Cost-Based Optimizer" (CBO)?

A) An optimizer that costs a lot of money

B) An optimizer that uses statistics (table size, histograms) to choose the best plan

C) An optimizer that reduces the cloud bill

D) A manual optimization process

Answer: B

105. What is the "Shuffle Service" (External)?

A) A music streaming app

B) A process that keeps shuffle files alive even if the executor dies

C) A tool to shuffle cards

D) A network speed test

Answer: B

106. What is "Sort-Merge Join" (SMJ)?

A) The default Spark join that sorts both tables and merges them (good for large data)

B) A very fast join for small data

C) A join that deletes data

D) A join that doesn't use a key

Answer: A

107. What is "Cartesian Join" (Cross Join)?

A) A join that combines every row of table A with every row of table B (very expensive)

B) A join for math scientists

C) A join that filters all rows

D) A join for small lists

Answer: A

108. What is "Data Locality: NODE_LOCAL"?

A) Data is on a different node

B) Data is on the same machine as the executor

C) Data is in a different city

D) Data is on a USB drive

Answer: B

109. What is "Data Locality: RACK_LOCAL"?

A) Data is on the same machine

B) Data is on a different machine but in the same rack

C) Data is in a different building

D) Data is on a different planet

Answer: B

110. How does "Dynamic Partition Pruning" work?

A) Deleting partitions randomly

B) Using results from one side of a join to skip partitions on the other side at runtime

C) Cutting branches off a tree

D) Renaming partitions dynamically

Answer: B

111. What is "Spark-on-Kubernetes"?

A) Running Spark on a ship

B) Using Kubernetes to manage Spark executors and driver containers

C) A Spark version for robots

D) A manual deployment method

Answer: B

112. What is "Memory Pressure" in Spark?

A) When the server is physically heavy

B) When the system is close to running out of RAM, causing GC issues or spills

C) High air pressure in the data center

D) Stress on the developers

Answer: B

113. What is "Off-Heap Memory"?

A) Memory outside the Java Virtual Machine's control

B) Memory on a different server

C) Memory on a hard drive

D) Memory used for the OS only

Answer: A

114. What is the "Block Manager"?

A) A person who manages a street block

B) A Spark component that manages the storage and retrieval of data blocks

C) A tool to block hackers

D) A folder organizer

Answer: B

115. What is "Task Deserialization"?

A) Deleting a task

B) The process of converting the byte stream back into a task object on the executor

C) Renaming a task

D) Scheduling a task

Answer: B

116. What is "Stage Skipping" in Spark?

A) When Spark skips work because it finds the data is already cached or computed

B) Skipping the testing phase

C) A bug in the code

D) A manual override

Answer: A

117. What is "Skew Join Optimization" in AQE?

A) Deleting skewed data

B) Automatically splitting skewed partitions into smaller sub-partitions to balance the load

C) Ignoring the skew

D) Buying a faster node

Answer: B

118. What is "Coalescing Partitions" in AQE?

A) Making partitions larger

B) Automatically merging small shuffle partitions at runtime to reduce overhead

C) Deleting partitions

D) Renaming partitions

Answer: B

119. What is "Task Metrics"?

A) The size of the monitor

B) Data about task execution (CPU time, shuffle read/write, memory used)

C) The number of tasks in a project

D) The price of a task

Answer: B

120. What is "Barrier Execution Mode"?

A) A mode for hackers

B) A mode used for deep learning where all tasks in a stage must start at the same time

C) A security wall

D) A slow execution mode

Answer: B

121. What is "Predicate" in Spark SQL?

A) A prediction of the future

B) The condition in a WHERE clause

C) A type of variable

D) A summary of the data

Answer: B

122. What is "Projection" in Spark SQL?

A) A 3D movie

B) Selecting specific columns from a dataset

C) Predicting the data size

D) Showing data on a wall

Answer: B

123. What is "User-Defined Aggregate Function" (UDAF)?

A) A UDF that works on groups of rows (like custom SUM or AVG)

B) A function to delete aggregates

C) A built-in function

D) A function for the user interface

Answer: A

124. What is "Apache Spark Connect"?

A) A social network for Spark

B) A thin client protocol that allows connecting to Spark from anywhere

C) A physical cable

D) A phone app

Answer: B

125. What is "Log4j" in Spark?

A) A Spark version

B) The logging framework used by Spark (famous for security vulnerabilities)

C) A database

D) A type of file

Answer: B

126. What is "Driver OOM" (Out of Memory)?

A) When the worker node crashes

B) When the driver process runs out of RAM (often from calling collect() on huge data)

C) When the computer is unplugged

D) When the code is too long

Answer: B

127. What is "Executor OOM"?

A) When the user logs out

B) When an executor process runs out of RAM (often from huge shuffles or skew)

C) A hardware failure

D) A network error

Answer: B

128. What is "G1GC"?

A) A type of video card

B) A Garbage Collector recommended for Spark to reduce long pauses

C) A Spark component for graphs

D) A 5G internet connection

Answer: B

129. What is "Safe Mode" in Spark?

A) A mode with no errors

B) It doesn't officially exist; it's a HDFS concept

C) A mode for kids

D) A trial version

Answer: B

130. What is "Catalyst Hint"?

A) A tip for the user

B) A way to manually tell the Spark optimizer to use a specific join (e.g., /*+ BROADCAST */)

C) A riddle

D) A search suggestion

Answer: B

131. What is "Schema on Read" in Spark?

A) Forcing data to fit a schema while writing

B) Figuring out the data structure when reading it (like JSON or CSV)

C) Reading a book about schemas

D) Deleting the schema

Answer: B

132. What is "Partition Discovery"?

A) Finding a lost partition

B) Automatically detecting folders like /year=2023/ as table partitions

C) Creating new partitions

D) Renaming folders

Answer: B

133. What is "Push-based Shuffle"?

A) Manually moving files

B) A Spark 3.2+ feature to improve shuffle performance on large clusters

C) Deleting shuffle data

D) Slowing down the shuffle

Answer: B

134. What is "Data Skipping"?

A) Skipping the data engineering phase

B) Using file statistics (min/max) to skip reading irrelevant files (like in Delta Lake)

C) Deleting data

D) Randomly selecting data

Answer: B

135. What is "Storage Level: OFF_HEAP"?

A) Storing data in the cloud

B) Storing data in memory outside the JVM's GC management

C) Storing data on paper

D) Storing data on a USB

Answer: B

136. What is "UnsafeRow" in Spark?

A) A row with incorrect data

B) The internal binary format Spark uses to represent rows in memory efficiently

C) A deleted row

D) A row with a virus

Answer: B

137. What is "Code Obfuscation"?

A) Making code clear

B) Making code difficult to read (not typical for Spark itself)

C) Deleting code

D) Moving code to a secret server

Answer: B

138. What is "Thread Dump"?

A) Deleting a thread

B) A snapshot of all active threads in the JVM (used for debugging hangs)

C) A forum for developers

D) A type of cable

Answer: B

139. What is "Heap Dump"?

A) A snapshot of everything in memory (used to find memory leaks)

B) Deleting the RAM

C) A hardware trash bin

D) A backup of the OS

Answer: A

140. What is "Garbage Collection Pause"?

A) When the trash collector stops at your house

B) When the JVM stops execution to clean up unused objects in memory

C) A pause in the music

D) A power outage

Answer: B

141. What is "Spark History Server"?

A) A server that stores old Spark versions

B) A web UI to view logs and details of completed Spark applications

C) A museum for Spark

D) A backup server

Answer: B

142. What is "Structured Streaming Output Mode: Append"?

A) Deleting old data

B) Only new rows are written to the sink

C) Overwriting everything

D) Updating existing rows

Answer: B

143. What is "Structured Streaming Output Mode: Complete"?

A) The whole result table is rewritten to the sink every time

B) The job is finished

C) No rows are written

D) The data is deleted

Answer: A

144. What is "Structured Streaming Output Mode: Update"?

A) Only the rows that were updated are written to the sink

B) Deleting updated rows

C) Adding new rows only

D) Overwriting the entire sink

Answer: A

145. What is "Zstandard" (zstd) in Spark?

A) A type of battery

B) A high-performance compression algorithm supported by Spark

C) A Spark security protocol

D) A manual way to compress files

Answer: B

146. What is "PySpark Gateway"?

A) A physical gate in the office

B) The process that allows communication between Python and the Spark JVM

C) A cloud portal

D) A type of internet router

Answer: B

147. What is "Spark-Shell"?

A) A physical shell from the ocean

B) An interactive environment to run Spark code (Scala) in real-time

C) A protective case for the server

D) A type of database

Answer: B

148. What is "Job Grouping"?

A) Hiring a group of people

B) Assigning multiple Spark jobs to a logical group for better management/cancellation

C) Organizing folders

D) Deleting jobs

Answer: B

149. What is "Fair Scheduler" in Spark?

A) A person who is very nice

B) A scheduling mode that gives all jobs an equal share of resources

C) A random scheduler

D) A scheduler for weekend work

Answer: B

150. What is "FIFO Scheduler"?

A) First-In, First-Out (the default Spark scheduler)

B) A scheduler for pets

C) A very complex scheduler

D) A scheduler that deletes tasks

Answer: A

Subtopic 5: Real-time Data Streaming (Full 150)

Phase 1: Easy Level (1–50)

1. What is the main difference between Batch and Streaming?

A) Batch processes data in one go; Streaming processes data continuously

B) Batch is faster

C) Streaming only happens at night

D) Batch is only for images

Answer: A

2. Which of these is a popular open-source event streaming platform?

A) Microsoft Word

B) Apache Kafka

C) Adobe Photoshop

D) Google Chrome

Answer: B

3. In streaming, what is a "Producer"?

A) The person who buys the data

B) An application that sends data to the stream

C) A storage disk

D) A type of computer monitor

Answer: B

4. In streaming, what is a "Consumer"?

A) The application that reads and processes data from the stream

B) A person shopping online

C) The power supply

D) A deleted file

Answer: A

5. What is a "Message" in a stream?

A) A phone call

B) The unit of data (record) being sent

C) An email subject line

D) A physical letter

Answer: B

6. What is a "Topic" in Kafka?

A) A book title

B) A category or feed name where records are stored

C) A search engine

D) A chat room

Answer: B

7. "Real-time" usually implies a latency of:

A) Days

B) Months

C) Milliseconds to seconds

D) Years

Answer: C

8. What is "Near Real-time"?

A) Data that arrives exactly on time

B) Data processed with a slight delay (seconds to minutes)

C) Data from the future

D) Manual data entry

Answer: B

9. Which of these is a "Streaming" use case?

A) Monthly payroll processing

B) Credit card fraud detection (instant)

C) Annual tax filing

D) Printing a book

Answer: B

10. What does "Pub-Sub" stand for?

A) Public-Subnet

B) Publish-Subscribe

C) Power-Subscription

D) Push-Submit

Answer: B

11. What is a "Stream"?

A) A stationary file

B) An unbounded sequence of data ordered in time

C) A small river

D) A type of video game

Answer: B

12. What is "Latency"?

A) The weight of the server

B) The time delay between data generation and processing

C) The cost of the data

D) The number of users

Answer: B

13. What is "Throughput"?

A) The amount of data processed in a given time period

B) The physical size of the cable

C) The speed of light

D) The number of files deleted

Answer: A

14. Which cloud service is used for streaming on AWS?

A) Amazon Kinesis

B) Amazon S3

C) Amazon EC2

D) Amazon Prime

Answer: A

15. Which cloud service is used for streaming on Google Cloud (GCP)?

A) Cloud Storage

B) Pub/Sub

C) BigQuery

D) Cloud Run

Answer: B

16. Which cloud service is used for streaming on Azure?

A) Event Hubs

B) Virtual Machines

C) SQL Database

D) Paint

Answer: A

17. What is an "Event"?

A) A party

B) A change in state or a specific action (e.g., a click)

C) A scheduled meeting

D) A hardware upgrade

Answer: B

18. What is a "Broker" in a streaming system?

A) A person selling stocks

B) A server that manages the storage and delivery of messages

C) A broken computer

D) A type of internet browser

Answer: B

19. What is "Data Ingestion"?

A) Deleting data

B) The process of bringing data into the streaming system

C) Printing data

D) Selling data

Answer: B

20. Is "Spark Streaming" part of the Apache Spark ecosystem?

A) Yes

B) No

Answer: A

21. What is "Flink"?

A) A brand of camera

B) A powerful open-source stream processing framework

C) A small bird

D) A social media app

Answer: B

22. What is a "Partition" in a Kafka topic?

A) A room divider

B) A subset of the topic used to scale and parallelize data

C) A deleted message

D) A user password

Answer: B

23. What is an "Offset" in Kafka?

A) A discount price

B) A unique ID for a message within a partition

C) A time zone

D) A type of cable

Answer: B

24. What is "Retention Period"?

A) How long data is kept in the stream before deletion

B) The time it takes to hire a developer

C) The speed of the processor

D) The length of a contract

Answer: A

25. Can you "Replay" data in Kafka?

A) Yes, by resetting the offset

B) No, once read it is gone

Answer: A

26. What is "At-least-once" delivery?

A) Messages might be duplicated but never lost

B) Messages are sent only on Mondays

C) Messages are deleted immediately

D) Messages are never delivered

Answer: A

27. What is "Exactly-once" delivery?

A) Each message is processed exactly one time

B) Messages are sent once a year

C) No messages are sent

D) Messages are sent twice

Answer: A

28. What is "At-most-once" delivery?

A) Messages are never duplicated but might be lost

B) Messages are sent many times

C) Messages are saved forever

D) Messages are encrypted

Answer: A

29. What is "Backpressure"?

A) Pushing a button hard

B) A mechanism where a consumer signals the producer to slow down

C) A plumbing issue

D) High wind in the data center

Answer: B

30. What is "Windowing" in streaming?

A) Looking out a window

B) Grouping data points into time-based buckets for analysis

C) Installing Windows OS

D) Deleting data blocks

Answer: B

31. What is a "Tumbling Window"?

A) A window that falls over

B) Fixed-size, non-overlapping time intervals

C) A window that changes shape

D) A circular window

Answer: B

32. What is a "Sliding Window"?

A) A window that moves

B) Overlapping time intervals that move with data

C) A glass door

D) A window that closes automatically

Answer: B

33. What is a "Connector" (e.g., Kafka Connect)?

A) A physical plug

B) A tool to move data between external systems and the stream

C) A person who introduces people

D) A Wi-Fi router

Answer: B

34. Is streaming data usually structured or semi-structured (JSON)?

A) Semi-structured (JSON)

B) Only Paper

C) Only Binary

D) Only Audio

Answer: A

35. What is "CDC" (Change Data Capture)?

A) Capturing a CD

B) Tracking and streaming changes from a database in real-time

C) Changing the data center

D) Calculating data costs

Answer: B

36. What is "Zookeeper" used for in older Kafka versions?

A) Managing animals

B) Managing cluster metadata and leader election

C) Cleaning the servers

D) Monitoring the temperature

Answer: B

37. What is "KRaft" in newer Kafka?

A) A type of cheese

B) A protocol that replaces Zookeeper for internal management

C) A small boat

D) A drawing tool

Answer: B

38. What is "Replication" in streaming?

A) Copying data to multiple brokers for safety

B) Deleting data

C) Renaming data

D) Selling data

Answer: A

39. What is a "Consumer Group"?

A) A group of people buying things

B) A set of consumers working together to read from a topic

C) A list of customer emails

D) A marketing team

Answer: B

40. What is "Lag"?

A) A type of computer virus

B) The gap between the latest message and where the consumer is currently reading

C) The weight of a file

D) The speed of the internet

Answer: B

41. What is "Schema Registry"?

A) A list of building plans

B) A service to manage and version data formats in a stream

C) A guest book

D) A database of passwords

Answer: B

42. Which format is commonly used with Schema Registry?

A) Avro

B) MP3

C) JPG

D) EXE

Answer: A

43. What is "Stateful Processing"?

A) Processing based on the current and previous events (e.g., running total)

B) Processing only in one state/country

C) Processing data for a government

D) Processing data very slowly

Answer: A

44. What is "Stateless Processing"?

A) Each event is processed independently

B) Processing with no laws

C) Processing without a computer

D) Processing data that is free

Answer: A

45. What is "IoT" (Internet of Things) data?

A) Data from sensors and smart devices

B) Data from old books

C) Data from manual entry

D) Data from a radio

Answer: A

46. Can a stream be "Unbounded"?

A) Yes (no defined end)

B) No (everything must end)

Answer: A

47. Can a batch be "Unbounded"?

A) No (batches have a start and end)

B) Yes

Answer: A

48. What is "Micro-batching" in Spark?

A) Processing data in very small batches to mimic streaming

B) Processing very tiny files

C) Deleting small data

D) Manual processing

Answer: A

49. What is "Apache NiFi"?

A) A streaming video player

B) A tool for data routing, transformation, and system mediation logic

C) A brand of car

D) A programming language

Answer: B

50. What is a "Dead Letter Queue" (DLQ)?

A) A post office for ghosts

B) A place where failed or unprocessable messages are sent

C) A deleted folder

D) A high-speed stream

Answer: B

Phase 2: Medium Level (51–100)

51. What is a "Session Window"?

A) A window that opens for a meeting

B) Windows based on activity periods followed by a gap of inactivity

C) A window for the browser

D) A fixed 1-hour window

Answer: B

52. What is "Event Time" vs "Ingestion Time"?

A) Event time is when it happened; Ingestion time is when it reached the system

B) They are the same

C) Event time is the future

D) Ingestion time is the past

Answer: A

53. Why is "Event Time" preferred for accurate analytics?

A) It accounts for network delays and out-of-order data

B) It is easier to calculate

C) It is cheaper

D) It doesn't require a clock

Answer: A

54. What is "Out-of-order" data?

A) Data that is broken

B) Data that arrives in a different sequence than it was generated

C) Data that is forbidden

D) Data that is not alphabetical

Answer: B

55. What is a "Watermark" in stream processing?

A) A digital logo

B) A threshold that tells the system how long to wait for late data

C) The depth of the data

D) A cooling system

Answer: B

56. What is "Log Compaction" in Kafka?

A) Compressing files to save space

B) Keeping only the latest value for each key in a topic

C) Deleting all logs

D) Encrypting logs

Answer: B

57. What is "Fan-out" architecture?

A) Using a fan to cool the server

B) Sending a single message to multiple different destination streams/systems

C) Turning off the server

D) Deleting the stream

Answer: B

58. What is "Idempotent Producer"?

A) A producer that makes mistakes

B) A producer that ensures the same message is not written twice even on retries

C) A producer that works only once

D) A producer that writes random data

Answer: B

59. What is "KSQL" or "ksqlDB"?

A) A database for kids

B) An SQL-like interface for processing Kafka streams

C) A new programming language

D) A way to delete Kafka

Answer: B

60. What is "Stream-Table Duality"?

A) A stream can be viewed as a table (and vice-versa) over time

B) Streams and Tables are enemies

C) Using a table as a monitor

D) Deleting tables

Answer: A

61. What is "SerDe"?

A) Serializer and Deserializer

B) Server and Desktop

C) Service and Delivery

D) Secure and Developed

Answer: A

62. What is "Apache Beam"?

A) A laser beam

B) A unified model for defining both Batch and Streaming pipelines

C) A type of wood

D) A light bulb

Answer: B

63. What is a "Side Input" in a streaming pipeline?

A) Plugging in a USB

B) Additional data used for enrichment that is not the main stream

C) A second power supply

D) A secret code

Answer: B

64. What is "Acknowledge" (ACK) in Kafka?

A) Saying hello

B) The producer waiting for a signal that the broker received the message

C) Deleting a message

D) Renaming a message

Answer: B

65. What does acks=all mean?

A) The producer waits for all replicas to confirm receipt

B) The producer waits for no one

C) The producer waits for 1 second

D) The producer sends everything at once

Answer: A

66. What is "Consumer Offset Committing"?

A) Deleting the consumer

B) Saving the position of the last read message so the consumer can resume

C) Buying more storage

D) Changing the password

Answer: B

67. What happens if a Consumer Group has more consumers than partitions?

A) Some consumers will be idle (doing nothing)

B) The system crashes

C) Data is duplicated

D) Speed increases

Answer: A

68. What is "Leader Election"?

A) Voting for a president

B) Choosing which broker is responsible for a partition

C) Choosing the best developer

D) Choosing a new cloud provider

Answer: B

69. What is "In-Sync Replica" (ISR)?

A) A replica that is up-to-date with the leader

B) A replica that is broken

C) A replica in a different city

D) A replica on a CD

Answer: A

70. What is "Stream Enrichment"?

A) Making the stream expensive

B) Combining a stream with metadata (e.g., adding user names to user IDs)

C) Deleting the stream

D) Adding water to the server

Answer: B

71. What is "Joining" two streams?

A) Connecting two cables

B) Correlating events from two different streams (e.g., Clicks and Purchases)

C) Deleting two streams

D) Merging two companies

Answer: B

72. What is "Data Gravity"?

A) Data falling down

B) The idea that data and applications are attracted to each other

C) The weight of a hard drive

D) The price of storage

Answer: B

73. What is "Message Ordering"?

A) Sorting mail

B) Ensuring messages are processed in the sequence they were produced

C) Ordering data from a menu

D) Deleting messages

Answer: B

74. Kafka guarantees ordering within a _____.

A) Cluster

B) Partition

C) Broker

D) Network

Answer: B

75. What is "Parallelism" in streaming?

A) Running tasks on multiple CPU cores/nodes simultaneously

B) Two streams side-by-side

C) Working on two projects

D) Deleting two files

Answer: A

76. What is "Horizontal Scaling"?

A) Adding more servers to the cluster

B) Making the server taller

C) Buying a bigger monitor

D) Increasing the price

Answer: A

77. What is "Vertical Scaling"?

A) Adding more CPU/RAM to an existing server

B) Stacking servers on top of each other

C) Adding more users

D) Making the code longer

Answer: A

78. What is "Message Durability"?

A) How strong the cable is

B) Ensuring data is not lost even if a server fails

C) How long a message is

D) The price of a message

Answer: B

79. What is "Exactly-once Processing" (EOSP)?

A) The result of the computation reflects each input exactly once

B) Processing only one file

C) Deleting the result

D) Manual processing

Answer: A

80. What is "Checkpointing" in Flink?

A) Taking a snapshot of the distributed data stream state

B) Checking the time

C) Stopping the job

D) Deleting the logs

Answer: A

81. What is "Savepoint" in Flink?

A) A place to save a game

B) A manually triggered checkpoint used for updates or migrations

C) A deleted file

D) A backup disk

Answer: B

82. What is "Unordered" processing?

A) Processing data as fast as possible regardless of sequence

B) Data that is messy

C) Deleting data

D) Alphabetical sorting

Answer: A

83. What is "Zero-copy" transfer in Kafka?

A) Not copying data

B) Moving data from disk to network without copying to application memory

C) Deleting the copy

D) Manual data entry

Answer: B

84. What is "Compaction" vs "Retention"?

A) Compaction keeps latest keys; Retention deletes by time

B) They are the same

C) Compaction is for images; Retention is for text

D) Compaction is manual; Retention is automatic

Answer: A

85. What is "Broker Metadata"?

A) The price of the broker

B) Information about the cluster structure (topics, partitions, leaders)

C) The broker's name

D) The broker's address

Answer: B

86. What is "Kafka Connect Source"?

A) Importing data into Kafka

B) Exporting data from Kafka

C) Deleting Kafka

D) A programming language

Answer: A

87. What is "Kafka Connect Sink"?

A) Exporting data from Kafka to an external system (e.g., S3 or DB)

B) Importing data

C) Washing data

D) Deleting data

Answer: A

88. What is "Exactly-once Kafka Transactions"?

A) Allowing atomic writes to multiple topics and partitions

B) Selling Kafka

C) Deleting Kafka

D) Using Kafka once

Answer: A

89. What is "High Watermark" in Kafka replication?

A) The deepest part of the lake

B) The offset of the last message successfully replicated to all followers

C) A security logo

D) A flood warning

Answer: B

90. What is "Rebalance" in a consumer group?

A) Moving consumers to a new room

B) Redistributing partitions among consumers when a member leaves or joins

C) Deleting consumers

D) Balancing the server weight

Answer: B

91. What is "Static Membership" in Kafka?

A) Consumers that never move

B) A feature to reduce rebalances when consumers restart

C) A fixed price

D) No users allowed

Answer: B

92. What is "Streaming ETL"?

A) Transforming data as it flows through the stream

B) Moving data by hand

C) Deleting data in the stream

D) Batch processing

Answer: A

93. What is "Fraud Detection" in streaming?

A) Finding fake people

B) Using stream processing to identify suspicious patterns in milliseconds

C) Deleting bank accounts

D) Printing money

Answer: B

94. What is "Anomaly Detection"?

A) Finding normal things

B) Identifying rare items or events that differ from the norm

C) Deleting data

D) Renaming data

Answer: B

95. What is "Operational Monitoring"?

A) Watching a surgery

B) Streaming logs and metrics to track system health in real-time

C) Deleting logs

D) Hiring a guard

Answer: B

96. What is "Aggregating" in a stream?

A) Splitting data

B) Calculating values like Sum, Count, or Average over a window

C) Deleting data

D) Moving data

Answer: B

97. What is "Materialized View" in ksqlDB?

A) A view that stays updated as new events arrive in the stream

B) A view of a mountain

C) A deleted view

D) A paper report

Answer: A

98. What is "Push Query"?

A) Pushing a button

B) A query that sends updates to the client in real-time

C) A search query

D) A query that deletes data

Answer: B

99. What is "Pull Query"?

A) Pulling a lever

B) A traditional request/response query for the current state

C) A query for old data

D) A deleted query

Answer: B

Phase 3: Master Level (101–150)

101. What is "Chandy-Lamport" algorithm?

A) A way to cook food

B) A classic algorithm for taking consistent snapshots in distributed systems

C) A coding style

D) A hardware design

Answer: B

102. In Flink, what is "Managed State"?

A) State managed by the user

B) State managed by the Flink runtime (e.g., ValueState, ListState)

C) State managed by the government

D) No state

Answer: B

103. What is "RocksDB" in the context of streaming?

A) A collection of rocks

B) A popular embedded key-value store used for large state in Flink/Kafka Streams

C) A music player

D) A type of cloud storage

Answer: B

104. What is "LSM Tree" (Log-Structured Merge-Tree)?

A) A physical tree

B) The data structure behind RocksDB optimized for high-write throughput

C) A family tree

D) A type of network

Answer: B

105. What is "Exact Scalar Temporal Join"?

A) Joining data based on exact time matches

B) Joining data from two different galaxies

C) Deleting time

D) Manual joining

Answer: A

106. What is "Two-Phase Commit" (2PC) in streaming?

A) Committing to two projects

B) A protocol to ensure atomicity across distributed systems (used for Exactly-Once)

C) Deleting data twice

D) A coding shortcut

Answer: B

107. What is "Barrier" in Flink snapshots?

A) A wall

B) A special marker injected into the stream to trigger state snapshots

C) A security gate

D) A stop sign

Answer: B

108. What is "Skewed Stream"?

A) A stream that is tilted

B) A stream where one key has significantly more data than others, causing bottlenecks

C) A fast stream

D) A deleted stream

Answer: B

109. What is "Event-Time Skew"?

A) When event time and processing time diverge significantly

B) A bug in the clock

C) A time zone change

D) A leap year

Answer: A

110. What is "Lateness" in streaming?

A) Being late for a meeting

B) The delay between an event's timestamp and when it's processed

C) A deleted event

D) A slow computer

Answer: B

111. What is "Allowed Lateness" in Apache Beam?

A) Letting employees be late

B) How much time after a watermark the system will still accept data for a window

C) Deleting late data

D) Ignoring all data

Answer: B

112. What is "Broadcast State" in Flink?

A) Sending a state to a TV

B) A state that is replicated across all parallel tasks (useful for config/rules)

C) A public state

D) A deleted state

Answer: B

113. What is "Keyed State"?

A) State partitioned by a specific key (e.g., UserID)

B) State locked with a key

C) State for the keyboard

D) A hidden state

Answer: A

114. What is "Side Output" for late data?

A) A secret exit

B) Redirecting data that arrived too late for a window to a separate stream

C) Deleting late data

D) Moving data to the cloud

Answer: B

115. What is "Queryable State"?

A) State you can ask questions to

B) Allowing external applications to query the internal state of a running stream job

C) A public database

D) A deleted state

Answer: B

116. What is "Network Buffer" in streaming?

A) A speed test

B) Memory used to batch records before sending them over the network

C) A router

D) A firewall

Answer: B

117. What is "Serialized Throughput"?

A) Speed of a TV show

B) The rate at which data can be converted to bytes and sent

C) Deleting data

D) The number of users

Answer: B

118. What is "Schema Evolution: Backward Compatibility"?

A) New code can read old data

B) Old code can read new data

C) Deleting old data

D) Using paper records

Answer: A

119. What is "Schema Evolution: Forward Compatibility"?

A) Old code can read new data

B) New code can read old data

C) Deleting new data

D) Future data

Answer: A

120. What is "Full Compatibility" in schemas?

A) Both forward and backward compatible

B) All data is the same

C) No data is deleted

D) Everything is free

Answer: A

121. What is "Controller" in Kafka?

A) A game controller

B) A broker responsible for managing states of partitions and replicas

C) The CEO of Kafka

D) A remote control

Answer: B

122. What is "Preferred Leader Election"?

A) Voting for a favorite

B) Automatically moving leadership back to the "best" broker once it recovers

C) Deleting the leader

D) Hiring a new leader

Answer: B

123. What is "Under-replicated Partitions"?

A) Partitions with fewer copies than required (danger of data loss)

B) Small partitions

C) Fast partitions

D) Deleted partitions

Answer: A

124. What is "Segment" in Kafka?

A) A piece of a circle

B) The actual physical file on disk where Kafka messages are stored

C) A part of a company

D) A type of code

Answer: B

125. What is "Zero-copy" (mmap)?

A) Not using memory

B) Using memory-mapped files to bypass the kernel-user space copy

C) Deleting maps

D) A paper map

Answer: B

126. What is "Network Partition" (Split Brain)?

A) A brain surgery

B) When nodes in a cluster can't see each other and think they are both the leader

C) A fast network

D) A deleted network

Answer: B

127. What is "Quorum" in a distributed system?

A) A type of bird

B) The minimum number of nodes that must be online for the cluster to work

C) A secret meeting

D) A type of data

Answer: B

128. What is "Raft" consensus algorithm?

A) A boat race

B) A protocol for managing a replicated log (used in KRaft)

C) A coding style

D) A hardware standard

Answer: B

129. What is "Vector Clocks"?

A) A wall clock

B) A mechanism for tracking the partial ordering of events in a distributed system

C) A fast clock

D) A type of math

Answer: B

130. What is "Exactly-once Sink"?

A) A sink that only takes one item

B) A destination system that can handle duplicates to ensure exactly-once output

C) A washing sink

D) A deleted sink

Answer: B

131. What is "Transactional Messaging"?

A) Buying and selling messages

B) Ensuring a set of messages are either all produced or none are

C) Sending an email

D) Deleting messages

Answer: B

132. What is "Kafka Tiered Storage"?

A) Moving old Kafka data to cheaper storage (like S3) automatically

B) Storing Kafka on a shelf

C) Deleting Kafka

D) Using many disks

Answer: A

133. What is "State Backend"?

A) The back of a server

B) The storage system used to hold the state (e.g., Memory vs RocksDB)

C) A government office

D) A hidden file

Answer: B

134. What is "Point-in-time" recovery?

A) Recovering a lost watch

B) The ability to restore a stream state to a specific historical moment

C) Deleting the past

D) Fast recovery

Answer: B

135. What is "Data Enrichment at Scale"?

A) Adding data manually

B) Using high-speed lookups (e.g., Redis) to add context to a million events/sec

C) Making data bigger

D) Selling data

Answer: B

136. What is "Event Sourcing"?

A) Finding the source of a river

B) Storing every change to application state as a sequence of events

C) Deleting events

D) Manual state entry

Answer: B

137. What is "CQRS" (Command Query Responsibility Segregation)?

A) Separating read and write operations (common in streaming architectures)

B) A coding error

C) A database type

D) A networking tool

Answer: A

138. What is "Lambda Architecture"?

A) Using Greek letters

B) Combining Batch and Streaming to get both accuracy and speed

C) A type of server

D) A cloud provider

Answer: B

139. What is "Kappa Architecture"?

A) Using Batch only

B) Removing Batch and doing everything via Streaming

C) A type of clothing

D) A coding language

Answer: B

140. What is "Window Trigger"?

A) A gun trigger

B) The condition that decides when a window should be "fired" (result output)

C) A window that breaks

D) A manual button

Answer: B

141. What is "Global Window"?

A) A window for the whole world

B) A window that covers all time for a specific key

C) A very large window

D) A deleted window

Answer: B

142. What is "Accumulating Fired Panes"?

A) Saving old window results

B) Keeping previous results and adding to them in the next trigger

C) Deleting results

D) Printing results

Answer: B

143. What is "Discarding Fired Panes"?

A) Deleting window results after they are sent

B) Saving results forever

C) Moving results to the cloud

D) Renaming results

Answer: A

144. What is "Group-By-Key" (GBK) in streaming?

A) Organizing data by a shared attribute

B) Deleting keys

C) Sorting keys alphabetically

D) Randomizing keys

Answer: A

145. What is "Co-grouping"?

A) Grouping two different streams by a common key

B) Grouping with a friend

C) Deleting a group

D) A social media group

Answer: A

146. What is "Streaming Join Latency"?

A) The delay added by waiting for the "other side" of a join to arrive

B) The speed of the join

C) The cost of the join

D) The number of joins

Answer: A

147. What is "Uptime" in streaming systems?

A) The time the system is operational and processing data

B) The time you spend at work

C) The time it takes to boot

D) The time until deletion

Answer: A

148. What is "Auto-scaling" in Flink?

A) Changing the font size

B) Dynamically adjusting the number of task managers based on load (Reactive Mode)

C) Buying more servers

D) Deleting idle nodes

Answer: B

149. What is "Operator" in a streaming DAG?

A) A person working on a machine

B) A single processing step (e.g., Map, Filter, Window)

C) A math symbol

D) A phone company

Answer: B

150. What is "Sink" in streaming?

A) Where data goes to die (be deleted)

B) The final destination of the processed stream (e.g., DB, Lake, Dashboard)

C) A kitchen sink

D) A type of error

Answer: B

Subtopic 6: Cloud Storage Optimization (Full 150)

Phase 1: Easy Level (1–50)

1. Which storage tier is the cheapest for data you rarely access?

A) Standard

B) Archive / Cold

C) Premium

D) SSD

Answer: B

2. What is "Object Storage"?

A) Storing data as files in a hierarchy

B) Storing data as discrete units with metadata and a unique ID

C) Storing data on a physical CD

D) Storing data in a relational table only

Answer: B

3. In cloud storage, what is a "Bucket"?

A) A physical container in a data center

B) A logical container for storing objects

C) A tool for cleaning servers

D) A type of network cable

Answer: B

4. Which of these is an example of cloud object storage?

A) Amazon S3

B) Microsoft Word

C) Windows 11

D) RAM

Answer: A

5. What does "Durability" mean in cloud storage?

A) How fast the data is

B) The probability that data will not be lost

C) The price of the storage

D) The color of the server

Answer: B

6. What is "Availability" in cloud storage?

A) The total size of the disk

B) The percentage of time the data is accessible when needed

C) The name of the user

D) The number of files

Answer: B

7. "Standard" storage tiers are best for:

A) Data accessed frequently

B) Data you never want to see again

C) Deleted files

D) Backups from 10 years ago

Answer: A

8. Which action usually costs money in cloud storage?

A) Looking at the bucket name

B) Data egress (moving data out of the cloud)

C) Deleting a bucket

D) Creating a folder

Answer: B

9. What is "Encryption at Rest"?

A) Encrypting data while it is being sent

B) Encrypting data while it is stored on the disk

C) Encrypting data while the server is turned off

D) Not encrypting data at all

Answer: B

10. What is "Encryption in Transit"?

A) Encrypting data while it is stored

B) Encrypting data while it moves across the network

C) Encrypting data on a USB drive

D) Manual encryption

Answer: B

11. What is a "Lifecycle Policy"?

A) A set of rules to automatically move or delete data after a certain time

B) A rule about human health

C) A contract for the cloud provider

D) A way to restart the server

Answer: A

12. Compressing files (like using GZIP) helps to:

A) Increase the cost

B) Reduce storage space and save money

C) Make the data more colorful

D) Speed up the CPU

Answer: B

13. What is a "Prefix" in S3?

A) A title for the user

B) A string of characters at the start of an object name (acting like a folder)

C) A type of encryption

D) A backup file

Answer: B

14. Which file format is better for big data storage optimization?

A) CSV

B) Parquet

C) TXT

D) DOCX

Answer: B

15. What is "Versioning"?

A) Keeping multiple versions of an object in the same bucket

B) Changing the name of the cloud provider

C) Deleting old data

D) Buying a new computer

Answer: A

16. What is "Multi-factor Authentication (MFA) Delete"?

A) Deleting data automatically

B) Requiring an extra code to delete objects for security

C) Deleting the user's account

D) Deleting the cloud bill

Answer: B

17. What is "Metadata" in cloud storage?

A) The actual content of the file

B) Information about the file (size, type, creation date)

C) The cloud provider's address

D) The price of the file

Answer: B

18. What is "Egress"?

A) Data coming into the cloud

B) Data leaving the cloud (often incurs costs)

C) Data being deleted

D) Data being encrypted

Answer: B

19. What is "Ingress"?

A) Data entering the cloud (usually free)

B) Data leaving the cloud

C) Data being renamed

D) Data being compressed

Answer: A

20. Which service is used for long-term "Cold" archive on AWS?

A) S3 Standard

B) S3 Glacier

C) EC2

D) RDS

Answer: B

21. What is "Storage Class Analysis"?

A) A tool to help decide when to move data to a cheaper tier

B) A school for storage

C) Deleting all data

D) Comparing different cloud providers

Answer: A

22. "S3 Intelligent-Tiering" automatically:

A) Deletes your data

B) Moves data between tiers based on access patterns to save money

C) Encrypts your data

D) Renames your files

Answer: B

23. What is a "Soft Delete" in Azure/GCP?

A) Deleting data gently

B) A period where deleted data can still be recovered before it's gone forever

C) Deleting only small files

D) Hiding files from view

Answer: B

24. What is "Object Lock"?

A) Locking the server room

B) Preventing an object from being deleted or overwritten for a fixed time

C) Forgetting the password

D) Locking the user account

Answer: B

25. Which is faster to retrieve data from?

A) Standard Storage

B) Deep Archive Storage

Answer: A

26. What is "Latent" data?

A) Data that is always fast

B) Data that takes a long time to retrieve

C) Deleted data

D) Random data

Answer: B

27. What is "Throughput" in cloud storage?

A) The amount of data transferred per second

B) The total storage limit

C) The weight of the disk

D) The cost per GB

Answer: A

28. What is "Region" in cloud storage?

A) A specific country

B) A geographic area containing multiple data centers

C) A type of folder

D) A security level

Answer: B

29. Why would you store data in multiple regions?

A) To make it more expensive

B) For disaster recovery and lower latency for global users

C) To confuse hackers

D) To use different languages

Answer: B

30. What is a "Public" bucket?

A) A bucket everyone on the internet can see (dangerous if not intentional)

B) A bucket owned by the government

C) A bucket that is free

D) A bucket with no data

Answer: A

31. What is "Access Control List" (ACL)?

A) A list of items to buy

B) A way to define who has access to specific buckets or objects

C) A list of deleted files

D) A type of folder

Answer: B

32. What is "Server-Side Encryption" (SSE)?

A) The user encrypts the data

B) The cloud provider encrypts the data on its servers

C) No encryption used

D) Encrypting the keyboard

Answer: B

33. What is "Client-Side Encryption"?

A) The user encrypts data before uploading it to the cloud

B) The cloud provider encrypts it

C) The server encrypts it

D) The internet provider encrypts it

Answer: A

34. What is a "Multipart Upload"?

A) Uploading a file in many small parts to improve speed and reliability

B) Uploading many files at once

C) Uploading to multiple clouds

D) Deleting while uploading

Answer: A

35. What is "Transfer Acceleration"?

A) Moving the server closer

B) Using a specialized network (like CDN) to speed up uploads over long distances

C) Upgrading the user's Wi-Fi

D) Deleting small files

Answer: B

36. What is "Cold Storage"?

A) Storage in a refrigerator

B) Low-cost storage for data that is rarely accessed

C) Storage that is turned off

D) Storage for video games

Answer: B

37. What is "Hot Storage"?

A) Storage that is on fire

B) High-performance storage for data accessed frequently

C) Storage in a desert

D) Expensive storage

Answer: B

38. What is a "Key" in object storage?

A) A physical key

B) The unique name/identifier for an object

C) A password

D) A type of encryption

Answer: B

39. What is "Data Sovereignty"?

A) Ensuring data stays within a specific geographic/legal boundary

B) Data owned by a king

C) Data that is very fast

D) Free data

Answer: A

40. Which cost is usually higher in "Archive" tiers?

A) Storage cost per GB

B) Retrieval cost (cost to get the data back)

C) Initial setup cost

D) Account fee

Answer: B

41. What is "Inventory" in S3?

A) A list of all objects in a bucket for auditing/management

B) A shop for data

C) A place to buy disks

D) A type of backup

Answer: A

42. What is "Bucket Policy"?

A) A set of rules in JSON format that controls access to a bucket

B) A contract with the cloud company

C) A list of prices

D) A way to rename buckets

Answer: A

43. What is "Cross-Region Replication" (CRR)?

A) Deleting data in two regions

B) Automatically copying every object uploaded in one region to another region

C) Moving the server to another country

D) Changing the data language

Answer: B

44. What is "IAM" in cloud storage?

A) Identity and Access Management

B) International Account Manager

C) Internal Archive Memory

D) Internet Access Mode

Answer: A

45. What is "Throttling"?

A) Speeding up the server

B) Intentionally slowing down requests to stay within limits

C) Deleting fast requests

D) Turning off the internet

Answer: B

46. What is "Bit Rot"?

A) Data becoming old

B) Slow deterioration of data on physical storage media

C) A computer virus

D) Deleting data by mistake

Answer: B

47. Cloud providers handle "Bit Rot" by:

A) Asking the user to re-upload

B) Automatically checking and repairing data integrity (Self-healing)

C) Deleting the data

D) Charging more money

Answer: B

48. What is "Cloud Storage Gateway"?

A) A physical bridge

B) A service connecting on-premises applications to cloud storage

C) A type of firewall

D) A web browser

Answer: B

49. What is "Data Migration"?

A) Moving data from one system/location to another

B) Deleting data

C) Renaming data

D) Encrypting data

Answer: A

Phase 2: Medium Level (51–100)

51. Why is Columnar Storage (Parquet) better for cloud costs?

A) It is more colorful

B) It allows the engine to read only the specific columns needed, reducing I/O costs

C) It is easier to read for humans

D) It stores data in alphabetical order

Answer: B

52. What is "Pre-signed URL"?

A) A URL signed by a celebrity

B) A temporary link that gives someone access to an object without needing an account

C) A link to a website

D) A deleted link

Answer: B

53. What is "S3 Select"?

A) Choosing which S3 bucket to buy

B) A feature that lets you use SQL to pull only a subset of data from an object

C) A way to select files with a mouse

D) A specialized cloud tier

Answer: B

54. What is "Storage Class Transition"?

A) Moving from one cloud provider to another

B) Moving an object from one storage class (e.g., Standard) to another (e.g., Glacier)

C) Renaming a storage class

D) Deleting a storage class

Answer: B

55. What is "Provisioned Throughput"?

A) Paying for a guaranteed level of data transfer speed

B) Free data transfer

C) Deleting slow data

D) Moving data by hand

Answer: A

56. What is "Consistent Hashing" in distributed storage?

A) Making sure all data is the same

B) A way to distribute data across nodes so that adding/removing nodes causes minimal data movement

C) Encrypting data twice

D) A type of sorting

Answer: B

57. What is "Data Locality" in a cloud context?

A) Keeping data in the same region as the compute to avoid egress charges

B) Storing data in your own house

C) Storing data in a local language

D) Moving data to a USB

Answer: A

58. What is "ETag" (Entity Tag)?

A) A price tag

B) A unique hash (usually MD5) used to check if an object's content has changed

C) The user's name

D) A security code

Answer: B

59. What is "In-Place Transformation"?

A) Transforming data without moving it out of the storage layer

B) Changing data in a different cloud

C) Deleting data and re-uploading

D) Renaming a folder

Answer: A

60. What is "Immutable Storage"?

A) Storage that can't be read

B) Storage that prevents data from being modified or deleted (WORM - Write Once Read Many)

C) Storage that changes every day

D) Free storage

Answer: B

61. What is "Block Storage" (EBS/Disk) vs "Object Storage" (S3)?

A) Block is like a hard drive for a VM; Object is for web-scale files

B) They are the same

C) Block is for photos; Object is for databases

D) Block is cheap; Object is expensive

Answer: A

62. What is "File Storage" (EFS/Filestore) vs "Object Storage"?

A) File storage uses a traditional folder hierarchy (NFS); Object uses a flat namespace

B) File storage is only for text files

C) Object storage is only for computers

D) File storage is always in the cloud

Answer: A

63. What is "Path-style" vs "Virtual-hosted-style" URLs?

A) Different ways to format the web link to an S3 object

B) Different speeds of light

C) Different prices of data

D) Different languages

Answer: A

64. What is "Bucket Sprawl"?

A) Having too many unnecessary buckets, making management difficult

B) A bucket that is very large

C) A bucket that is leaking data

D) A bucket in the desert

Answer: A

65. What is "Data Scrubbing"?

A) Cleaning the server with soap

B) Background process that checks data for errors and repairs them

C) Deleting bad data

D) Renaming data

Answer: B

66. What is "Event Notifications" in cloud storage?

A) Sending an email when someone logs in

B) Triggering a process (like a Lambda) when a new file is uploaded

C) A calendar of events

D) A news feed

Answer: B

67. What is "Multipart Upload Abort"?

A) Crashing the computer

B) A policy to delete incomplete file uploads to save money

C) Stopping the internet

D) Deleting the whole bucket

Answer: B

68. What is "Range Request" (Byte-range)?

A) Requesting a range of prices

B) Downloading only a specific part of a file (e.g., first 100 bytes)

C) Requesting a range of dates

D) Selecting multiple files

Answer: B

69. What is "Query Latency"?

A) The time it takes for a database to return results from storage

B) The speed of the user typing

C) The cost of the query

D) The number of queries

Answer: A

70. What is "Direct Connect" / "ExpressRoute"?

A) A faster Wi-Fi

B) A dedicated physical network connection from an office to the cloud provider

C) A direct phone line

D) A fast courier service

Answer: B

71. What is "Snowball" / "Snowmobile" (AWS)?

A) A toy for the snow

B) Physical hardware used to move petabytes of data into the cloud

C) A cooling system

D) A type of cloud tier

Answer: B

72. What is "Coldline" / "Nearline" in GCP?

A) Different tiers of low-cost storage

B) Different types of phone lines

C) Different programming languages

D) Different cloud providers

Answer: A

73. What is "Blob" in Azure?

A) A mistake in the code

B) Binary Large Object (Azure's version of object storage)

C) A type of folder

D) A deleted file

Answer: B

74. What is "Hot Path" vs "Cold Path" data?

A) Fast data for real-time; Slow data for batch/archiving

B) Data in summer; Data in winter

C) Encrypted data; Unencrypted data

D) Large files; Small files

Answer: A

75. What is "Storage Over-provisioning"?

A) Buying more storage than you actually need (wasting money)

B) Not having enough storage

C) Sharing storage with a friend

D) Deleting storage

Answer: A

76. What is "Data Compaction"?

A) Making data files physically smaller on disk

B) Combining multiple small files into one larger file for efficiency

C) Deleting small files

D) Encrypting files

Answer: B

77. What is "Small File Problem" in Cloud Storage?

A) Files are too tiny to read

B) Having millions of tiny files leads to high API costs and slow processing

C) Files are automatically deleted

D) Files take up no space

Answer: B

78. What is "HDFS" vs "S3" for big data?

A) HDFS is a local cluster file system; S3 is a remote cloud object store

B) They are the same

C) HDFS is for images; S3 is for text

D) HDFS is faster for everything

Answer: A

79. What is "S3 Strong Consistency"?

A) Every read after a write returns the newest data immediately

B) Data is always the same color

C) Data is never lost

D) Data is encrypted strongly

Answer: A

80. What is "Eventual Consistency"?

A) Data will be updated everywhere eventually, but maybe not immediately

B) Data is deleted eventually

C) Data is never updated

D) Data is always wrong

Answer: A

81. What is "Audit Logging" (CloudTrail)?

A) Logging into a computer

B) Recording every API call made to your storage for security and compliance

C) A list of employees

D) A backup of the cloud

Answer: B

82. What is "VPC Endpoint"?

A) The end of a cable

B) A private connection that keeps storage traffic within the cloud network (no internet)

C) A type of router

D) A security firewall

Answer: B

83. What is "Cross-Account Access"?

A) Letting a friend use your account

B) Giving a different cloud account permission to read/write to your bucket

C) Hacking an account

D) Sharing a password

Answer: B

84. What is "Pre-warming" storage?

A) Putting the disk in a heater

B) Asking the provider to prepare for a massive surge in traffic

C) Deleting cold data

D) Running a script to touch all files

Answer: B

85. What is "Data Sanitization"?

A) Washing the disks

B) Permanently erasing data so it cannot be recovered

C) Cleaning the code

D) Renaming sensitive data

Answer: B

86. What is "Storage Quota"?

A) A limit on how much data a user or bucket can store

B) A price discount

C) A type of folder

D) A security level

Answer: A

87. What is "Standard-IA" (Infrequent Access)?

A) Cheap storage for data accessed once or twice a month

B) Standard storage for images

C) Expensive storage

D) Storage for internet apps

Answer: A

88. What is "One Zone-IA"?

A) Storage kept in only one data center (cheaper but less durable against disasters)

B) Storage for only one user

C) Storage for one file

D) Storage in one country

Answer: A

89. What is "EFS" (Elastic File System)?

A) A cloud version of a shared network drive (NAS)

B) A flexible hard drive

C) A type of object storage

D) A backup service

Answer: A

90. What is "Glacier Deep Archive"?

A) The cheapest tier, retrieval can take 12-48 hours

B) Storage under the ocean

C) A deleted archive

D) A very fast archive

Answer: A

91. What is "Storage Class Pricing"?

A) Paying more for speed and less for size

B) Different rates per GB depending on the tier (Standard vs Glacier)

C) One price for everything

D) Free storage for everyone

Answer: B

92. What is "Cost Explorer"?

A) A person who finds cheap things

B) A tool to visualize and manage your cloud spending

C) A way to search for files

D) A discount shop

Answer: B

93. What is "Lifecycle Transition"?

A) Moving from childhood to adulthood

B) Automated movement of data between storage classes based on age

C) Deleting old data

D) Renaming old data

Answer: B

94. What is "Object Metadata"?

A) Tags and key-value pairs assigned to an object

B) The file extension

C) The file size

D) The file content

Answer: A

95. What is "KMS" (Key Management Service)?

A) A tool to manage encryption keys

B) A service for physical keys

C) A manager for keyboard shortcuts

D) A security guard

Answer: A

96. What is "Bucket Tagging"?

A) Labeling buckets for cost allocation and tracking

B) Putting a sticker on a bucket

C) Deleting a bucket

D) Renaming a bucket

Answer: A

97. What is "CloudFront" (CDN)?

A) A weather app

B) A service that caches storage content globally to speed up access for users

C) The front door of the data center

D) A security firewall

Answer: B

98. What is "Origin Access Identity" (OAI)?

A) A secret ID for a user

B) A way to restrict S3 access so only CloudFront can see the files

C) A type of encryption

D) A backup system

Answer: B

99. What is "Signed Cookie"?

A) A snack

B) A way to give access to multiple private files using a web cookie

C) A type of encryption

D) A deleted file

Answer: B

100. What is "Field-Level Encryption"?

A) Encrypting only specific parts of a file (like credit card numbers)

B) Encrypting a whole field of servers

C) Encrypting a file name

D) No encryption

Answer: A

Phase 3: Master Level (101–150)

[Image explaining Erasure Coding vs Replication for data durability]

101. What is "Erasure Coding"?

A) Deleting code

B) A method of data protection where data is broken into fragments and encoded with redundant data pieces

C) A type of encryption

D) A coding language

Answer: B

102. Why is Erasure Coding better than Replication for large-scale storage?

A) It uses much less storage overhead for the same level of durability

B) It is faster

C) It is more colorful

D) It is free

Answer: A

103. What is "Write-Through Cache"?

A) Writing data to the cache and the permanent storage at the same time

B) Writing data quickly

C) Deleting the cache

D) Writing data on paper

Answer: A

104. What is "Write-Back Cache"?

A) Writing only to the cache first (faster) and updating storage later

B) Writing data in reverse

C) Deleting data

D) Writing to a USB

Answer: A

105. What is "Cache Eviction Policy" (LRU)?

A) Deleting the oldest data from cache to make room for new data

B) Keeping everything forever

C) Deleting random data

D) Expanding the cache

Answer: A

106. What is "Hotspotting" in a distributed bucket?

A) A Wi-Fi hotspot

B) When a single prefix/object gets too much traffic, causing performance degradation

C) A fire in the server

D) A popular file

Answer: B

107. How do you prevent "Hotspotting" in S3?

A) By using random prefixes or hashing object names to distribute data

B) By deleting the file

C) By buying more servers

D) By turning off the bucket

Answer: A

108. What is "Byzantine Fault Tolerance"?

A) A failure in an old empire

B) The ability of a system to work correctly even if some nodes act maliciously or fail

C) A network error

D) A power outage

Answer: B

109. What is "CAP Theorem" (Consistency, Availability, Partition Tolerance)?

A) You can only have 2 out of the 3 in a distributed system

B) You must have all 3

C) It only applies to local drives

D) It is a marketing term

Answer: A

110. What is "PACELC Theorem"?

A) An extension of CAP theorem focusing on Latency and Consistency

B) A theorem about math

C) A cloud provider name

D) A security protocol

Answer: A

111. What is "Geo-Redundancy"?

A) Storing data in two different cities

B) Storing data on two different disks

C) Deleting data in one country

D) Moving to the cloud

Answer: A

112. What is "Point-in-Time Restore" (PITR)?

A) Restoring data to a specific second in the past (using versioning/logs)

B) Restoring a watch

C) Deleting the past

D) Fast restoration

Answer: A

113. What is "Data Gravity" in storage optimization?

A) The cost of data

B) The tendency for data and applications to be pulled toward the largest data source

C) Data falling off a disk

D) Heavy hard drives

Answer: B

114. What is "Compute-Storage Decoupling"?

A) Scaling servers and disks independently (saves money and improves flexibility)

B) Putting servers and disks in different rooms

C) Deleting the storage

D) Using only one server

Answer: A

115. What is "Storage Density"?

A) How heavy a disk is

B) The amount of data stored per unit of physical space or cost

C) The color of the disk

D) The number of users

Answer: B

116. What is "I/O Wait"?

A) Waiting for the computer to start

B) Time the CPU spends waiting for storage to respond

C) Waiting for a download

D) A broken keyboard

Answer: B

117. What is "Data Tiering" vs "Data Archiving"?

A) Tiering moves data to slower live storage; Archiving moves data to offline/near-line storage

B) They are the same

C) Tiering is for images; Archiving is for text

D) Tiering is manual; Archiving is automatic

Answer: A

118. What is "Object Versioning Overhead"?

A) The cost of the computer

B) The extra storage costs incurred by keeping every old version of a file

C) The time to name a file

D) The security risk

Answer: B

119. What is "Data Parity"?

A) Equality for data

B) Redundant data used to reconstruct lost information in a RAID or Erasure Coding system

C) A data party

D) A type of encryption

Answer: B

120. What is "Re-hydration" in cloud storage?

A) Giving water to a server

B) The process of moving an object from an Archive tier back to a Hot/Standard tier

C) Cleaning the disks

D) Turning on the power

Answer: B

121. What is "Expedited Retrieval"?

A) Getting data from Glacier very fast (minutes) for a higher cost

B) A fast delivery service

C) Deleting data quickly

D) A security check

Answer: A

122. What is "Bulk Retrieval"?

A) Getting massive amounts of data from Glacier slowly and cheaply (hours/days)

B) Buying a lot of storage

C) Deleting many files

D) A large backup

Answer: A

123. What is "Bit-level Integrity"?

A) Being an honest person

B) Ensuring that not a single bit has changed since the data was written

C) A fast bit

D) A type of code

Answer: B

124. What is "Checksum" (e.g., CRC32, SHA256)?

A) A way to pay for storage

B) A value used to verify that data hasn't been corrupted during transfer

C) A list of files

D) A security password

Answer: B

125. What is "Fan-out" in storage notifications?

A) Cooling the bucket

B) Sending one event (e.g., file upload) to multiple subscribers/systems

C) Deleting many files

D) Turning off the notification

Answer: B

126. What is "Storage Gateway - Tape Gateway"?

A) A gateway made of tape

B) Replacing physical tape backups with cloud storage while keeping the same software interface

C) A very old computer

D) A type of network

Answer: B

127. What is "Direct Upload to S3"?

A) Sending data from a browser directly to S3 (bypassing the app server)

B) Uploading via a USB

C) Moving data from one bucket to another

D) A manual process

Answer: A

128. What is "S3 Access Points"?

A) Physical locations

B) Specific hostnames with customized policies to manage access for different applications

C) A Wi-Fi name

D) A backup gate

Answer: B

129. What is "Storage Lens"?

A) A magnifying glass for disks

B) A tool that provides organization-wide visibility into object storage usage and activity

C) A security camera

D) A type of encryption

Answer: B

130. What is "Cloud Storage Bursting"?

A) A disk exploding

B) Temporarily using cloud storage to handle overflow from on-premises systems

C) A fast upload

D) Deleting data

Answer: B

131. What is "Data Sovereignty Compliance"?

A) Following laws about where data is stored

B) Being the king of data

C) Deleting illegal data

D) Free data for all

Answer: A

132. What is "Cross-Origin Resource Sharing" (CORS)?

A) Sharing resources with a friend

B) Allowing a webpage on one domain to access storage on another domain

C) A type of encryption

D) A network error

Answer: B

133. What is "Object Tagging" for Cost Allocation?

A) Using tags to see exactly how much money each project is spending on storage

B) Labeling files with prices

C) Deleting expensive files

D) Selling tags

Answer: A

134. What is "Bucket Replication Policy"?

A) Rules defining what, where, and how data is copied to another bucket

B) A rule to delete the bucket

C) A price list

D) A security check

Answer: A

135. What is "Bucket Owner Enforced" (S3)?

A) The person who pays owns the data, even if another account uploaded it

B) A manager for the bucket

C) A security password

D) A deleted setting

Answer: A

136. What is "Transfer Family" (AWS)?

A) A family of cloud users

B) A managed service for transferring files via SFTP, FTPS, and FTP directly to S3

C) A group of servers

D) A backup plan

Answer: B

137. What is "Storage-Class Override"?

A) Forcing a specific tier even if a policy says otherwise

B) Deleting a tier

C) Renaming a tier

D) A security error

Answer: A

138. What is "In-Region Transfer Cost"?

A) Moving data within the same region (usually $0)

B) The price of a plane ticket

C) The cost of a new disk

D) A hidden fee

Answer: A

139. What is "Inter-Region Transfer Cost"?

A) The cost of moving data between different cloud regions (can be expensive)

B) Moving data for free

C) Deleting data in two regions

D) A type of storage tier

Answer: A

140. What is "Request Pays" bucket?

A) A bucket that asks for money

B) A bucket where the person downloading the data pays the transfer cost, not the owner

C) A free bucket

D) A deleted bucket

Answer: B

141. What is "S3 Batch Operations"?

A) Working on many objects at once (e.g., tagging or copying millions of files)

B) A type of cake

C) A slow way to upload

D) A manual task

Answer: A

142. What is "Pre-replication Metadata"?

A) Data about data before it is copied

B) A price tag

C) A deleted file

D) A security code

Answer: A

143. What is "Replication Lag" in cloud storage?

A) The time it takes for a file to appear in the destination region

B) A network error

C) The weight of the file

D) A slow computer

Answer: A

144. What is "Cloud Storage Mirroring"?

A) Seeing your reflection in the server

B) Keeping two identical copies of data in different locations for high availability

C) A type of encryption

D) A backup disk

Answer: B

145. What is "Cold Storage Retrieval Time"?

A) The time it takes to "wake up" data from archive tiers

B) The speed of the internet

C) The time to build a data center

D) A security check

Answer: A

146. What is "Lifecycle Rule Filter"?

A) Applying rules only to specific files (e.g., only files ending in .log)

B) A water filter

C) A way to delete everything

D) A security wall

Answer: A

147. What is "Storage Cost Optimization Audit"?

A) Checking your storage to see where you can save money

B) Paying your cloud bill

C) Deleting your account

D) Buying more disks

Answer: A

148. What is "EFS Lifecycle Management"?

A) Moving old files from EFS Standard to EFS Infrequent Access

B) Deleting files on a network drive

C) Renaming a drive

D) A security protocol

Answer: A

149. What is "Object Lambda"?

A) Using code to change data "on the fly" as it is retrieved from storage

B) A type of file

C) A security guard

D) A cloud provider

Answer: A

150. What is the ultimate goal of Cloud Storage Optimization?

A) To spend as much money as possible

B) To balance high performance/durability with the lowest possible cost

C) To delete all data

D) To use only one cloud provider

Answer: B

Subtopic 7: Data Version Control (Full 150)

Phase 1: Easy Level (1–50)

1. What is the primary purpose of Data Version Control (DVC)?

A) To make data larger

B) To track changes in datasets and models just like code

C) To delete old data automatically

D) To hide data from other users

Answer: B

2. Which popular tool is often used alongside DVC to manage the "code" part of a project?

A) Microsoft Paint

B) Git

C) Adobe Reader

D) Spotify

Answer: B

3. Why can't we just use standard Git to track 10GB datasets?

A) Git is too colorful

B) Git is not designed to handle large binary files efficiently

C) Git only works on Windows

D) Large files make Git free

Answer: B

4. In DVC, where is the actual large data usually stored?

A) Inside the .git folder

B) In a remote storage (S3, GCS, Azure Blob, or local server)

C) In the computer's RAM only

D) On a physical floppy disk

Answer: B

5. What is a "Commit" in version control?

A) A promise to work harder

B) A snapshot of the current state of the project

C) Deleting the entire project

D) Sending an email to the boss

Answer: B

6. What does "Reproducibility" mean in data science?

A) Creating more data

B) The ability to get the same results using the same data and code versions

C) Copying someone else's work

D) Printing the results

Answer: B

7. What is a "Data Pipeline" in the context of DVC?

A) A physical pipe for water

B) A series of steps that transform raw data into a final output

C) A list of employees

D) A type of network cable

Answer: B

8. Which command is used to start a DVC project?

A) dvc start

B) dvc init

C) dvc go

D) dvc new

Answer: B

9. What file does DVC create to "track" a large data file in Git?

A) A .dvc file (containing the hash/metadata)

B) A .txt file with the data

C) An .exe file

D) A .jpg image

Answer: A

10. What is "Remote Storage" in DVC?

A) A computer in another room

B) An external location (like a cloud bucket) where the actual data files are kept

C) A hidden folder on the desktop

D) A backup CD

Answer: B

11. Which of these is a DVC command to send data to remote storage?

A) dvc pull

B) dvc push

C) dvc send

D) dvc upload

Answer: B

12. Which command is used to get data from remote storage to your local machine?

A) dvc push

B) dvc pull

C) dvc fetch

D) dvc get

Answer: B

13. What is an "Experiment" in data version control?

A) A science lab test

B) A specific run of code with specific data and parameters

C) A random guess

D) A mistake in the code

Answer: B

14. What is a "Hash" (e.g., MD5)?

A) A type of food

B) A unique digital fingerprint of a file's content

C) A secret password

D) The file's name

Answer: B

15. If I change one single byte in a 1GB file, will the DVC hash change?

A) Yes

B) No

Answer: A

16. What is "Metadata"?

A) Data about data (size, hash, path)

B) Expensive data

C) Deleted data

D) Data stored in a metal box

Answer: A

17. Why is versioning important for Machine Learning models?

A) To make the models look better

B) To track which version of data and code produced a specific model

C) To increase the model speed

D) To make the model free

Answer: B

18. What is "DVC Checkout"?

A) Paying for the software

B) Restoring the data files that match the current .dvc files

C) Leaving the office

D) Deleting the data

Answer: B

19. What is "Collaboration" in DVC?

A) Working alone

B) Multiple people sharing the same data versions via remote storage

C) Competing with others

D) Hiding your code

Answer: B

20. Can DVC track directories (folders) as well as single files?

A) Yes

B) No

Answer: A

21. What is "Data Lineage"?

A) The family history of the developer

B) The history of how data was moved and transformed over time

C) The size of the dataset

D) The physical path of the cables

Answer: B

22. Is DVC open-source?

A) Yes

B) No

Answer: A

23. Which command is used to record a data transformation step in DVC?

A) dvc run

B) dvc transform

C) dvc step

D) dvc record

Answer: A

24. What is a "DAG" in DVC?

A) A type of dog

B) Directed Acyclic Graph (representing pipeline dependencies)

C) A data archive group

D) A digital access gate

Answer: B

25. What is "Dependency" in a DVC pipeline?

A) Relying on a boss

B) An input file or code that a stage needs to run

C) A bug in the system

D) A type of cloud storage

Answer: B

26. What is an "Output" in a DVC pipeline?

A) A printer

B) The result (file/folder) produced by a pipeline stage

C) The computer screen

D) A user's comment

Answer: B

27. What is "dvc.yaml"?

A) A file that describes the pipeline stages and their connections

B) A file for passwords

C) A list of team members

D) A manual for DVC

Answer: A

28. What is "dvc.lock"?

A) A physical lock for the computer

B) A file that records the exact hashes of data at a specific state

C) A way to prevent others from coding

D) A security password

Answer: B

29. What is "Data Governance"?

A) Managing data according to rules and policies

B) A government for data

C) Deleting all data

D) Selling data to the government

Answer: A

30. Which of these is a remote storage provider DVC supports?

A) Amazon S3

B) Microsoft Paint

C) Google Search

D) Netflix

Answer: A

31. What is "Storage Cost Optimization" in DVC?

A) Deleting all data

B) Only storing unique versions of data (deduplication)

C) Buying cheaper disks

D) Not using a remote

Answer: B

32. What is "Deduplication"?

A) Making data smaller

B) Ensuring only one copy of a unique file is stored, even if used in many places

C) Deleting all files

D) Renaming files

Answer: B

33. What is "Audit Trail"?

A) A path in the woods

B) A record of who changed what data and when

C) A list of computer prices

D) A backup of the OS

Answer: B

34. What is a "Branch" in version control?

A) A part of a tree

B) A separate line of development (e.g., testing a new cleaning method)

C) A company office

D) A type of folder

Answer: B

35. What is "Merging"?

A) Combining changes from one branch into another

B) Deleting a branch

C) Renaming a branch

D) Moving to a new office

Answer: A

36. What is "Conflict" in version control?

A) An argument between developers

B) When two people change the same part of a file differently

C) A computer crash

D) A power outage

Answer: B

37. Does DVC require a special server to run?

A) No, it works on your local machine

B) Yes, you must buy a DVC server

Answer: A

38. What is "DVC Add"?

A) Adding numbers

B) Telling DVC to start tracking a specific data file

C) Adding a new user

D) Buying more storage

Answer: B

39. Should you add large data files to .gitignore when using DVC?

A) Yes, because DVC handles them separately

B) No, Git should track them too

Answer: A

40. What is "Cache" in DVC?

A) Money

B) A hidden local folder where DVC stores all versions of your data

C) A deleted folder

D) A type of RAM

Answer: B

41. What is "Pulling" in DVC?

A) Pulling a lever

B) Downloading data from the remote to your local workspace

C) Deleting data

D) Renaming data

Answer: B

42. What is "Pushing" in DVC?

A) Pushing a button

B) Uploading your local data versions to the remote storage

C) Turning off the computer

D) Sending an email

Answer: B

43. What is "DVC Status"?

A) Your job title

B) A command to see if your data has changed or needs to be pushed/pulled

C) The computer's temperature

D) The time of day

Answer: B

44. What is "Version History"?

A) A list of past versions of a file

B) A history book

C) The file's creation date

D) The user's login history

Answer: A

45. What is "Rollback"?

A) Rolling a ball

B) Returning to a previous version of data or code

C) Deleting the history

D) Renaming the version

Answer: B

46. Can DVC be used for non-ML data projects?

A) Yes, any project with large files

B) No, only for Neural Networks

Answer: A

47. What is "Granularity"?

A) The level of detail at which data is versioned

B) The weight of the data

C) The price of the data

D) The number of users

Answer: A

48. Is "LakeFS" another data versioning tool?

A) Yes

B) No

Answer: A

49. Is "Git LFS" (Large File Storage) similar to DVC?

A) Yes, they solve similar problems in different ways

B) No, they are for completely different things

Answer: A

50. What is the main benefit of Data Version Control?

A) Making data bigger

B) Ensuring transparency, reproducibility, and safety for data assets

C) Increasing the cost of storage

D) Deleting old data

Answer: B

Phase 2: Medium Level (51–100)

51. What is the difference between Git LFS and DVC?

A) LFS is part of Git; DVC is an independent tool that manages pipelines too

B) LFS is faster

C) DVC is only for images

D) LFS is for code only

Answer: A

52. How does DVC handle "Data Pipelines" differently than a shell script?

A) DVC tracks dependencies and only re-runs steps if inputs change

B) DVC is more colorful

C) Shell scripts are faster

D) DVC cannot run scripts

Answer: A

53. What is "DVC Import"?

A) Buying data from another country

B) Tracking a data file from another DVC/Git repository as a dependency

C) Deleting data

D) Moving data to a USB

Answer: B

54. What is "DVC Get"?

A) Buying a new computer

B) Downloading a file from a DVC repository without tracking it

C) Getting a job

D) Finding a password

Answer: B

55. What is the purpose of the .dvc/config file?

A) To store remote storage settings (URL, credentials)

B) To store the actual data

C) To store the code

D) To store the developer's name

Answer: A

56. What is "Hard Links" vs "Symlinks" in DVC?

A) Different ways DVC connects the workspace files to the local cache

B) Different speeds of light

C) Different prices of storage

D) Different names for folders

Answer: A

57. Why would you use "Reflink" (on supported file systems)?

A) To save space and time by creating instant copies that only use space when changed

B) To make data look better

C) To delete files faster

D) To encrypt data

Answer: A

58. What is "DVC Metrics"?

A) Measuring the size of the desk

B) Tracking performance values (like Accuracy or Loss) associated with a version

C) Counting the number of files

D) Measuring the internet speed

Answer: B

59. What is "DVC Plots"?

A) Plotting a secret plan

B) Visualizing metrics over time or across different experiments

C) Drawing on the screen

D) Printing graphs on paper

Answer: B

60. What is "Feature Store" vs "Data Version Control"?

A) Feature Store manages features for production; DVC versions raw/processed files

B) They are exactly the same

C) Feature Store is for shopping

D) DVC is for physical stores

Answer: A

61. What is "Time Travel" in data versioning?

A) Going back to the dinosaur age

B) Switching the entire project (code + data) to exactly how it looked on a specific date

C) Speeding up the clock

D) Deleting the future

Answer: B

62. What is "Sharding" in data versioning?

A) Breaking a glass

B) Splitting large datasets into smaller pieces for easier tracking and processing

C) Deleting part of the data

D) Renaming files

Answer: B

63. What is "DVC repro"?

A) Reproducing a bug

B) Command to automatically re-run the pipeline to ensure results are up-to-date

C) Deleting the pipeline

D) Copying the pipeline

Answer: B

64. What happens if you run dvc repro and nothing has changed?

A) It re-runs everything anyway

B) It skips all stages because they are "up to date" (caching)

C) It deletes the project

D) It asks for a password

Answer: B

65. What is "Data Drift"?

A) Data moving slowly

B) When the statistical properties of data change over time, making models less accurate

C) Data being deleted

D) Data being renamed

Answer: B

66. How does versioning help with "Data Drift"?

A) It stops the drift

B) It allows you to compare current data versions with historical ones to identify changes

C) It makes the data static

D) It deletes the drifting data

Answer: B

67. What is "CI/CD for Machine Learning" (CML)?

A) Using robots to code

B) Automatically training and testing models whenever code or data is updated

C) A type of database

D) A security protocol

Answer: B

68. What is "DVC GC" (Garbage Collection)?

A) Picking up trash in the office

B) Removing old, unused data versions from the local cache to save space

C) Deleting the whole project

D) Restarting the computer

Answer: B

69. What is "Immutability" in the DVC cache?

A) Files in the cache should not be changed directly by the user

B) Files change every second

C) Files are invisible

D) Files are free

Answer: A

70. What is "Data Partitioning" for versioning?

A) Organizing data into folders (e.g., by date) to version small chunks instead of one giant file

B) Deleting partitions

C) Combining all data into one row

D) Encrypting data

Answer: A

71. What is "WandB" (Weights & Biases)?

A) A gym for data

B) A popular tool for experiment tracking often used alongside DVC

C) A type of scale

D) A programming language

Answer: B

72. What is "MLflow"?

A) A flow of water

B) An open-source platform for managing the ML lifecycle (including versioning)

C) A dance move

D) A cloud provider

Answer: B

73. What is "Pachyderm"?

A) A type of elephant

B) A data versioning and pipeline tool built on Kubernetes

C) A security system

D) A hardware server

Answer: B

74. What is "Semantic Versioning" (SemVer)?

A) Versioning based on meaning (e.g., v1.0.0)

B) Versioning using random numbers

C) Versioning using colors

D) No versioning

Answer: A

75. What is "Gold/Silver/Bronze" data layers?

A) Different levels of data quality/processing in a Data Lakehouse

B) Actual medals for data

C) The price of data

D) The color of the servers

Answer: A

76. How do you version "Gold" data vs "Bronze" data?

A) You track them as different outputs in the DVC pipeline

B) You only version Gold

C) You delete Bronze

D) You keep them on different computers

Answer: A

77. What is "Delta Sharing"?

A) Sharing a secret

B) An open protocol for secure data sharing across different versioned datasets

C) Deleting data

D) Sharing a USB drive

Answer: B

78. What is "Schema Enforcement"?

A) Ensuring data follows a specific structure before it is versioned

B) Forcing a user to code

C) Deleting the schema

D) Renaming the data

Answer: A

79. What is "Data Quality Testing" in a pipeline?

A) Checking if data is "good" (no nulls, correct ranges) before committing it

B) Asking the data how it feels

C) Deleting bad data

D) Printing the data

Answer: A

80. What is "DVC Freeze"?

A) Putting the computer in the freezer

B) Preventing a specific pipeline stage from being re-run, even if inputs change

C) Stopping the internet

D) Deleting the stage

Answer: B

81. What is "DVC Unfreeze"?

A) Warming up the computer

B) Allowing a frozen stage to be re-run again

C) Starting the internet

D) Renaming the stage

Answer: B

82. What is "DVC Move"?

A) Moving to a new house

B) Renaming or moving a DVC-tracked file and updating the metadata

C) Deleting the file

D) Copying the file

Answer: B

83. What is "DVC Remove"?

A) Removing a person from the team

B) Stop tracking a file with DVC and remove its .dvc file

C) Deleting the whole repo

D) Closing the program

Answer: B

84. What is "External Dependencies" in DVC?

A) Depending on a different company

B) Tracking data stored in S3 or HTTP directly without moving it into the repo

C) A bug in the cloud

D) A security risk

Answer: B

85. What is "Cloud Agnostic"?

A) A tool that doesn't believe in the cloud

B) A tool that works the same way on AWS, Azure, or Google Cloud

C) A tool for only one cloud

D) A free cloud

Answer: B

86. What is "Version Collision"?

A) Two versions crashing

B) When two different versions are mistakenly given the same ID/tag

C) A network error

D) A deleted file

Answer: B

87. What is "State-based" vs "Change-based" versioning?

A) Tracking the whole file vs. tracking only what changed

B) Tracking the mood of the data

C) Tracking the price

D) Tracking the user

Answer: A

88. Why is "State-based" (snapshots) easier for large binary data?

A) Because calculating "diffs" for binary files is very difficult and slow

B) Because it is faster to name

C) Because it uses less RAM

D) It's not easier

Answer: A

89. What is "DVC Data Validation"?

A) Checking if the data hash matches the .dvc file record

B) Asking the data for ID

C) Deleting invalid data

D) Printing the hash

Answer: A

90. What is "DVC List"?

A) A shopping list

B) Command to list all tracked files in a DVC project/remote

C) A list of users

D) A list of commands

Answer: B

91. What is "Commit Message" best practice?

A) Typing "fixed"

B) Describing exactly what changed in the data or parameters

C) Leaving it blank

D) Writing a poem

Answer: B

92. What is "Squashing" commits?

A) Stepping on the computer

B) Combining multiple minor version updates into one single significant version

C) Deleting the versions

D) Renaming the versions

Answer: B

93. What is "Data Pipeline Orchestration" (e.g., Airflow) vs DVC?

A) Orchestration handles the "when"; DVC handles the "what" and "history"

B) They are the same

C) Orchestration is for music

D) DVC is for video

Answer: A

94. What is "Model Registry"?

A) A list of fashion models

B) A central place to manage, version, and deploy trained Machine Learning models

C) A shop for models

D) A deleted file

Answer: B

95. What is "Provenance" in data?

A) The birth certificate of a dataset (origin, transformations, ownership)

B) The price of data

C) The weight of data

D) The security level

Answer: A

96. What is "Reproducible Research"?

A) Research that anyone can run and get the same results

B) Research that is copied

C) Research that is never finished

D) Research that is free

Answer: A

97. What is "DVC Pipeline Graph"?

A) A drawing of a pipe

B) A visual representation of all data stages and their connections

C) A list of stages

D) A pie chart of data

Answer: B

98. What is "Lazy Pulling" (DVC)?

A) Not pulling data at all

B) Downloading data only when it is actually needed by a specific stage

C) Pulling data slowly

D) Pulling data at night

Answer: B

99. What is "Data Integrity"?

A) Data being honest

B) Ensuring that data has not been altered or corrupted over time

C) Data being fast

D) Data being free

Answer: B

100. What is "DVC Config Remote"?

A) Setting the TV remote

B) Command to set up where DVC should store the actual data files

C) Renaming the remote

D) Deleting the remote

Answer: B

Phase 3: Master Level (101–150)

101. What is "Content-Addressable Storage" (CAS)?

A) Storing data based on its file name

B) Storing data based on its hash value rather than its location

C) Storing data in a post office

D) Storing data on a USB

Answer: B

102. In CAS, if two files have identical content but different names, how many copies are stored?

A) Two

B) One (De-duplicated)

C) Zero

D) Four

Answer: B

103. What is "Merkle Tree"?

A) A physical tree

B) A data structure where every leaf node is a hash and every non-leaf is a hash of its children

C) A family tree

D) A type of network

Answer: B

104. How do Git and DVC use Merkle Trees?

A) To efficiently verify large data structures and find exactly what changed

B) To make data look like a tree

C) To delete old nodes

D) To grow the project

Answer: A

105. What is "Atomic Commit" in a data context?

A) A tiny commit

B) A commit that either succeeds completely or fails completely (no partial updates)

C) An explosive commit

D) A fast commit

Answer: B

106. What is "Git-flow" for Data Science?

A) A river of data

B) Using specific branching strategies (Feature, Develop, Main) for data projects

C) A type of code

D) A manual process

Answer: B

107. What is "Data as Code"?

A) Writing data in Python

B) Treating datasets with the same rigor, versioning, and testing as software code

C) Deleting code

D) Moving data to the cloud

Answer: B

108. What is "Cold/Warm/Hot" data versioning?

A) Versioning based on how often the data version is accessed

B) Versioning based on temperature

C) Versioning based on file size

D) Versioning based on price

Answer: A

109. What is "Snapshot Isolation"?

A) Isolating a person

B) A technique to ensure a transaction sees a consistent state of the database from a point in time

C) Deleting a snapshot

D) A security wall

Answer: B

110. What is "Write-Ahead Logging" (WAL)?

A) Writing data before logging in

B) Recording changes in a log before they are applied to the main data (for recovery)

C) Deleting logs

D) A fast way to write

Answer: B

111. What is "Vectorized Data Versioning"?

A) Versioning using drawings

B) Versioning embeddings or vector representations of data (common in AI/LLMs)

C) A fast versioning

D) A manual versioning

Answer: B

112. What is "DVC API" (Python)?

A) A way to use DVC inside your Python scripts without the command line

B) A website for DVC

C) A type of database

D) A security code

Answer: A

113. What is "dvc.api.get_url()"?

A) Getting a website link

B) Function to get the direct cloud path of a versioned file for programmatic access

C) Deleting a URL

D) Renaming a URL

Answer: B

114. What is "Data Governance Compliance" (GDPR/HIPAA)?

A) Ensuring versioned data follows privacy laws (e.g., right to be forgotten)

B) A list of laws

C) Deleting all data

D) Selling data legally

Answer: A

115. How do you handle "Right to be Forgotten" in a versioned dataset?

A) You must delete the individual's data from ALL historical versions/backups

B) You just delete the current version

C) You ignore it

D) You rename the person

Answer: A

116. What is "Hard Delete" in DVC?

A) Deleting data gently

B) Permanently removing a specific version from the remote storage and cache

C) Deleting the whole project

D) Closing the program

Answer: B

117. What is "Shallow Clone"?

A) A clone that is not deep

B) Downloading only the latest version of data without the full history

C) A mistake in the code

D) A fake version

Answer: B

118. What is "Sparse Checkout"?

A) Checking out only some parts of a large dataset instead of the whole thing

B) Buying some items

C) Deleting parts of the data

D) Checking out slowly

Answer: A

119. What is "Data Lineage Graph"?

A) A family tree

B) A map showing how data moved from raw -> silver -> gold -> model

C) A list of files

D) A pie chart

Answer: B

120. What is "Upstream" dependency?

A) A fish swimming up

B) A file or process that comes BEFORE the current stage

C) A future task

D) A deleted stage

Answer: B

121. What is "Downstream" dependency?

A) A file or process that comes AFTER the current stage

B) A past task

C) A deleted stage

D) A hidden task

Answer: A

122. What is "DVC Metrics Diff"?

A) A math problem

B) Comparing performance metrics between two different versions or branches

C) Deleting metrics

D) Renaming metrics

Answer: B

123. What is "DVC Experiments" (exp)?

A) A science test

B) A feature to run hundreds of versions with different parameters without making new Git commits

C) A mistake in the code

D) A manual task

Answer: B

124. What is "dvc exp run"?

A) Running a science test

B) Executing a specific experiment and tracking all its inputs/outputs automatically

C) Deleting an experiment

D) Renaming an experiment

Answer: B

125. What is "dvc exp show"?

A) Showing a science test

B) Displaying a table comparing all results from different experiment runs

C) A TV show

D) A hidden file

Answer: B

126. What is "Distributed Training" versioning?

A) Versioning data used across multiple GPUs or nodes

B) Training people to version

C) Deleting training data

D) Using many versions at once

Answer: A

127. What is "Multi-Cloud Mirroring" in DVC?

A) Storing data in two clouds

B) Using multiple remotes (e.g., S3 and Azure) for the same project for redundancy

C) A type of encryption

D) A security check

Answer: B

128. What is "IAM Role" based access for DVC?

A) Using passwords

B) Using cloud-native identity roles instead of secret keys to access remote storage

C) A security guard

D) A list of roles

Answer: B

129. What is "Symlink" vs "Copy" mode in DVC?

A) Symlink creates a pointer; Copy makes a physical second file

B) They are the same

C) Symlink is for images; Copy is for text

D) Symlink is slow; Copy is fast

Answer: A

130. Why use "Symlink" mode?

A) To save massive amounts of disk space in the workspace

B) To make data look better

C) To delete data faster

D) To encrypt data

Answer: A

131. What is "Project Isolation"?

A) Working alone

B) Ensuring that changes in one project's data do not affect another's

C) A security wall

D) A deleted folder

Answer: B

132. What is "Data Sandbox"?

A) A place for kids

B) A temporary version of data used for safe testing without changing the "main" data

C) A deleted database

D) A physical box of sand

Answer: B

133. What is "Schema Drift"?

A) When the structure of data changes (e.g., a new column is added) unexpectedly

B) Data moving slowly

C) Deleting a schema

D) Renaming a schema

Answer: A

134. What is "DVC Pipeline Callbacks"?

A) Calling a friend

B) Running specific scripts automatically when a stage fails or succeeds

C) A return phone call

D) A deleted script

Answer: B

135. What is "Pre-commit Hooks" for DVC?

A) Scripts that run automatically before you allow a git commit (e.g., checking data size)

B) Fishing hooks

C) A security gate

D) A deleted hook

Answer: A

136. What is "Artifact" in version control?

A) An old object in a museum

B) Any versioned output (dataset, model, plot) generated by a pipeline

C) A mistake in the data

D) A user's comment

Answer: B

137. What is "DVC Data Registry"?

A) A central DVC repository used as a single source of truth for all datasets in a company

B) A list of names

C) A database of prices

D) A security guard

Answer: A

138. What is "Cross-Project Dependency"?

A) Depending on another company

B) When a pipeline in Project B depends on a versioned artifact from Project A

C) A bug in the code

D) A security risk

Answer: B

139. What is "Monorepo" vs "Polyrepo" for data?

A) Putting all projects in one Git repo vs. separate repos

B) Different types of money

C) Different programming languages

D) Different cloud providers

Answer: A

140. What is "DVC-Docker" integration?

A) Running DVC inside a container to ensure the processing environment is also versioned

B) Running DVC on a ship

C) A type of network

D) A manual task

Answer: A

141. What is "Parameter Sweeping"?

A) Sweeping the floor

B) Running a pipeline many times with different inputs to find the best settings

C) Deleting parameters

D) Renaming parameters

Answer: B

142. What is "Cold Start" in a data pipeline?

A) Starting in winter

B) Running a pipeline for the first time when no cache exists

C) A slow computer

D) A deleted pipeline

Answer: B

143. What is "Cache Hit"?

A) Hitting the computer

B) When DVC finds that a stage doesn't need to run because the result is already in the cache

C) A success in a game

D) A deleted file

Answer: B

144. What is "Cache Miss"?

A) Missing a target

B) When DVC has to re-run a stage because the result is not in the cache

C) An error in the code

D) A deleted file

Answer: B

145. What is "DVC Remotes per-user"?

A) Every user has their own remote

B) A setting where users can have private remotes for their own experiments

C) A remote control

D) A security password

Answer: B

146. What is "Data Encryption Key" (DEK) vs "Master Key" (MK)?

A) DEK encrypts data; MK encrypts the DEK

B) They are the same

C) DEK is for images; MK is for text

D) DEK is slow; MK is fast

Answer: A

147. What is "Zero-Trust" data versioning?

A) Not trusting anyone

B) A security model where every access request to versioned data must be verified

C) Deleting all data

D) No security

Answer: B

148. What is "Metadata Store"?

A) A shop for data

B) A specialized database (like SQL) used to store version hashes for fast searching

C) A hidden folder

D) A backup disk

Answer: B

149. What is "Large Scale Data Catalog"?

A) A big book of data

B) A system that indexes millions of versioned files across the whole company

C) A list of employees

D) A search engine

Answer: B

150. Why is "Version Control" the foundation of MLOps?

A) Because you cannot manage or improve what you cannot track and reproduce

B) Because it is a law

C) Because it is free

D) Because it is popular

Answer: A

Subtopic 8: Data Governance Standards (Full 150)

Phase 1: Easy Level (1–50)

1. What is the primary goal of Data Governance?

A) To delete as much data as possible

B) To ensure data is high quality, secure, and used correctly

C) To hide data from the CEO

D) To make data more expensive

Answer: B

2. What does GDPR stand for?

A) General Data Protection Regulation

B) Global Data Privacy Rule

C) General Digital Privacy Reform

D) Google Data Protection Role

Answer: A

3. Which of these is a key principle of Data Privacy?

A) Sharing data with everyone

B) Collecting only the data that is necessary (Data Minimization)

C) Keeping data forever

D) Selling data to the highest bidder

Answer: B

4. What is "PII"?

A) Personal Internet Information

B) Personally Identifiable Information

C) Private Internal Infrastructure

D) Public Identification Index

Answer: B

5. Which of the following is considered PII?

A) Weather data

B) Social Security Number

C) Stock market prices

D) A list of car models

Answer: B

6. What is a "Data Steward"?

A) A person who cleans the data center

B) A person responsible for the daily management and quality of data

C) The owner of the cloud company

D) A type of database software

Answer: B

7. What is "Data Quality"?

A) How fast the data moves

B) The accuracy, completeness, and reliability of data

C) The physical weight of the disks

D) The number of colors in a chart

Answer: B

8. What does "Compliance" mean in data governance?

A) Following laws, regulations, and internal policies

B) Making everyone use the same laptop

C) Deleting data every Friday

D) Moving all data to the cloud

Answer: A

9. What is "Data Lineage"?

A) The family tree of the data scientists

B) The map of where data comes from and where it goes

C) The cost of data over time

D) The speed of the data pipeline

Answer: B

10. What is "Metadata"?

A) Data about data

B) Data about the weather

C) Data that is very heavy

D) Data stored in a metal box

Answer: A

11. What is "Data Encryption"?

A) Deleting data

B) Scrambling data so only authorized people can read it

C) Renaming files to be shorter

D) Compressing files

Answer: B

12. What is an "Access Policy"?

A) A rule about who can enter the office

B) A rule defining who is allowed to view or edit specific data

C) A list of computer passwords

D) A plan for buying new servers

Answer: B

13. What is "Data Masking"?

A) Putting a mask on the server

B) Hiding sensitive parts of data (e.g., showing only last 4 digits of a card)

C) Deleting data

D) Making data unreadable forever

Answer: B

14. What is a "Data Catalog"?

A) A shopping magazine for data

B) An organized inventory of all data assets in a company

C) A list of employees

D) A folder on a desktop

Answer: B

15. What is the "Right to be Forgotten" (under GDPR)?

A) The right to forget your password

B) The right for an individual to have their personal data deleted

C) The right to quit a job

D) The right to stop using the internet

Answer: B

16. What is "Data Silo"?

A) A large storage tank for data

B) Data that is trapped in one department and not shared with others

C) A backup system

D) A very fast database

Answer: B

17. Why are Data Silos bad?

A) They take up too much physical space

B) They lead to inconsistent data and poor decision-making

C) They are too expensive to build

D) They are always empty

Answer: B

18. What is "RBAC"?

A) Role-Based Access Control

B) Real-time Backup and Compression

C) Remote Binary Access Code

D) Rule-Based Account Center

Answer: A

19. What is "Least Privilege" principle?

A) Giving everyone access to everything

B) Giving users only the minimum access they need to do their job

C) Giving the boss no access

D) Giving new employees the most access

Answer: B

20. What is "Data Auditing"?

A) Calculating the price of data

B) Reviewing who accessed or changed data to ensure rules were followed

C) Deleting old files

D) Buying more storage

Answer: B

21. What is "Data Sovereignty"?

A) The idea that data is subject to the laws of the country it is stored in

B) Data that is owned by a king

C) Data that never leaves the server

D) Free data for all

Answer: A

22. What is "Data Profiling"?

A) Drawing a picture of data

B) Analyzing data to understand its structure, quality, and content

C) Hiring a data scientist

D) Renaming columns in a table

Answer: B

23. What is "Data Classification"?

A) Sorting data into categories (e.g., Public, Internal, Confidential)

B) Putting data into alphabetical order

C) Deleting data that is not useful

D) Selling data to different classes

Answer: A

24. What is "Anonymization"?

A) Removing PII so individuals cannot be identified from the data

B) Making data secret

C) Deleting data

D) Writing data in a code language

Answer: A

25. What is "Pseudonymization"?

A) Replacing names with fake identifiers (IDs) that can be reversed if needed

B) Using a pen name

C) Deleting half the data

D) Moving data to a secret folder

Answer: A

26. What is "Data Retention Policy"?

A) A rule about how long data must be kept before it is deleted

B) A rule about how many people to hire

C) A plan to buy new disks

D) A security password

Answer: A

27. What is "Data Disposal"?

A) Recycling old computers

B) Securely deleting data when it is no longer needed

C) Giving data away

D) Storing data in a bin

Answer: B

28. What is "HIPAA" related to?

A) Space exploration

B) Healthcare and medical data privacy

C) Banking and finance

D) Social media apps

Answer: B

29. What is "PCI DSS" related to?

A) Protecting credit card and payment data

B) Protecting the environment

C) Programming in C++

D) Public computer insurance

Answer: A

30. What is a "Data Breach"?

A) A data pipeline breaking

B) Unauthorized access or theft of sensitive data

C) A server losing power

D) A computer screen cracking

Answer: B

31. What is "Data Ethics"?

A) Making sure data is profitable

B) Using data in a way that is fair, moral, and respects privacy

C) The laws about data

D) Using data to win arguments

Answer: B

32. What is "BCDR" (Business Continuity and Disaster Recovery)?

A) A plan to keep data safe and accessible after a failure or disaster

B) A new type of hard drive

C) A programming language

D) A list of employees

Answer: A

33. What is "RPO" (Recovery Point Objective)?

A) The maximum amount of data loss you can tolerate (measured in time)

B) The price of a backup

C) The name of the server

D) The location of the office

Answer: A

34. What is "RTO" (Recovery Time Objective)?

A) The maximum time it should take to get systems back online after a failure

B) The speed of the internet

C) The number of disks

D) The date of the next audit

Answer: A

35. What is "Data Accuracy"?

A) Data that is very fast

B) Data that is correct and reflects reality

C) Data that is stored in a table

D) Data that is free of errors

Answer: B

36. What is "Data Completeness"?

A) Having all the required data elements with no missing values

B) Having a lot of data

C) Finishing a data project

D) Closing a database

Answer: A

37. What is "Data Consistency"?

A) Data that looks the same across different systems

B) Data that is stored on the same disk

C) Data that is processed daily

D) Data that is easy to read

Answer: A

38. What is "Master Data Management" (MDM)?

A) Managing the CEO's data

B) Creating a single, reliable source of truth for core business data

C) Deleting all data except one file

D) Hiring a master developer

Answer: B

39. What is "Reference Data"?

A) Data used to classify other data (e.g., country codes, zip codes)

B) Data used in a book

C) A letter of recommendation

D) A link to a website

Answer: A

40. What is a "Data Quality Dashboard"?

A) A screen showing how fast the servers are

B) A visual tool to monitor data quality metrics and errors

C) A control panel for a car

D) A list of passwords

Answer: B

41. What is "Data Standardization"?

A) Making data very boring

B) Ensuring data follows the same format (e.g., YYYY-MM-DD for dates)

C) Using the same brand of computers

D) Deleting non-standard data

Answer: B

42. What is "Data Enrichment"?

A) Making data more valuable by adding information from external sources

B) Making data scientists rich

C) Increasing the file size

D) Renaming the data files

Answer: A

43. What is "Data Deduplication"?

A) Removing duplicate copies of data to save space and improve quality

B) Making two copies of everything

C) Deleting half of the data

D) Renaming files to match

Answer: A

44. What is "Self-Service Data Governance"?

A) Making users govern themselves

B) Providing tools so users can find and use data according to policies easily

C) Deleting the governance team

D) Paying users to check data

Answer: B

45. What is "Data Ownership"?

A) Owning a hard drive

B) The legal and functional responsibility for a specific dataset

C) Knowing the database password

D) Working for a data company

Answer: B

46. What is "Data Sovereignty Compliance"?

A) Ensuring data is stored in the correct geographic location to meet local laws

B) Being the king of the data center

C) Deleting illegal data

D) Free data for everyone

Answer: A

47. What is a "Shadow IT"?

A) IT support that works at night

B) Systems and data used within a company without the IT department's knowledge

C) A dark mode for the computer

D) A backup of the IT systems

Answer: B

48. Why is Shadow IT a risk for Data Governance?

A) It makes the office too dark

B) It circumvents security and privacy policies

C) It is too fast

D) It costs nothing

Answer: B

49. What is "Data Asset"?

A) A person who is good at data

B) A collection of data that has value to the organization

C) A computer screen

D) A bank account for data

Answer: B

50. What is "Data Culture"?

A) Eating lunch with data scientists

B) An organizational mindset that values data-driven decisions and governance

C) The history of data

D) A museum for old computers

Answer: B

Phase 2: Medium Level (51–100)

51. What is "Fine-Grained Access Control"?

A) Access control for small files

B) Restricting access at the row or column level rather than the whole table

C) Controlling access for small people

D) Giving access very slowly

Answer: B

52. What is "Coarse-Grained Access Control"?

A) Access control for large files

B) Restricting access at the database or table level

C) Rough access rules

D) Giving access all at once

Answer: B

53. What is "ABAC" (Attribute-Based Access Control)?

A) Access based on characteristics (e.g., user department, time of day, location)

B) Access based on the alphabet

C) Access for a specific person named Abac

D) Automated Backup and Coding

Answer: A

54. What is "Data Entropy"?

A) Data that is very energetic

B) The tendency for data quality and organization to degrade over time

C) A type of encryption

D) The speed of data

Answer: B

55. What is a "Data Dictionary"?

A) A book that defines the meaning and format of data elements (columns)

B) A translation tool for data

C) A list of common data errors

D) A search engine for data

Answer: A

56. What is "Technical Metadata" vs "Business Metadata"?

A) Tech is for machines (types, sizes); Business is for people (definitions, owners)

B) They are the same

C) Tech is expensive; Business is free

D) Tech is for text; Business is for numbers

Answer: A

57. What is "Operational Metadata"?

A) Information about data processing (run times, row counts, error logs)

B) Information about surgery

C) Data about office supplies

D) A list of phone numbers

Answer: A

58. What is "Data Provenance"?

A) The birth certificate of data (who created it, where, and why)

B) The price of data

C) The location of the data center

D) A type of security

Answer: A

59. What is "Zero Trust" security architecture?

A) Trusting no one and verifying every request for access

B) Not trusting the cloud

C) Having no security

D) Deleting all user accounts

Answer: A

60. What is "DLP" (Data Loss Prevention)?

A) Tools that stop sensitive data from leaving the company network

B) A way to recover lost data

C) A backup disk

D) A security guard

Answer: A

61. What is "Data Minimization"?

A) Making data files smaller

B) Deleting all data

C) Collecting only the minimum amount of data required for a specific task

D) Hiring fewer data scientists

Answer: C

62. What is "Differential Privacy"?

A) Being private about your math skills

B) Adding "noise" to data so individual identities are hidden while patterns stay visible

C) Hiding data from different people

D) Having different privacy for different files

Answer: B

63. What is "K-Anonymity"?

A) A brand of cereal

B) Ensuring that any individual in a dataset cannot be distinguished from at least k-1 others

C) Deleting k files

D) Making data secret for k days

Answer: B

64. What is "Data Quality Rule"?

A) A math formula to check if data is valid (e.g., "Age must be > 0")

B) A law about data

C) A command to delete data

D) A way to rename data

Answer: A

65. What is "Self-Healing Data"?

A) Data that fixes its own errors using automated scripts

B) Data that is healthy

C) Deleting broken data

D) Moving data to a clean server

Answer: A

66. What is "Dynamic Data Masking"?

A) Masking data only when it is being viewed in real-time

B) Masking data very fast

C) Changing the mask every day

D) Masking only moving data

Answer: A

67. What is "Static Data Masking"?

A) Permanently changing the data in the database to a masked version

B) Masking data that doesn't move

C) A boring mask

D) Masking data on paper

Answer: A

68. What is "Data Governance Maturity Model"?

A) A way to measure how advanced an organization's governance practices are

B) A school for data governance

C) A list of governance prices

D) A history of governance

Answer: A

69. What is "Data Marketplace"?

A) A place to buy data

B) An internal portal where users can "shop" for approved, governed datasets

C) A stock market for data

D) A shop for data scientists

Answer: B

70. What is "Federated Data Governance"?

A) Governance controlled by a central government

B) A model where central policies exist, but local teams manage their own data

C) Deleting the governance team

D) Governance for one country only

Answer: B

71. What is "Centralized Data Governance"?

A) One single team makes all the rules for the whole company

B) Governance in the center of the office

C) Governance for the central database only

D) No governance at all

Answer: A

72. What is "Agile Data Governance"?

A) Governance that moves very fast

B) An iterative approach to governance that adapts to changing business needs

C) Governance for athletes

D) Governance without rules

Answer: B

73. What is "Data Leakage"?

A) Data dripping out of a pipe

B) When sensitive data is accidentally exposed to unauthorized people

C) Losing data during a move

D) Deleting data by mistake

Answer: B

74. What is "Infoworks" or "Collibra"?

A) Examples of popular Data Governance and Metadata platforms

B) Programming languages

C) Cloud storage providers

D) Types of databases

Answer: A

75. What is "Apache Atlas"?

A) A map of the stars

B) An open-source framework for data governance and metadata management

C) A type of airplane

D) A coding language

Answer: B

76. What is "Object-Level Security" (OLS)?

A) Security for physical objects

B) Restricting access to specific tables or views

C) A security camera

D) A locked server room

Answer: B

77. What is "Row-Level Security" (RLS)?

A) Security for people in a row

B) Restricting which rows of data a user can see (e.g., only their own customers)

C) A security fence

D) Locked files

Answer: B

78. What is "Data Sanitization"?

A) Cleaning the disks with alcohol

B) Removing all sensitive information from a device or dataset permanently

C) Cleaning the code

D) Renaming files

Answer: B

79. What is "Data Obfuscation"?

A) Making data difficult to understand without authorization (similar to masking)

B) Making data clear

C) Deleting data

D) Printing data

Answer: A

80. What is "Synthetic Data"?

A) Fake data

B) Artificially generated data that mimics the properties of real data (safe for testing)

C) Data made of plastic

D) Data from a computer

Answer: B

81. What is "Data Subject"?

A) A person whose personal data is being processed

B) A topic in a data book

C) A data scientist

D) A database table

Answer: A

82. What is "Data Processor"?

A) A computer chip

B) An entity that processes data on behalf of a Data Controller

C) A person who types data

D) A type of software

Answer: B

83. What is "Data Controller"?

A) A game controller

B) The entity that decides how and why personal data is processed

C) The manager of the data team

D) A security firewall

Answer: B

84. What is "Consent Management"?

A) Managing a team

B) Tracking and respecting users' choices about their data being used

C) Saying yes to everything

D) Deleting data

Answer: B

85. What is "Data Privacy Impact Assessment" (DPIA)?

A) A test of how fast data is

B) A process to identify and minimize privacy risks in a project

C) An exam for data scientists

D) A backup plan

Answer: B

86. What is "Governance as Code"?

A) Writing governance rules in Python or SQL scripts for automation

B) Writing laws on a computer

C) Deleting the governance team

D) Using code to hide data

Answer: A

87. What is "Data Contracts"?

A) A legal agreement with a vendor

B) A technical agreement between data producers and consumers about data format/quality

C) A job contract for a developer

D) A list of prices

Answer: B

88. Why are Data Contracts important?

A) They make data more expensive

B) They prevent pipeline breakages when source data changes

C) They are required by law

D) They are easy to write

Answer: B

89. What is "Data Redaction"?

A) Adding data

B) Blacking out or removing sensitive information from a document or image

C) Renaming a file

D) Moving data to a new folder

Answer: B

90. What is "Separation of Duties"?

A) People working in different rooms

B) Ensuring that no single person has control over all parts of a critical process

C) Deleting the team

D) Everyone doing the same job

Answer: B

91. What is "Key Rotation"?

A) Turning a physical key

B) Regularly changing the encryption keys used to protect data

C) Rotating the server

D) Changing the password every hour

Answer: B

92. What is "HSM" (Hardware Security Module)?

A) A physical device that manages and protects digital keys

B) A heavy security monitor

C) A high-speed modem

D) A hidden security map

Answer: A

93. What is "BYOK" (Bring Your Own Key)?

A) Bringing your house key to work

B) Allowing a company to use their own encryption keys in a cloud provider's system

C) A security password

D) A type of lock

Answer: B

94. What is "Data Lakehouse Governance"?

A) Managing a house full of data

B) Applying governance rules consistently across both data lakes and warehouses

C) Deleting the lake

D) Building a fence around the server

Answer: B

95. What is "Data Integrity Constraint"?

A) A rule that prevents bad data from entering a database (e.g., Primary Key)

B) A law about data

C) A speed limit for data

D) A file size limit

Answer: A

96. What is "Temporal Data"?

A) Data that is angry

B) Data that includes time-based information (history of changes)

C) Temporary data

D) Data about the weather

Answer: B

97. What is "Non-Repudiation"?

A) Not being able to deny that you sent or changed something

B) Not being able to say no

C) A security error

D) A deleted log

Answer: A

98. What is "Data Localization"?

A) Keeping data in a specific city

B) A requirement that data must be stored and processed within a specific country

C) Translating data

D) Renaming data files

Answer: B

99. What is "Information Lifecycle Management" (ILM)?

A) Managing data from creation to deletion

B) A school for data

C) A type of software

D) A security plan

Answer: A

100. What is "Data Governance Council"?

A) A group of senior leaders who set the governance strategy

B) A secret meeting

C) A government for data

D) A training class

Answer: A

Phase 3: Master Level (101–150)

[Image explaining Differential Privacy showing how adding 'Laplace Noise' to a query result protects individual identity while maintaining aggregate accuracy]

101. What is "Data Mesh"?

A) A network of cables

B) An architectural approach where data is owned and governed by decentralized domain teams

C) A messy database

D) A type of encryption

Answer: B

102. In Data Mesh, what does "Data as a Product" mean?

A) Selling data to customers

B) Treating datasets as products that must be high quality and easy for others to use

C) Putting data in a box

D) Deleting bad data

Answer: B

103. What is "Computational Federated Governance"?

A) Using AI to govern data

B) Automating governance rules globally across a decentralized Data Mesh

C) A government for computers

D) A very fast governance

Answer: B

104. What is "Homomorphic Encryption"?

A) A type of code

B) Encryption that allows data to be processed/computed without being decrypted first

C) A secret password

D) Encryption for humans

Answer: B

105. Why is Homomorphic Encryption powerful for privacy?

A) It is very fast

B) It allows for secure analysis on sensitive data (like medical) while keeping it secret

C) it is free

D) It is easy to use

Answer: B

106. What is "Differential Privacy Epsilon" (ϵ)?

A) The Greek letter E

B) A parameter that measures the privacy loss/risk in a dataset

C) A speed limit

D) A type of disk

Answer: B

107. What is "Secure Multi-Party Computation" (SMPC)?

A) Multiple people sharing one computer

B) A way for multiple parties to jointly compute a function while keeping their inputs private

C) A secret club

D) A type of network

Answer: B

108. What is "Tokenization" (Security)?

A) Giving everyone a coin

B) Replacing sensitive data with a non-sensitive "token" (reference)

C) Deleting data

D) A gambling game

Answer: B

109. What is "Granular Data Lineage"?

A) Lineage that shows changes down to the individual column or cell level

B) Lineage for small files

C) A rough map of data

D) Lineage for sand

Answer: A

110. What is "Data Observability"?

A) Watching the data center

B) Monitoring the health, patterns, and quality of data in real-time

C) Looking at data charts

D) A security camera

Answer: B

111. What is "Data Drift" in governance?

A) When data moves slowly

B) When the statistical properties or structure of data change unexpectedly

C) Deleting drifting data

D) Renaming files

Answer: B

112. What is "Metadata Harvesting"?

A) Picking data from a farm

B) Automatically collecting metadata from various systems into a central catalog

C) Deleting metadata

D) Selling metadata

Answer: B

113. What is "Semantic Layer"?

A) A layer of meaning

B) A bridge between technical data and business terms for consistent reporting

C) A type of cloud storage

D) A security wall

Answer: B

114. What is "Knowledge Graph" for Governance?

A) A drawing of a brain

B) A network that represents relationships between data assets, people, and processes

C) A search engine

D) A list of books

Answer: B

115. What is "COBIT"?

A) A brand of computer

B) A framework for IT governance and management

C) A type of code

D) A security virus

Answer: B

116. What is "NIST Cybersecurity Framework"?

A) A set of standards to help organizations manage and reduce cybersecurity risk

B) A security software

C) A list of passwords

D) A government office

Answer: A

117. What is "ISO/IEC 27001"?

A) An international standard for information security management systems

B) A type of camera

C) A computer model

D) A phone number

Answer: A

118. What is "Data Governance Scorecard"?

A) A grade for the governance team

B) A set of KPIs used to track the progress and effectiveness of governance

C) A sports card

D) A list of prices

Answer: B

119. What is "Immutability" in data audit?

A) Ensuring that audit logs cannot be changed or deleted

B) Logs that change every day

C) Logs that are invisible

D) Free logs

Answer: A

120. What is "Air-Gapped" storage?

A) Storage with fans

B) A computer or network that is physically isolated from the internet for extreme security

C) Storage under water

D) A broken server

Answer: B

121. What is "Shadow Data"?

A) Dark data

B) Data that is collected but not governed, often stored in unofficial locations

C) A backup of data

D) Data from the past

Answer: B

122. What is "Dark Data"?

A) Data in the dark

B) Information collected during regular business activities but not used for anything

C) Data from the "Dark Web"

D) Deleted data

Answer: B

123. What is "Data Fabric"?

A) A type of cloth

B) An architecture that connects different data sources and provides integrated governance

C) A net for data

D) A type of network

Answer: B

124. What is "Auto-Classification"?

A) Sorting data with a car

B) Using AI to automatically tag and classify sensitive data

C) Deleting random data

D) Renaming files

Answer: B

125. What is "Policy Enforcement Point" (PEP)?

A) The location where security rules are actually applied to a request

B) A physical gate

C) A meeting room

D) A button on the screen

Answer: A

126. What is "Policy Decision Point" (PDP)?

A) The system that evaluates a request against rules to decide if access is allowed

B) A person who says yes

C) A security camera

D) A list of rules

Answer: A

127. What is "Entitlement"?

A) Feeling special

B) The specific permissions a user has been granted

C) A job title

D) A salary

Answer: B

128. What is "Micro-segmentation"?

A) Small pieces of data

B) Breaking a network into small, isolated security zones to contain breaches

C) Deleting segments

D) A type of folder

Answer: B

129. What is "Security-by-Design"?

A) Making security look good

B) Building security and privacy into a system from the very beginning of its development

C) A security poster

D) A fancy lock

Answer: B

130. What is "Privacy-by-Design"?

A) Hiding the design

B) Integrating privacy protections into every stage of a data product's lifecycle

C) A private meeting

D) A secret plan

Answer: B

131. What is "Data Parity" (Governance)?

A) Data equality

B) Ensuring that copies of data match the original source in quality and state

C) A data party

D) A type of math

Answer: B

132. What is "Cold Storage Auditing"?

A) Checking files in the refrigerator

B) Verifying that archived data is still readable and secure after many years

C) Deleting old archives

D) Moving archives to the cloud

Answer: B

133. What is "Data Lineage Reconstruction"?

A) Building a bridge

B) Rebuilding the history of data by analyzing logs when direct maps are missing

C) Deleting the history

D) Renaming the data

Answer: B

134. What is "Graph-Based Lineage"?

A) A line chart

B) Using graph database technology to map complex relationships in data pipelines

C) Drawing on paper

D) A type of math

Answer: B

135. What is "Semantic Search" in a Data Catalog?

A) Searching by meaning and intent rather than just keywords

B) Searching in a different language

C) Searching for names

D) A fast search

Answer: A

136. What is "Data Governance as a Service" (DGaaS)?

A) Hiring a company to manage your governance for you

B) A cloud-based platform for governance

C) Deleting the governance team

D) Governance for one app only

Answer: B

137. What is "Compliance Drift"?

A) When a system's settings slowly change until it is no longer following rules

B) Moving to another country

C) Deleting rules

D) Renaming a policy

Answer: A

138. What is "Penetration Testing"?

A) Testing a pen

B) Hiring ethical hackers to try and break into your data systems to find weaknesses

C) Breaking the server

D) A security check

Answer: B

139. What is "Data Custodian"?

A) A person who cleans the office

B) A technical person (like a DBA) who carries out the data owner's security rules

C) The CEO

D) A security guard

Answer: B

140. What is "Schema Evolution Governance"?

A) Managing changes to the structure of data without breaking downstream apps

B) Changing the data color

C) Deleting the schema

D) Renaming a table

Answer: A

141. What is "In-Transit Decryption"?

A) Decrypting data while it moves (usually a security risk if unauthorized)

B) Deleting moving data

C) Renaming moving data

D) A fast move

Answer: A

142. What is "OIDC" (OpenID Connect)?

A) A protocol for user authentication often used for secure data access

B) A type of code

C) An internet company

D) A security password

Answer: A

143. What is "SAML" (Security Assertion Markup Language)?

A) A language for security

B) A standard for exchanging authentication data between parties (SSO)

C) A type of file

D) A coding shortcut

Answer: B

144. What is "SSO" (Single Sign-On)?

A) Signing your name once

B) Logging into multiple systems with just one set of credentials

C) A security password

D) A deleted account

Answer: B

145. What is "Entitlement Management"?

A) Managing happy people

B) The process of granting, reviewing, and revoking user access rights

C) A job promotion

D) A list of roles

Answer: B

146. What is "Continuous Governance"?

A) Governance that never stops

B) Automating governance checks so they happen 24/7, not just during audits

C) A long meeting

D) A manual task

Answer: B

147. What is "Data Residency"?

A) Where data lives physically

B) A legal requirement to keep data within a certain border

C) A house for data

D) A security wall

Answer: B

148. What is "Governance ROI"?

A) Return on Investment for governance (how much money/risk was saved)

B) The price of a policy

C) A bonus for the team

D) A deleted file

Answer: A

149. What is "Data Quality ROI"?

A) The financial value gained from having cleaner, more accurate data

B) The number of errors fixed

C) The speed of the database

D) A discount on storage

Answer: A

150. Why is Data Governance important in the age of AI?

A) Because AI is dangerous

B) Because AI is only as good as the data it is trained on (Garbage In, Garbage Out)

C) Because it is a law

D) Because it is free

Answer: B

Subtopic 9: BigQuery Fundamentals

Phase 1: Easy Level (1–50)

1. What is Google BigQuery?

A) A physical hardware server

B) A serverless, highly scalable cloud data warehouse

C) A social media platform

D) A web browser

Answer: B

2. BigQuery is "Serverless." What does this mean for the user?

A) You have to buy your own servers

B) Google manages the underlying infrastructure and scaling

C) It doesn't use any computers

D) It only works offline

Answer: B

3. Which language is primarily used to query data in BigQuery?

A) Python

B) SQL (Standard SQL)

C) Java

D) HTML

Answer: B

4. What is a "Dataset" in BigQuery?

A) A single row of data

B) A container that holds tables and views

C) A type of chart

D) A list of users

Answer: B

5. In BigQuery, data is stored in which format?

A) Row-based

B) Columnar

C) PDF

D) Plain Text

Answer: B

6. Why is columnar storage better for analytical queries?

A) It makes the screen look better

B) It allows BigQuery to read only the specific columns needed, saving time and cost

C) It is older technology

D) It stores data in alphabetical order

Answer: B

7. What is a "Project" in BigQuery?

A) A school assignment

B) The top-level container for organizing Google Cloud resources and billing

C) A single SQL query

D) A folder on your desktop

Answer: B

8. What is the BigQuery "Sandbox"?

A) A place for kids to play

B) A free tier that lets you explore BigQuery without a credit card

C) A security feature to block hackers

D) A type of data format

Answer: B

9. How does BigQuery charge for queries by default?

A) By the hour

B) By the number of bytes processed

C) By the number of rows in the table

D) It is always free

Answer: B

10. What is a "View" in BigQuery?

A) A picture of the data

B) A virtual table defined by a SQL query

C) A way to delete data

D) A physical copy of a table

Answer: B

11. What is an "External Table"?

A) A table kept outside the building

B) A table that points to data stored in Cloud Storage without importing it

C) A table owned by another company

D) A table made of wood

Answer: B

12. Which of these can be a source for BigQuery data?

A) CSV files

B) JSON files

C) Parquet files

D) All of the above

Answer: D

13. What is "Slot" in BigQuery?

A) A place to put a coin

B) A unit of computational capacity (CPU/RAM) used to execute queries

C) A row in a table

D) A time of day

Answer: B

14. What happens when you run SELECT * on a 1TB table?

A) It is very cheap

B) It processes the entire 1TB and can be very expensive

C) It only reads the first row

D) It deletes the table

Answer: B

15. What is "Partitioning"?

A) Splitting a table into smaller segments (e.g., by date) to improve performance

B) Deleting half of the table

C) Renaming the columns

D) Sharing the table with a friend

Answer: A

16. What is "Clustering"?

A) Grouping similar data together within a partition to speed up filters

B) Putting servers in a cluster

C) Deleting duplicate rows

D) A type of encryption

Answer: A

17. What is the "BigQuery Storage Write API"?

A) A way to write emails

B) A high-performance way to stream data into BigQuery

C) A tool for drawing charts

D) A security password

Answer: B

18. What is "Dry Run" in BigQuery?

A) Running a query without processing data to see how much it will cost

B) Drying the servers

C) A query that fails

D) A test for the internet speed

Answer: A

19. What is a "Temporary Table"?

A) A table that deletes itself after the session ends

B) A table that is only half-finished

C) A table stored on a USB drive

D) A table for testing only

Answer: A

20. What is "Data Expiration"?

A) When the data becomes old and smells

B) A setting that automatically deletes data after a certain number of days

C) A bug in the system

D) A security lock

Answer: B

21. What is "Information Schema"?

A) A blueprint of the data center

B) System tables that provide metadata about your datasets and tables

C) A list of employees

D) A help manual

Answer: B

22. Which BigQuery feature allows you to run Machine Learning using SQL?

A) BigQuery ML

B) BigQuery AI

C) SQL Learn

D) Google Brain

Answer: A

23. What is "BigQuery Omni"?

A) A type of car

B) A multi-cloud analytics solution (run BQ on AWS or Azure)

C) A very large monitor

D) A search engine

Answer: B

24. What is "Public Datasets" in BigQuery?

A) Datasets anyone can edit

B) A collection of free-to-query datasets provided by Google

C) Datasets about the government

D) Social media data

Answer: B

25. What is "Federated Query"?

A) A query about the government

B) A query that pulls data from a different database (like Cloud SQL) in real-time

C) A very fast query

D) A query that uses multiple languages

Answer: B

26. What is "Job ID"?

A) Your employee number

B) A unique identifier for every query or task run in BigQuery

C) The name of a table

D) A password

Answer: B

27. What is "Streaming Buffer"?

A) A video loading screen

B) A place where recently streamed data is held before being fully committed

C) A type of cache

D) A network error

Answer: B

28. What is "Identity and Access Management" (IAM) in BigQuery?

A) Managing passwords

B) Controlling who can see or edit datasets and tables

C) A security camera

D) A list of user names

Answer: B

29. What is "Query History"?

A) A list of every query you have run in the past

B) A history of the SQL language

C) A backup of your data

D) A list of deleted tables

Answer: A

30. What is "Table Snapshot"?

A) A photo of the computer screen

B) A point-in-time copy of a table that saves storage costs vs a full copy

C) A small version of a table

D) A deleted table

Answer: B

31. What is "Wildcard Table" query?

A) A query that uses * to search across multiple tables with similar names

B) A random query

C) A query that is difficult to write

D) A query that fails

Answer: A

32. What is "Billing Project"?

A) The project that is charged for the query costs

B) A project for accountants

C) A free project

D) A project that has no data

Answer: A

33. What is "DML" in BigQuery?

A) Data Manipulation Language (INSERT, UPDATE, DELETE)

B) Digital Mode Logic

C) Data Mining Link

D) Direct Mail List

Answer: A

34. What is "DDL" in BigQuery?

A) Data Definition Language (CREATE, ALTER, DROP)

B) Digital Design Level

C) Data Delivery Link

D) Direct Data Log

Answer: A

35. What is "BigQuery Studio"?

A) A place to record music

B) A unified interface for data engineering and analytics in the GCP console

C) A training room

D) A type of computer

Answer: B

36. Can BigQuery handle Petabytes of data?

A) Yes

B) No, only up to 1GB

Answer: A

37. What is "Standard SQL" vs "Legacy SQL"?

A) Standard is the modern, preferred version; Legacy is the old version

B) They are the same

C) Legacy is faster

D) Standard is only for small data

Answer: A

38. What is "BigQuery Reservations"?

A) Booking a table at a restaurant

B) Buying a fixed amount of Slot capacity for a predictable monthly cost

C) Saving a name for a table

D) A security feature

Answer: B

39. What is "On-Demand" pricing?

A) Paying for queries based on the amount of data they read

B) Paying for a monthly subscription

C) Free for the first year

D) Paying for the number of users

Answer: A

40. What is "Load Job"?

A) A job that is very heavy

B) A process to move data into a BigQuery table

C) A training class

D) A way to delete data

Answer: B

41. What is "Export Job"?

A) Moving data from BigQuery to Cloud Storage

B) Selling data to another country

C) Deleting data

D) A type of query

Answer: A

42. What is "UDF" (User Defined Function)?

A) A custom function written in SQL or JS to extend capabilities

B) A user's name

C) A system error

D) A type of table

Answer: A

43. What is "BigQuery Data Transfer Service"?

A) A physical truck moving disks

B) A service that automatically imports data from apps like Google Ads

C) A network cable

D) A manual upload

Answer: B

44. What is "Materialized View"?

A) A view that actually stores the data on disk to speed up queries

B) A view you can touch

C) A view that is very large

D) A deleted view

Answer: A

45. What is "Schema"?

A) A plan to hack the system

B) The structure of a table (column names, types, and modes)

C) The color of the UI

D) The location of the data center

Answer: B

46. What is "Nullable" mode in a schema?

A) The column can have empty (NULL) values

B) The column must be empty

C) The column is invisible

D) The column is deleted

Answer: A

47. What is "Required" mode in a schema?

A) Every row must have a value in that column

B) The column is optional

C) The column is for passwords only

D) The column is for numbers only

Answer: A

48. What is "Repeated" mode (Array)?

A) A column that can hold multiple values (a list) in a single row

B) A column that repeats the same word

C) A column that is copied twice

D) A column for dates

Answer: A

49. What is "Casting" in SQL?

A) Throwing a fishing line

B) Changing a value from one data type to another (e.g., String to Integer)

C) Deleting a column

D) Sorting a table

Answer: B

50. What is "BigQuery Cache"?

A) Storing results of previous queries so running them again is free and instant

B) Money kept in the data center

C) A hidden dataset

D) A type of disk

Answer: A

Phase 2: Medium Level (51–100)

51. How does Clustering differ from Partitioning?

A) Partitioning uses strict boundaries; Clustering sorts data within those boundaries

B) They are exactly the same

C) Clustering is only for text; Partitioning is for numbers

D) Partitioning is free; Clustering is expensive

Answer: A

52. What is the limit for the number of partitions per table?

A) 10

B) 4,000

C) Unlimited

D) 100

Answer: B

53. What is "BigQuery BI Engine"?

A) An engine for a car

B) An in-memory service that speeds up dashboard queries to sub-second levels

C) A tool for building websites

D) A security firewall

Answer: B

54. What is "Time Travel" in BigQuery?

A) Viewing data as it existed at any point in the last 7 days

B) Predicting future data

C) Deleting old data

D) Moving to a new time zone

Answer: A

55. How do you query a table using Time Travel?

A) Using the FOR SYSTEM_TIME AS OF clause

B) Using the BACK TO THE FUTURE command

C) Using a special time-machine button

D) You can't query it

Answer: A

56. What is "Nested and Repeated" data?

A) Data that is annoying

B) Data structured in JSON-like format (Structs and Arrays) within a row

C) Data that is copied many times

D) Data in a secret folder

Answer: B

57. What is the benefit of "Nested" data in BigQuery?

A) It avoids the need for expensive JOINs by keeping related data together

B) It makes the table smaller

C) It is easier for Excel to read

D) It is more colorful

Answer: A

58. What is "UNNEST" in SQL?

A) Deleting a bird's nest

B) A function used to flatten an Array into individual rows for querying

C) Combining two tables

D) Creating a new array

Answer: B

59. What is "Dremel"?

A) A power tool

B) The underlying execution engine that powers BigQuery queries

C) A type of database

D) A Google employee

Answer: B

60. What is "Colossus"?

A) A large statue

B) Google’s distributed file system where BigQuery data is physically stored

C) A type of server

D) A very long query

Answer: B

61. What is "Jupiter" in the BigQuery context?

A) A planet

B) The high-speed network that connects BigQuery compute to storage

C) A programming language

D) A type of cloud region

Answer: B

62. What is "Query Execution Plan"?

A) A plan to learn SQL

B) A visual breakdown of the steps BigQuery takes to run your query

C) A list of project goals

D) A security map

Answer: B

63. What is "Shuffle" in a query plan?

A) Moving data between different slots to perform joins or aggregations

B) Randomizing the rows

C) Deleting data

D) Sorting data by hand

Answer: A

64. What is "Slot Contention"?

A) When a query is fighting for enough computational power to run

B) An argument between developers

C) A network error

D) A slow database

Answer: A

65. What is "BigQuery ML: CREATE MODEL"?

A) A command to train a machine learning model using SQL syntax

B) Creating a 3D model

C) Hiring a data scientist

D) Renaming a table

Answer: A

66. Which model type does BigQuery ML support?

A) Linear Regression

B) Logistic Regression

C) K-means Clustering

D) All of the above

Answer: D

67. What is "Approximate Aggregate Functions"?

A) Functions that give an estimated answer very quickly for massive data

B) Functions that are always wrong

C) Functions for small data

D) Functions that are very slow

Answer: A

68. What is "Authorized View"?

A) A view that allows users to query results without access to the source tables

B) A view with a password

C) A view for the boss only

D) A view that is public

Answer: A

69. What is "Column-level Security"?

A) Hiding certain columns from specific users based on policy tags

B) Locking the column names

C) Deleting columns

D) Renaming columns for security

Answer: A

70. What is "Data Policy Tag"?

A) A label used to control access to sensitive columns (e.g., PII)

B) A price tag for data

C) A tag used for sorting

D) A type of metadata

Answer: A

71. What is "BigQuery Search Index"?

A) A way to find files on the computer

B) A feature to speed up finding specific strings in large logs

C) A search engine like Google

D) A list of table names

Answer: B

72. What is "Change Data Capture" (CDC) in BigQuery?

A) Catching hackers

B) Processing only the changes (inserts/updates) from a source database

C) Deleting old data

D) Renaming tables

Answer: B

73. What is "BigQuery Connection"?

A) A social network for data engineers

B) An object used to connect to external sources like Cloud SQL

C) A physical cable

D) An internet link

Answer: B

74. What is "BigQuery Object Table"?

A) A table for physical objects

B) A table that represents unstructured data (images/PDFs) in Cloud Storage

C) A table for 3D models

D) A table that cannot be edited

Answer: B

75. What is "BigLake"?

A) A large body of water

B) A storage engine that unifies data lakes and warehouses

C) A type of cloud storage

D) A fast network

Answer: B

76. What is "Scalar Function"?

A) A function that operates on a single value and returns a single value

B) A function for weight scales

C) A function that returns a table

D) A function that deletes data

Answer: A

77. What is "Aggregate Function"?

A) A function that takes many rows and returns one value (e.g., SUM, AVG)

B) A function that makes data larger

C) A function for dates

D) A function for text

Answer: A

78. What is "Window Function" (OVER clause)?

A) A function that looks out a window

B) A function that performs calculations across a set of rows related to the current row

C) A function that opens a new tab

D) A function that only works on Windows OS

Answer: B

79. What is "PARTITION BY" in a Window Function?

A) Dividing the result set into groups to apply the function separately

B) Deleting partitions

C) Creating a new table

D) Sorting data

Answer: A

80. What is "BigQuery GeoViz"?

A) A tool for visualizing geographic/map data from BigQuery

B) A weather app

C) A map of the data center

D) A GPS for data

Answer: A

81. What is "Geography" data type?

A) Data about schools

B) A type used for points, lines, and polygons (latitude/longitude)

C) Data about countries

D) A list of addresses

Answer: B

82. What is "Safe Prefix" (e.g., SAFE_CAST)?

A) A way to prevent a query from failing if there is an error

B) A security lock

C) A way to encrypt data

D) A fast way to cast

Answer: A

83. What is "MERGE" statement?

A) Combining two tables into one

B) An atomic operation to perform INSERT, UPDATE, and DELETE in one go

C) Deleting duplicate rows

D) Renaming a dataset

Answer: B

84. What is "BigQuery Job Priority" (Interactive vs Batch)?

A) Interactive runs immediately; Batch runs when resources are free

B) Interactive is for games; Batch is for work

C) Interactive is for small data; Batch is for large data

D) They are the same

Answer: A

85. What is "BigQuery ML: ML.EVALUATE"?

A) A function to check how accurate your model is after training

B) A function to delete a model

C) A function to name a model

D) A function to buy a model

Answer: A

86. What is "BigQuery ML: ML.PREDICT"?

A) A function to use your trained model to make predictions on new data

B) A function to guess the weather

C) A function to see the future

D) A function to rename data

Answer: A

87. What is "Scripting" in BigQuery SQL?

A) Writing code in Python

B) Using control flow (IF, WHILE) and variables inside SQL

C) A list of movie scripts

D) A way to automate the keyboard

Answer: B

88. What is a "Stored Procedure"?

A) A saved block of SQL code that can be called with parameters

B) A rule for the office

C) A backup of the database

D) A way to store files

Answer: A

89. What is "BigQuery Table Clone"?

A) An identical copy of a table

B) A lightweight, writable copy of a table that only charges for changed data

C) A deleted table

D) A fake table

Answer: B

90. What is "Customer-Managed Encryption Keys" (CMEK)?

A) Letting the customer hold the physical keys to the building

B) Allowing the user to manage the keys that encrypt BigQuery data

C) A type of password

D) No encryption

Answer: B

91. What is "Query Tail Latency"?

A) The delay for the slowest 1% of queries

B) The time it takes to type a query

C) The length of a query

D) A network error

Answer: A

92. What is "Export to Dataframe" (Python)?

A) Using the BigQuery Python client to pull results into a Pandas dataframe

B) Sending an email with data

C) Deleting a dataframe

D) Printing data in a table

Answer: A

93. What is "BigQuery ML: Hyperparameter Tuning"?

A) Automatically finding the best settings for an ML model to improve accuracy

B) Tuning a radio

C) Speeding up the CPU

D) Deleting parameters

Answer: A

94. What is "BigQuery Information Schema: JOBS_BY_USER"?

A) A table that shows every query a specific person has run

B) A list of jobs at Google

C) A way to find new employees

D) A password list

Answer: A

95. What is "Query Plan Stage"?

A) A step in the execution process (e.g., S01: Input, S02: Aggregate)

B) A physical location

C) A time of day

D) A level in a game

Answer: A

96. What is "Wait Time" in a query plan?

A) The time a slot spends waiting for data or other slots to finish

B) The time a user waits for results

C) The time the computer is turned off

D) A network delay

Answer: A

97. What is "Read Ratio"?

A) The amount of data read vs data stored

B) The speed of reading

C) The cost of reading

D) A math problem

Answer: A

98. What is "BigQuery Data Clean Room"?

A) A place to wash servers

B) A secure environment for multiple parties to analyze shared data safely

C) A deleted dataset

D) A security firewall

Answer: B

99. What is "Analytics Hub"?

A) A central place to share and discover datasets within or across organizations

B) A news website

C) A type of network

D) A search engine

Answer: A

100. What is the primary advantage of BigQuery over traditional MySQL for big data?

A) It is cheaper for small data

B) It can scale to petabytes and perform massive parallel processing instantly

C) It is easier to install on a laptop

D) It doesn't use SQL

Answer: B

Phase 3: Master Level (101–150)

101. What is the "Capacitor" format?

A) A component in a computer

B) BigQuery’s proprietary columnar storage format

C) A type of battery

D) A data transmission protocol

Answer: B

102. How does "Capacitor" handle semi-structured data like JSON?

A) It converts it to a flat string

B) It encodes nested fields into their own columns for efficient access

C) It deletes it

D) It stores it as an image

Answer: B

103. What is "Slot Auto-scaling"?

A) Automatically adding more slots to a query as it becomes more complex

B) Changing the size of the computer

C) Deleting slow slots

D) A pricing model for small data

Answer: A

104. What is "BigQuery Flex Slots"?

A) Slots that can bend

B) Short-term slot commitments for burst workloads

C) Free slots

D) Slots that only work at night

Answer: B

105. What is "Query Complexity" in terms of slots?

A) The more JOINs and GROUP BYs, the more slots and shuffle required

B) The number of words in the query

C) The font size of the SQL

D) The user's experience level

Answer: A

106. What is "Skew" in BigQuery data?

A) When data is perfectly balanced

B) When one partition has significantly more data than others, causing slow slots

C) A type of encryption

D) A mathematical error

Answer: B

107. How do you handle "Data Skew" in a JOIN?

A) Using a "Map-side join" or pre-aggregating the skewed values

B) Deleting the skewed data

C) Using more servers

D) Renaming the columns

Answer: A

108. What is "BigQuery ML: ML.FEATURES_AT_TIME"?

A) A function to get feature values as they existed at a specific point in time

B) A way to see future features

C) A list of features

D) A security check

Answer: A

109. What is "External Query" vs "Federated Query"?

A) External refers to the table definition; Federated refers to the act of querying it

B) They are the same

C) External is for images; Federated is for text

D) Federated is faster

Answer: A

110. What is "BigQuery Remote Function"?

A) Using a TV remote to query

B) Calling a Cloud Function (Python/Node) directly from inside a SQL query

C) A function on a different planet

D) A deleted function

Answer: B

111. What is "BigQuery ML: Model Export"?

A) Exporting a trained model to Vertex AI or Cloud Storage as TF/ONNX

B) Selling a model

C) Deleting a model

D) Renaming a model

Answer: A

112. What is "Table Decorator" (Legacy)?

A) A way to make tables look pretty

B) An old way to perform time-travel queries (e.g., table@timestamp)

C) A type of metadata

D) A security tag

Answer: B

113. What is "BigQuery Write API: Pending Mode"?

A) Data is written but not visible until the entire batch is committed

B) Data is waiting in a queue

C) Data is deleted

D) A network error

Answer: A

114. What is "BigQuery Write API: Committed Mode"?

A) Data is immediately visible as soon as it is written

B) Data is promised to be written later

C) Data is encrypted

D) A pricing plan

Answer: A

115. What is "BigQuery Write API: Buffered Mode"?

A) Data is flushed to the table periodically or when the buffer is full

B) A video loading screen

C) Data is deleted

D) A security feature

Answer: A

116. What is "Query Metadata Cache"?

A) BigQuery using metadata to answer queries (like COUNT(*)) without reading rows

B) A hidden file

C) A list of users

D) A type of RAM

Answer: A

117. What is "Input Bytes Read" vs "Input Bytes Billed"?

A) Read is what the engine touches; Billed is what you pay for (min 10MB)

B) They are the same

C) Billed is always higher

D) Read is always higher

Answer: A

118. What is "Partition Elimination" (Pruning)?

A) BigQuery skipping partitions that don't match the WHERE clause

B) Deleting partitions

C) Creating new partitions

D) Renaming partitions

Answer: A

119. What is "Clustering Pruning"?

A) Skipping blocks of data within a partition that don't match the filter

B) Deleting clusters

C) Creating clusters

D) Renaming clusters

Answer: A

120. What is "BigQuery Storage Read API"?

A) A high-speed, parallelized way for external tools (like Spark) to read BQ data

B) A way to read books

C) A security check

D) A pricing plan

Answer: A

121. What is "Arrow" format in BigQuery?

A) A type of weapon

B) An in-memory columnar format used to transfer data fast to Python/R

C) A directional sign

D) A type of network

Answer: B

122. What is "BigQuery ML: ML.PRINCIPAL_COMPONENT_ANALYSIS"?

A) A model type used for dimensionality reduction

B) Analyzing the principal of a school

C) A security check

D) A type of database

Answer: A

123. What is "BigQuery ML: ML.TRANSFORM"?

A) A clause to perform feature engineering inside the model definition

B) A robot movie

C) Deleting data

D) Renaming data

Answer: A

124. What is "BigQuery ML: ML.RECOMENDER"?

A) A model type for building recommendation systems

B) A person who gives advice

C) A list of movies

D) A security rule

Answer: A

125. What is "Incremental Materialized View"?

A) A view that only updates the changed parts to save compute power

B) A view that grows forever

C) A view that is deleted daily

D) A view for small data

Answer: A

126. What is "BigQuery Search Index: Text Analyzer"?

A) A setting that determines how text is broken into searchable tokens

B) A person who reads text

C) A tool for translating languages

D) A security firewall

Answer: A

127. What is "BigQuery ML: ML.FORECAST"?

A) A function for time-series forecasting (ARIMA+)

B) Predicting the weather

C) Guessing the stock price

D) A future date

Answer: A

128. What is "Recursive CTE" in BigQuery?

A) A Common Table Expression that refers to itself

B) A query that never ends

C) A type of table

D) A security error

Answer: A

129. What is "BigQuery Reservation: Baseline vs Autoscale"?

A) Baseline is the minimum guaranteed slots; Autoscale adds more as needed

B) Baseline is for small data; Autoscale is for large data

C) They are the same

D) Baseline is free

Answer: A

130. What is "BigQuery Resource Charts"?

A) Visual tools in the console to monitor slot usage and job concurrency

B) A list of employees

C) A map of the data center

D) A pricing table

Answer: A

131. What is "VPC Service Controls" for BigQuery?

A) Preventing data from being exfiltrated to unauthorized projects

B) A faster internet

C) A type of network cable

D) A security camera

Answer: A

132. What is "BigQuery Data Governance Policy"?

A) A set of rules for data quality and access

B) A law about data

C) A list of users

D) A backup plan

Answer: A

133. What is "BigQuery Table Copy" vs "Clone"?

A) Copy is a full new copy (double cost); Clone is a pointer (saves cost)

B) They are the same

C) Clone is faster

D) Copy is for small data

Answer: A

134. What is "BigQuery Snapshot Expiration"?

A) When a snapshot is automatically deleted to save money

B) When a photo fades

C) A bug in the system

D) A security lock

Answer: A

135. What is "BigQuery ML: ML.EXPLAIN_PREDICT"?

A) A function that tells you why a model made a specific prediction

B) A dictionary for ML

C) A training class

D) A security check

Answer: A

136. What is "BigQuery ML: ML.GLOBAL_EXPLAIN"?

A) A function to see which features are most important for the model overall

B) A map of the world

C) A news report

D) A list of features

Answer: A

137. What is "BigQuery Routine"?

A) A daily task

B) An object that encompasses both Stored Procedures and UDFs

C) A type of workout

D) A manual process

Answer: B

138. What is "BigQuery Table Schema Evolution"?

A) Automatically adding new columns to a table when new data is loaded

B) Changing the table color

C) Deleting the table

D) Renaming the table

Answer: A

139. What is "BigQuery ML: ML.GENERATE_TEXT" (LLM)?

A) A function to call Large Language Models (like Gemini) directly from SQL

B) A type of typewriter

C) A way to write books

D) A security check

Answer: A

140. What is "BigQuery ML: Remote Model"?

A) A model hosted on Vertex AI but called from BigQuery SQL

B) A model on a different planet

C) A model with a remote control

D) A deleted model

Answer: A

141. What is "BigQuery ML: ML.UNDERLYING_TABLE"?

A) Finding the original data used for a view or snapshot

B) A table under a desk

C) A hidden table

D) A backup table

Answer: A

142. What is "BigQuery Query Execution: Parallelism"?

A) Running many parts of a query at the same time across thousands of slots

B) Running queries in alphabetical order

C) Running queries one by one

D) A security feature

Answer: A

143. What is "BigQuery Query Execution: Pipelining"?

A) Passing data from one stage to the next without writing to disk

B) A physical pipe

C) A type of network

D) A slow way to query

Answer: A

144. What is "BigQuery Distributed Join"?

A) Moving data across the network so two large tables can be joined

B) Joining a club

C) Combining two files

D) A manual task

Answer: A

145. What is "BigQuery Broadcast Join"?

A) Sending a small table to every slot to join with a large table (very fast)

B) A radio broadcast

C) A public query

D) A type of network

Answer: A

146. What is "BigQuery Slot Utilization"?

A) The percentage of purchased slots that are actually being used

B) The price of a slot

C) The size of a slot

D) The number of users

Answer: A

147. What is "BigQuery Data Lineage API"?

A) A way to track how data moves through BigQuery for governance

B) A family tree

C) A search engine

D) A list of files

Answer: A

148. What is "BigQuery Error: Resource Exceeded"?

A) Your query tried to use more memory/shuffle than a single slot allows

B) You ran out of money

C) The data center is full

D) Your computer is broken

Answer: A

149. What is "BigQuery Error: Billing Not Enabled"?

A) You haven't connected a credit card to the project

B) You are too rich

C) The project is free

D) A network error

Answer: A

150. Why is BigQuery considered the "Modern Data Stack" leader?

A) Because it is easy to use

B) Because it unifies storage, compute, ML, and multi-cloud analytics in one serverless tool

C) Because it is free

D) Because it is from Google

Answer: B

Subtopic 10: Workflow Orchestration,

Phase 1: Easy Level (1–50)

1. What is the primary purpose of workflow orchestration?

A) To write better code

B) To automate and manage the order of execution of different tasks

C) To delete data automatically

D) To make the computer faster

Answer: B

2. What does "DAG" stand for in orchestration?

A) Data Access Gate

B) Directed Acyclic Graph

C) Digital Archive Group

D) Direct Analysis Grid

Answer: B

3. In a DAG, what does "Acyclic" mean?

A) The tasks can run forever

B) There are no loops (tasks cannot circle back to themselves)

C) All tasks must be circular

D) Tasks are not connected

Answer: B

4. Which of these is a popular open-source tool for orchestration?

A) Microsoft Word

B) Apache Airflow

C) Spotify

D) Adobe Photoshop

Answer: B

5. What is a "Task" in a workflow?

A) A complete project

B) A single unit of work (e.g., running a SQL query or a Python script)

C) An email from a boss

D) A physical computer

Answer: B

6. What is "Scheduling" in orchestration?

A) Planning a meeting

B) Setting a specific time or frequency for a workflow to run

C) Deleting old tasks

D) Hiring new engineers

Answer: B

7. What is a "Dependency" between tasks?

A) When two tasks hate each other

B) When Task B cannot start until Task A successfully finishes

C) When tasks run at the exact same time

D) When tasks are in different projects

Answer: B

8. What is a "Trigger"?

A) An error message

B) An event that starts the execution of a workflow

C) A button to delete code

D) A type of database

Answer: B

9. What happens if a task "Fails"?

A) The computer explodes

B) The orchestration tool stops the downstream tasks and alerts the user

C) Nothing happens; it just continues

D) The whole project is deleted

Answer: B

10. What is a "Retry" policy?

A) Trying to get a new job

B) Automatically attempting to run a failed task again a certain number of times

C) Deleting the failed task

D) Asking a human to fix it immediately

Answer: B

11. What is "Monitoring" in orchestration?

A) Watching the computer screen

B) Tracking the status (Success, Failure, Running) of all workflows

C) Buying a new monitor

D) Typing code

Answer: B

12. What is an "Operator" in Apache Airflow?

A) A person who runs the computer

B) A template for a specific type of task (e.g., PythonOperator, BashOperator)

C) A mathematical symbol (+, -)

D) A security guard

Answer: B

13. What is a "Workflow"?

A) The time you spend at work

B) A sequence of tasks that move data from source to destination

C) A list of employees

D) A type of office chair

Answer: B

14. Why do we need orchestration instead of just "Cron" jobs for complex pipelines?

A) Cron is too expensive

B) Orchestration handles complex dependencies and retries much better than Cron

C) Orchestration is older

D) Cron only works on Windows

Answer: B

15. What is "Backfilling"?

A) Filling a hole with dirt

B) Running a workflow for dates in the past (e.g., catching up on missed data)

C) Deleting old data

D) Renaming files

Answer: B

16. What is an "Upstream" task?

A) A task that runs after the current one

B) A task that runs before the current one

C) A task that is very fast

D) A task in a different country

Answer: B

17. What is a "Downstream" task?

A) A task that runs after the current one

B) A task that runs before the current one

C) A task that is very slow

D) A task that is deleted

Answer: A

18. What is "Concurrency" in orchestration?

A) Running tasks one after another

B) Running multiple tasks or workflows at the same time

C) A type of currency

D) An error in the code

Answer: B

19. What is a "Sensor" in orchestration tools?

A) A camera on the computer

B) A special task that waits for something to happen (like a file appearing)

C) A way to measure temperature

D) A list of passwords

Answer: B

20. What is a "Hook" (in Airflow)?

A) A tool for fishing

B) An interface to interact with external systems (like a database or API)

C) A way to delete data

D) A security lock

Answer: B

21. What is "Parallelism"?

A) Drawing straight lines

B) The ability to execute many tasks simultaneously to save time

C) Writing code in two languages

D) Using two monitors

Answer: B

22. What is "GCP Cloud Composer"?

A) A tool for writing music

B) A managed version of Apache Airflow on Google Cloud

C) A type of cloud storage

D) A security service

Answer: B

23. What is "AWS Step Functions"?

A) A tool for exercising

B) A serverless orchestration service provided by Amazon

C) A type of database

D) A search engine

Answer: B

24. What is "Azure Data Factory"?

A) A physical factory in Seattle

B) A cloud-based data integration and orchestration service from Microsoft

C) A list of computers

D) A coding language

Answer: B

25. What is a "Timeout"?

A) A break for the engineer

B) A limit on how long a task is allowed to run before it is marked as failed

C) Turning off the computer

D) A fast task

Answer: B

26. What is "Idempotency"?

A) A medical condition

B) The property where running the same task multiple times produces the same result

C) Deleting data by mistake

D) Changing the code daily

Answer: B

27. What is a "Worker" in Airflow?

A) A manual laborer

B) The node or process that actually performs the task logic

C) A keyboard

D) A type of file

Answer: B

28. What is a "Log"?

A) A piece of wood

B) A text record of what happened during execution (useful for debugging)

C) A type of password

D) A database table

Answer: B

29. What is "Alerting"?

A) Shouting in the office

B) Sending notifications (Email, Slack) when a workflow fails

C) A type of code

D) A security firewall

Answer: B

30. What is a "Task Instance"?

A) A single specific run of a task for a specific date/time

B) A group of tasks

C) A deleted task

D) A person who writes tasks

Answer: A

31. What is "Execution Date"?

A) The day a task is deleted

B) The logical time for which a workflow is processing data

C) Today's date

D) The engineer's birthday

Answer: B

32. What is the "Metadata Database" in Airflow?

A) A place to store the actual big data

B) A database that tracks the status of jobs, tasks, and users

C) A list of book titles

D) A hidden folder

Answer: B

33. What is a "Worker" in orchestration?

A) A human employee

B) The component/node that actually executes the tasks

C) A desk in the office

D) A piece of code

Answer: B

34. What is the "Scheduler"?

A) A calendar

B) The component that decides which tasks need to run and when

C) A person who sets meetings

D) A type of CPU

Answer: B

35. What is the "Web UI"?

A) A type of internet

B) The visual interface used to view and manage workflows in a browser

C) A security password

D) A programming language

Answer: B

36. Can a DAG have a "Cycle" (A -> B -> A)?

A) Yes

B) No

Answer: B

37. What is "Priority Weight"?

A) The size of the code

B) A value that tells the scheduler which tasks to run first if resources are limited

C) The cost of the task

D) The number of users

Answer: B

38. What is a "Pool" in orchestration?

A) A swimming area

B) A way to limit the number of parallel tasks for a specific system

C) A group of engineers

D) A type of database

Answer: B

39. What is "XCom" (in Airflow)?

A) A type of alien

B) A mechanism for tasks to share small amounts of data/messages

C) A fast network

D) A security code

Answer: B

40. What is the "State" of a task?

A) The country it is in

B) Its current status (e.g., Queued, Running, Success, Failed)

C) The size of the file

D) The name of the user

Answer: B

41. What is "SLO" (Service Level Objective)?

A) A slow query

B) A goal for how fast or reliable a workflow should be

C) A type of task

D) A security rule

Answer: B

42. What is "Catchup" in Airflow?

A) A tomato sauce

B) A setting that tells Airflow to run all missed intervals when a DAG is enabled

C) A fast way to code

D) Deleting the past

Answer: B

43. What is an "External Task Sensor"?

A) A physical sensor on a machine

B) A task that waits for a task in a different DAG to complete

C) A security alarm

D) A way to measure light

Answer: B

44. What is "Jinja Templating"?

A) A type of tea

B) A way to dynamically change task parameters (like dates) using code

C) A drawing tool

D) A security password

Answer: B

45. What is a "Variable" in Airflow?

A) Something that changes

B) A way to store and retrieve runtime configuration (like API keys)

C) A type of math

D) A person's name

Answer: B

46. What is a "Connection" in Airflow?

A) A social link

B) Stored credentials used to talk to external systems

C) A physical cable

D) An internet speed

Answer: B

47. Is orchestration only for "Big Data"?

A) Yes

B) No, it can be used for any sequence of automated tasks

Answer: B

48. What is "Version Control" for DAGs?

A) Keeping track of changes to the pipeline code (usually in Git)

B) Renaming the DAG files

C) Deleting old versions

D) Moving files to a USB

Answer: A

49. What is "Orchestration Fatigue"?

A) When an engineer is tired

B) When there are too many complex, failing workflows to manage

C) A computer slowing down

D) A broken server

Answer: B

50. What is the main benefit of Workflow Orchestration?

A) It makes data bigger

B) It ensures complex data pipelines are reliable, predictable, and observable

C) It is free

D) It replaces the need for data engineers

Answer: B

Phase 2: Medium Level (51–100)

51. What is the role of the "Executor" in Airflow?

A) It writes the code

B) It determines how tasks are sent to workers (e.g., Celery, Kubernetes)

C) It deletes failed tasks

D) It manages the user interface

Answer: B

52. What is the "Celery Executor"?

A) An executor made of vegetables

B) A distributed executor that uses a message queue to handle tasks across multiple workers

C) A fast executor for small tasks

D) A way to delete data

Answer: B

53. What is the "Kubernetes Executor"?

A) An executor that runs each task in its own temporary Docker container (pod)

B) A very heavy executor

C) An executor that only works on Windows

D) A security guard

Answer: A

54. What is "Dynamic DAG generation"?

A) Creating DAGs that change every second

B) Writing Python code that creates multiple DAGs automatically based on a list or config

C) A DAG that moves on the screen

D) A random DAG

Answer: B

55. What is "SubDAG" (and why is it often avoided now)?

A) A DAG inside a DAG (often causes performance and scheduling issues)

B) A small DAG

C) A DAG for beginners

D) A deleted DAG

Answer: A

56. What is "Task Group"?

A) A group of engineers

B) A way to organize tasks visually in the UI without the problems of SubDAGs

C) A meeting of tasks

D) A type of folder

Answer: B

57. What is "BranchPythonOperator"?

A) A task that jumps from a tree

B) A task that allows the workflow to take different paths based on a condition

C) A way to delete a branch

D) A security lock

Answer: B

58. What is "Trigger Rule"?

A) A rule for a gun

B) Logic that defines when a task should run (e.g., "all_success", "one_failed", "always")

C) A way to start the internet

D) A password requirement

Answer: B

59. What is "all_failed" trigger rule used for?

A) To run a task only if everything else went wrong (e.g., an error handling task)

B) To stop the pipeline

C) To delete the code

D) To restart the computer

Answer: A

60. What is "one_success" trigger rule?

A) The task runs as soon as at least one upstream task finishes successfully

B) The task only runs if only one task succeeds

C) A task that runs for one person

D) A security rule

Answer: A

61. What is "ShortCircuitOperator"?

A) An electrical error

B) A task that stops the entire rest of the DAG if a condition is not met

C) A very fast task

D) A way to skip one task

Answer: B

62. What is "Data-Aware Scheduling" (Airflow 2.4+)?

A) Scheduling based on how much data costs

B) Automatically starting a DAG when a specific dataset is updated by another DAG

C) Scheduling data scientists

D) A type of calendar

Answer: B

63. What is a "Dataset" in Airflow orchestration?

A) A table in BigQuery

B) A logical reference (URI) used to trigger dependencies between different DAGs

C) A list of files

D) A backup folder

Answer: B

64. What is "Zombie Task"?

A) A task that comes back to life

B) A task that is running but the scheduler has lost track of it

C) A task that is deleted but still shows up

D) A very slow task

Answer: B

65. What is "Poison Pill" in a queue?

A) A virus

B) A special message that tells a worker to shut down

C) A task that always fails

D) A security breach

Answer: B

66. What is "Prefect"?

A) A school monitor

B) A modern orchestration tool that focuses on "code as a workflow" (Python-native)

C) A type of database

D) A security software

Answer: B

67. What is "Dagster"?

A) A type of dog

B) An orchestration tool that focuses on data assets and local development

C) A gaming platform

D) A list of DAGs

Answer: B

68. What is "Luigi"?

A) Mario's brother

B) An older Python-based orchestration framework originally from Spotify

C) A type of pizza

D) A coding language

Answer: B

69. What is "Event-Driven Orchestration"?

A) Orchestration for parties

B) Workflows that start in response to real-time events (like a Pub/Sub message)

C) A manual process

D) A scheduled task

Answer: B

70. What is "Error Handling" in a pipeline?

A) Ignoring errors

B) Predetermined steps to take (like cleanup or alerts) when a task fails

C) Deleting bad data

D) Renaming an error

Answer: B

71. What is "Atomicity" in a task?

A) A small task

B) A task should either complete fully or not at all (no partial changes)

C) A fast task

D) A security lock

Answer: B

72. What is "Task Rescheduling" vs "Up for Retry"?

A) Rescheduling releases worker resources while waiting; Retry keeps them

B) They are the same

C) Retry is for images

D) Rescheduling is for errors

Answer: A

73. What is "SLA Miss" (in Airflow)?

A) Missing a meeting

B) When a task takes longer than its defined Service Level Agreement

C) A security error

D) A deleted task

Answer: B

74. What is "DAG Run"?

A) A person running with code

B) A single execution of a DAG for a specific point in time

C) A list of tasks

D) A backup of the code

Answer: B

75. What is "Manual Trigger"?

A) A physical button

B) Starting a workflow by hand instead of waiting for the schedule

C) A mistake in the code

D) A type of task

Answer: B

76. What is "Clear" task instance?

A) Deleting the task

B) Resetting a task's status so the scheduler runs it again

C) Making the code easy to read

D) Hiding the task

Answer: B

77. What is "Mark Success"?

A) Congratulating a teammate

B) Manually telling the system a task is done so downstream tasks can start

C) Deleting the task

D) Renaming the task

Answer: B

78. What is "Cluster Policy" (Airflow)?

A) A rule for the office

B) Custom logic to enforce rules on all DAGs in a cluster

C) A security firewall

D) A list of servers

Answer: B

79. What is "Task Flow API" (Python decorators)?

A) A new way to write Airflow tasks using @task instead of traditional Operators

B) A way to draw flowcharts

C) A social media app

D) A security protocol

Answer: A

80. What is "Task Lifecycle"?

A) The time from a task's creation to its deletion

B) The sequence of states a task goes through (Queued -> Running -> Success)

C) The job of a developer

D) A type of database

Answer: B

81. What is "Worker Saturation"?

A) When workers have too much work and cannot handle new tasks

B) When workers are wet

C) When there are too many workers

D) A security error

Answer: A

82. What is "Backlog"?

A) A pile of wood

B) A queue of tasks waiting for an available worker

C) A list of past tasks

D) A deleted folder

Answer: B

83. What is "Throughput" in orchestration?

A) The number of tasks completed per unit of time

B) The speed of the internet

C) The size of the code

D) The cost of the project

Answer: A

84. What is "Latent Task"?

A) A task that is late

B) A task that has a delay between when it should run and when it actually runs

C) A hidden task

D) A deleted task

Answer: B

85. What is "State Persistence"?

A) Being stubborn

B) Ensuring the orchestration tool remembers where it left off even if it crashes

C) Keeping data forever

D) A security lock

Answer: B

86. What is "Log Aggregation"?

A) Collecting logs from many workers into one central place for easy searching

B) Deleting logs

C) Renaming logs

D) Printing logs on paper

Answer: A

87. What is "Fernet Key" (Airflow)?

A) A key to a door

B) An encryption key used to protect sensitive connection passwords in the database

C) A type of code

D) A security camera

Answer: B

88. What is "RBAC" in Airflow?

A) Role-Based Access Control (Admin vs Viewer vs User)

B) Real-time Backup and Coding

C) Remote Binary Access

D) Rule-Based Account Center

Answer: A

89. What is "External Link" in a task?

A) A link to a website

B) A button in the UI that takes you to the external system (e.g., BigQuery console)

C) A shortcut on the desktop

D) A security error

Answer: B

90. What is "Pool Slot"?

A) A place to swim

B) A unit of capacity in a pool; a task takes 1 or more slots to run

C) A row in a table

D) A password

Answer: B

91. What is "DAG Bag"?

A) A bag for your laptop

B) The collection of all DAGs that the Airflow scheduler is currently tracking

C) A deleted folder

D) A group of engineers

Answer: B

92. What is "Serialization" of a DAG?

A) Turning a DAG into a format (JSON) so the web server can read it from the DB

B) Watching a TV series

C) Deleting a DAG

D) Renaming a DAG

Answer: A

93. What is "LocalExecutor"?

A) An executor that runs tasks as processes on the same machine as the scheduler

B) An executor for local people

C) An executor that is very slow

D) A security guard

Answer: A

94. What is "SequentialExecutor"?

A) An executor that runs only one task at a time (mostly for testing)

B) A very fast executor

C) An executor that runs tasks in alphabetical order

D) A type of database

Answer: A

95. What is "Dry Run" in orchestration?

A) Testing if the DAG can be parsed without actually running the tasks

B) Drying the computer

C) A failed task

D) A fast task

Answer: A

96. What is "Airflow CLI"?

A) Command Line Interface to interact with Airflow from the terminal

B) A type of aircraft

C) A security code

D) A help manual

Answer: A

97. What is "Import Error" in a DAG?

A) When the Python code has a syntax error and cannot be loaded

B) When you buy the wrong data

C) When the computer is off

D) A security breach

Answer: A

98. What is "Task Decorator"?

A) A person who paints the office

B) Using @task in Python to turn a function into an Airflow task automatically

C) A way to make code look pretty

D) A security lock

Answer: B

99. What is "Workflow Portability"?

A) Being able to carry the computer

B) The ability to move a workflow from one environment to another easily

C) Deleting the workflow

D) Renaming the files

Answer: B

100. What is a "SLA" in orchestration?

A) Service Level Agreement: a time limit for a task to finish

B) A type of task

C) A security protocol

D) A programming language

Answer: A

Phase 3: Master Level (101–150)

101. What is "Dynamic Task Mapping"?

A) Creating a task that moves on a map

B) Generating a variable number of parallel tasks at runtime based on the output of a previous task

C) A random task

D) A security check

Answer: B

102. When would you use Dynamic Task Mapping?

A) When you don't know how many files you need to process until the pipeline starts

B) When you want to save money

C) When the code is too long

D) When you are using a phone

Answer: A

103. What is "Cold Start" in serverless orchestration?

A) Starting in winter

B) The delay when a new container/pod must be created before a task can run

C) A slow computer

D) A deleted task

Answer: B

104. What is "Orchestrator of Orchestrators"?

A) A manager of engineers

B) A top-level system that triggers workflows across different tools (e.g., Airflow to AWS)

C) A very big computer

D) A search engine

Answer: B

105. What is "Linear Scalability" in orchestration?

A) Scaling in a straight line

B) The ability to handle twice the work by adding exactly twice the resources

C) A type of math

D) A security rule

Answer: B

106. What is "Context" in an Airflow task?

A) The words around a task

B) A dictionary of metadata (dates, IDs, params) passed to the task at runtime

C) The office environment

D) A type of database

Answer: B

107. What is "XCom Backend"?

A) The back of a computer

B) A custom location (like S3 or GCS) to store XCom data if it's too big for the metadata DB

C) A security firewall

D) A type of network

Answer: B

108. What is "Custom Operator"?

A) A person with a unique job

B) Writing your own Python class to handle a specific, repeatable task type

C) A math symbol you invented

D) A deleted task

Answer: B

109. What is "Sensors: Poke vs Reschedule" mode?

A) Poke keeps the worker busy; Reschedule releases it while waiting

B) They are the same

C) Poke is for small tasks

D) Reschedule is for errors

Answer: A

110. What is "Deadlock" in orchestration?

A) A broken lock

B) When two tasks are waiting for each other, preventing anything from moving forward

C) A security breach

D) A computer crash

Answer: B

111. What is "Race Condition" in a pipeline?

A) Tasks running a race

B) When the outcome depends on the unpredictable timing of parallel tasks

C) A very fast pipeline

D) A deleted task

Answer: B

112. What is "Throttling" in orchestration?

A) Choking the computer

B) Intentionally slowing down task submission to avoid overwhelming a source system

C) Deleting tasks

D) Renaming files

Answer: B

113. What is "Backoff" strategy?

A) Moving away from the computer

B) Increasing the wait time between retries after each failure

C) Deleting the failed task

D) A security lock

Answer: B

114. What is "Jitter" in a retry policy?

A) Being nervous

B) Adding a small random amount of time to retries to prevent "thundering herd" issues

C) A network error

D) A fast retry

Answer: B

115. What is "Thundering Herd" problem?

A) A group of cows

B) When many tasks or clients retry at the exact same time, crashing the server

C) A security breach

D) A deleted database

Answer: B

116. What is "Side Effect" of a task?

A) A medical symptom

B) Any change a task makes outside of its return value (e.g., writing to a DB)

C) A mistake in the code

D) A type of task

Answer: B

117. What is "Task Idempotency" check?

A) Verifying if a task can be safely re-run without causing duplicate data

B) Checking the spelling of a task

C) Deleting a task

D) Renaming a task

Answer: A

118. What is "Graceful Shutdown"?

A) Closing the computer gently

B) Allowing running tasks to finish before a worker or system stops

C) A power outage

D) Deleting all tasks

Answer: B

119. What is "Hard Kill"?

A) Breaking the computer

B) Immediately stopping a task process without letting it clean up

C) Deleting the project

D) A security lock

Answer: B

120. What is "Zombie Cleanup"?

A) A horror movie

B) A process where the scheduler identifies and kills tasks that are no longer "heartbeating"

C) Cleaning the office

D) Deleting old code

Answer: B

121. What is "Heartbeat" in orchestration?

A) A person's pulse

B) A regular signal a worker sends to the scheduler to say "I am still alive"

C) A type of clock

D) A security check

Answer: B

122. What is "Orchestration as a Service" (OaaS)?

A) Hiring a company to run your DAGs

B) Cloud-managed orchestration (like Cloud Composer or Astronomer)

C) Deleting the orchestration team

D) A type of network

Answer: B

123. What is "Astronomer"?

A) A scientist who looks at stars

B) A popular commercial platform and managed service for Apache Airflow

C) A type of telescope

D) A search engine

Answer: B

124. What is "KEDA" (Kubernetes Event-Driven Autoscaling)?

A) A way to scale Airflow workers automatically based on the number of tasks in the queue

B) A type of food

C) A security firewall

D) A manual task

Answer: A

125. What is "Wait for Downstream" setting?

A) A task waits for its own previous run's downstream tasks to finish before starting

B) Waiting for a friend

C) A slow task

D) A security rule

Answer: A

126. What is "depends_on_past" setting?

A) A task only runs if its own run from the previous schedule was successful

B) A task that likes history

C) A deleted task

D) A security lock

Answer: A

127. What is "Execution Timeout"?

A) The time you spend at work

B) The maximum time a task can run; if exceeded, it is killed

C) Turning off the monitor

D) A fast task

Answer: B

128. What is "DAG Level Access Control"?

A) Restricting who can see or trigger specific DAGs within the UI

B) A password for the computer

C) A security gate

D) A list of users

Answer: A

129. What is "Service Account" for orchestration?

A) A bank account

B) A non-human identity used by the orchestrator to access cloud resources

C) A user's name

D) A security password

Answer: B

130. What is "Cross-DAG Dependency"?

A) When DAG A depends on the completion of DAG B

B) When two DAGs are in the same folder

C) A bug in the code

D) A security risk

Answer: A

131. What is "TriggerDagRunOperator"?

A) A task that manually starts another DAG

B) A task that deletes a DAG

C) A task that renames a DAG

D) A security lock

Answer: A

132. What is "ExternalTaskMarker"?

A) A tool for drawing

B) A way to clear downstream tasks across different DAGs simultaneously

C) A security alarm

D) A list of tasks

Answer: B

133. What is "Task Isolation"?

A) Working in a private room

B) Ensuring that one task's environment doesn't affect another's

C) A security wall

D) A deleted folder

Answer: B

134. What is "Shared File System" for DAGs?

A) A way to share photos

B) Ensuring all workers have access to the same DAG code files

C) A type of USB

D) A security breach

Answer: B

135. What is "DAG Parsing Time"?

A) How long it takes to read the code

B) The time the scheduler takes to evaluate the Python file and create the DAG object

C) The time to delete a DAG

D) The time to run a task

Answer: B

136. Why is high "DAG Parsing Time" bad?

A) It makes the code ugly

B) It slows down the scheduler and can cause tasks to be late

C) It is too expensive

D) It deletes the DAG

Answer: B

137. What is "Local Task Job"?

A) A person's job

B) The actual OS process that monitors a single running task

C) A type of software

D) A security check

Answer: B

138. What is "Inter-task communication"?

A) Tasks talking on the phone

B) Moving data between tasks (e.g., XComs, database writes, or files)

C) A meeting of tasks

D) A security breach

Answer: B

139. What is "Data Governance" in orchestration?

A) Tracking who ran what and what data was touched (Lineage)

B) Deleting all data

C) Renaming files

D) Buying more disks

Answer: A

140. What is "Observability" in a workflow?

A) Watching the screen

B) Having deep insights into why a pipeline is slow or failing (logs, metrics)

C) A security camera

D) A list of tasks

Answer: B

141. What is "Blue/Green Deployment" for DAGs?

A) Using colors for code

B) Testing a new version of a pipeline alongside the old one before switching

Answer: B

142. What is "Canary Deployment" for DAGs?

A) Testing a new DAG on a small percentage of data or users first

B) Using a bird to code

Answer: A

143. What is "DAG Serialization Lag"?

A) The delay between updating code and seeing it in the Web UI

B) A network error

Answer: A

144. What is "Multi-tenant" Airflow?

A) One Airflow instance shared by many different teams with isolated permissions

B) An apartment building for data

Answer: A

145. What is "Airflow as Code"?

A) Writing all configuration and pipelines in Python scripts

B) Writing code on paper

Answer: A

146. What is "Task Retry Delay"?

A) The time to wait before trying a failed task again

B) The time to delete a task

Answer: A

147. What is "Pool starvation"?

A) When a pool is empty

B) When all slots in a pool are full, so new tasks must wait in a queue

Answer: B

148. What is "DAG Dependency Graph"?

A) A drawing of a DAG

B) A visual tool to see how different DAGs relate to each other

Answer: B

149. What is "Workflow Versioning"?

A) Tagging your pipeline code with version numbers for rollback safety

B) Renaming the file

Answer: A

150. Why is Workflow Orchestration the "brain" of a Data Platform?

A) Because it is smart

B) Because it coordinates all the "muscles" (compute tools) to work together

Answer: B

Subtopic 11: Data Observability

1. What is Data Observability?

A) Watching a data scientist work

B) The ability to understand the internal state of data by looking at its external outputs (metrics, logs)

C) Deleting data that looks bad

D) Buying a new monitor for the database

Answer: B

2. Why is Data Observability important?

A) It makes the data bigger

B) It helps identify and fix "silent" data errors before they reach the business

C) It is a legal requirement

D) It makes SQL queries shorter

Answer: B

3. What is "Data Freshness"?

A) How cold the servers are

B) How up-to-date the data is (e.g., was it updated in the last hour?)

C) Using data for the first time

D) Data that has no smell

Answer: B

4. What is "Data Volume" monitoring?

A) Turning up the sound on the computer

B) Checking if the number of rows is within the expected range

C) Measuring the physical weight of the disks

D) Counting the number of users

Answer: B

5. What is "Data Distribution"?

A) Giving data to different departments

B) The spread and frequency of values within a column (e.g., mean, max, null count)

C) Moving data to the cloud

D) Sharing data on social media

Answer: B

6. What is "Schema Change" monitoring?

A) Changing the office floor plan

B) Detecting when a column is added, renamed, or deleted unexpectedly

C) A type of security guard

D) Renaming a project

Answer: B

7. What is "Data Lineage" in observability?

A) The history of the company

B) Tracking the flow of data from source to destination to find where it broke

C) A family tree of the CEO

D) A list of database passwords

Answer: B

8. What is a "Null Count" check?

A) Counting how many zeros are in a table

B) Monitoring the percentage of missing values in a critical column

C) Deleting empty rows

D) A type of math error

Answer: B

9. What is "Data Downtime"?

A) When the power goes out

B) Periods of time when data is partial, erroneous, or missing

C) When the data scientists are on holiday

D) Turning off the database for maintenance

Answer: B

10. What is an "Anomaly"?

A) A standard piece of data

B) A data point or pattern that deviates significantly from the norm

C) A type of encryption

D) A new database table

Answer: B

11. What is "Automated Alerting"?

A) Sending a Slack or Email notification as soon as a data quality rule fails

B) Shouting when you see an error

C) A security alarm

D) A manual report

Answer: A

12. What is "Data Profiling"?

A) Drawing a picture of data

B) Automatically generating statistics (min, max, mean) to understand a dataset

C) Hiring a new data engineer

D) A type of backup

Answer: B

13. What is "Data Quality"?

A) How fast the data moves

B) The degree to which data is accurate, complete, and reliable

C) The price of the data

D) The number of files in a folder

Answer: B

14. What is "Root Cause Analysis" (RCA)?

A) Finding the bottom of a tree

B) The process of finding the exact reason why a data incident happened

C) A type of SQL query

D) A security check

Answer: B

15. What is a "False Positive" in observability?

A) An alert that fires when nothing is actually wrong

B) A successful data load

C) A deleted table

D) A correct calculation

Answer: A

16. What is "Metadata"?

A) Data about data (e.g., table name, size, owner, last updated)

B) Data about the weather

C) A heavy metal disk

D) Secret data

Answer: A

17. What is "Data Validation"?

A) Proving data is expensive

B) Checking if data meets specific requirements (e.g., "Email must contain @")

C) Deleting old data

D) Renaming a dataset

Answer: B

18. What is a "Data SLA"?

A) A type of database

B) A Service Level Agreement defining the expected quality and timeliness of data

C) A security password

D) A job title

Answer: B

19. What is "Monitoring" vs "Observability"?

A) Monitoring tells you when something is wrong; Observability helps you find why

B) They are the same

C) Monitoring is for humans; Observability is for machines

D) Monitoring is free; Observability is expensive

Answer: A

20. What is "Data Health"?

A) The physical condition of the disks

B) A general measure of how reliable and accurate a dataset is

C) A medical term

D) A type of software

Answer: B

21. What is "Threshold-based Alerting"?

A) Alerting when a metric goes above or below a fixed number (e.g., > 10% Nulls)

B) Alerting everyone at once

C) A security lock

D) A manual check

Answer: A

22. What is "Data Discovery"?

A) Finding new data sources

B) Helping users find and understand the data they need through a catalog

C) Deleting hidden data

D) A search engine

Answer: B

23. What is "Completeness"?

A) Finishing a project

B) Ensuring all expected data is present (no missing rows or records)

C) A full hard drive

D) A long SQL query

Answer: B

24. What is "Accuracy"?

A) Fast data

B) Data that correctly represents the real-world values it is intended to show

C) Data that is stored in a table

D) A type of code

Answer: B

25. What is "Consistency"?

A) Data that stays the same across different systems

B) Data that is processed daily

C) A boring dataset

D) A list of names

Answer: A

26. What is "Validity"?

A) Data that follows the correct format and rules (e.g., US Zip code is 5 digits)

B) Data that is useful

C) Data that is new

D) A security password

Answer: A

27. What is "Uniqueness"?

A) Data that is special

B) Ensuring there are no duplicate records where there shouldn't be

C) A type of primary key

D) A secret file

Answer: B

28. What is "Timeliness"?

A) Using a clock

B) Data being available when it is needed for decision-making

C) A fast internet connection

D) A deleted task

Answer: B

29. What is "Data Reliability"?

A) Data that you can trust to be correct consistently over time

B) Data that never moves

C) A backup system

D) A type of disk

Answer: A

30. What is "Incident Management" for data?

A) Calling the police

B) The process of responding to, tracking, and resolving data quality issues

C) Deleting bad files

D) Buying more storage

Answer: B

31. What is a "Data Quality Dashboard"?

A) A screen in a car

B) A visual tool to see health metrics for all data pipelines at a glance

C) A list of passwords

D) A help manual

Answer: B

32. What is "Passive Observability"?

A) Not doing anything

B) Monitoring data health without interfering with the data pipeline itself

C) A slow query

D) A manual check

Answer: B

33. What is "Active Observability"?

A) Running tests and validation checks as part of the pipeline (e.g., stopping on error)

B) Moving fast

C) A security guard

D) A fast network

Answer: A

34. What is "Log-based Observability"?

A) Using logs from databases and tools to track data movement and errors

B) Reading a book

C) A type of password

D) A deleted file

Answer: A

35. What is "Metric-based Observability"?

A) Using numbers and graphs (like row counts) to track data health

B) A type of math

C) A measuring tape

D) A list of users

Answer: A

36. What is "Data Integrity"?

A) Data that is honest

B) Maintaining the accuracy and consistency of data throughout its entire lifecycle

C) A security firewall

D) A type of disk

Answer: B

37. What is "Downstream Impact"?

A) Data moving to a lower server

B) How a data error in one table affects reports and dashboards built on top of it

C) A waterfall

D) A deleted task

Answer: B

38. What is "Upstream Cause"?

A) A problem in the source system that led to a data error later on

B) A fish swimming

C) A new data source

D) A security lock

Answer: A

39. What is "Schema Drift"?

A) When a database moves slowly

B) When the source data structure changes without the data team being notified

C) A type of encryption

D) A deleted column

Answer: B

40. What is "Data Lineage Graph"?

A) A drawing of a family

B) A visual map showing how data transforms from source to final report

C) A pie chart

D) A list of names

Answer: B

41. What is "Monte Carlo" or "Bigeye"?

A) Famous casinos

B) Popular commercial platforms for Data Observability

C) Types of databases

D) Programming languages

Answer: B

42. What is "Great Expectations" (GE)?

A) A book by Charles Dickens

B) A popular open-source tool for validating and documenting data

C) A movie

D) A list of goals

Answer: B

43. What is "dbt Tests"?

A) Tests for a car

B) Built-in data quality checks (unique, not_null) in the dbt transformation tool

C) A type of password

D) A deleted project

Answer: B

44. What is "Data Profiling Report"?

A) A summary of a dataset's structure, quality, and statistical properties

B) A list of employees

C) A news article

D) A backup plan

Answer: A

45. What is "Slack Integration" in observability?

A) Chatting with data

B) Automatically sending data alerts to a Slack channel for the team to see

C) A type of code

D) A security firewall

Answer: B

46. What is "Data Stakeholder"?

A) A person who owns the data center

B) Anyone who relies on the data to do their job (analysts, managers, etc.)

C) A data engineer

D) A security guard

Answer: B

47. What is "Silent Failure"?

A) A computer that makes no noise

B) When a pipeline finishes successfully but the data inside is wrong

C) A deleted task

D) A quiet office

Answer: B

48. What is "Data Trust"?

A) A bank for data

B) The confidence that business users have in the accuracy of the data

C) A security password

D) A type of disk

Answer: B

49. What is "Data Cataloging"?

A) Organizing books

B) Creating a searchable inventory of all data assets with metadata and health status

C) A list of users

D) A search engine

Answer: B

50. What is the "Modern Data Stack" approach to quality?

A) Ignoring quality to move faster

B) Integrating automated observability and testing into the entire data lifecycle

C) Manual checks once a month

D) Deleting data daily

Answer: B

Phase 2: Medium Level (51–100)

51. What is "Machine Learning-based Anomaly Detection"?

A) Using AI to learn "normal" data patterns and alert when something looks strange

B) A robot cleaning the server

C) Manual SQL checks

D) Deleting random rows

Answer: A

52. What is a "Z-score" in data observability?

A) A type of encryption

B) A statistical measure used to detect how many standard deviations a value is from the mean

C) A grade in school

D) A database table

Answer: B

53. What is "Data Diffing"?

A) Finding the difference between two versions of a dataset to see what changed

B) Digging for data

C) Renaming files

D) A type of math

Answer: A

54. What is "Unit Testing" for data?

A) Testing one row

B) Testing a small, isolated piece of data logic or transformation code

C) A physical test

D) A security check

Answer: B

55. What is "Integration Testing" for data?

A) Testing how different parts of a data pipeline work together

B) Combining two databases

C) A social meeting

D) A security firewall

Answer: A

56. What is "End-to-End Testing"?

A) Testing only the ends of a table

B) Testing the entire pipeline from source ingestion to final dashboard output

C) A long SQL query

D) A type of code

Answer: B

57. What is "Data Lineage - Field Level"?

A) Tracking exactly which source column produced which target column

B) Lineage for farmers

C) A rough map of tables

D) A security lock

Answer: A

58. What is "Alert Fatigue"?

A) Being tired of working

B) When a team starts ignoring alerts because there are too many or too many are "false"

C) A broken alarm

D) A computer crash

Answer: B

59. What is "Signal-to-Noise Ratio" in alerting?

A) The volume of the speakers

B) The ratio of real, important alerts (signal) to false or unimportant alerts (noise)

C) A type of radio

D) A network speed

Answer: B

60. What is "Data Observability Platform"?

A) A dedicated software that unifies monitoring, lineage, and alerting for data

B) A type of table

C) A security wall

D) A list of users

Answer: A

61. What is "Self-Healing Data"?

A) Data that fixes its own health

B) Automated systems that can detect an error and trigger a fix (like a re-run)

C) A medical term

D) Deleting broken files

Answer: B

62. What is "Schema Enforcement"?

A) Forcing a database to only accept data that matches a strict, predefined structure

B) A law about data

C) A security gate

D) A type of code

Answer: A

63. What is "Data Governance" vs "Observability"?

A) Governance is the rules; Observability is the proof that the rules are being followed

B) They are the same

C) Governance is free; Observability is expensive

D) Governance is for text; Observability is for numbers

Answer: A

64. What is "Cardinality" monitoring?

A) Monitoring the number of unique values in a column (e.g., to detect data duplication)

B) Counting birds

C) A type of math

D) A security lock

Answer: A

65. What is "Mean Time to Detection" (MTTD)?

A) The average time it takes to find a data issue after it happens

B) A fast query

C) The time to delete a file

D) A security check

Answer: A

66. What is "Mean Time to Resolution" (MTTR)?

A) The average time it takes to fix a data issue once it has been found

B) A high-resolution screen

C) A slow repair

D) A type of code

Answer: A

67. What is "Post-Mortem" for a data incident?

A) A funeral for data

B) A meeting to review what went wrong, why, and how to prevent it in the future

C) Deleting the evidence

D) A security check

Answer: B

68. What is "Data Contracts"?

A) A legal document

B) A technical agreement between a data provider and a data consumer on quality standards

C) A job contract

D) A list of prices

Answer: B

69. What is "Data Contract Violation"?

A) Breaking the law

B) When a data provider sends data that does not meet the agreed-upon schema or quality

C) A security breach

D) A deleted file

Answer: B

70. What is "Lineage Impact Analysis"?

A) Using lineage to see which specific users or tools will be affected if a table is changed

B) Drawing on a graph

C) A type of math

D) A security lock

Answer: A

71. What is "Statistical Process Control" (SPC) for data?

A) Using statistical methods to monitor if a data process is "in control" or unstable

B) A remote control

C) A type of code

D) A manual check

Answer: A

72. What is "Data Reconciliation"?

A) Making data friends again

B) Comparing data across different stages of a pipeline to ensure it matches exactly

C) Deleting duplicates

D) Renaming files

Answer: B

73. What is "Metadata Harvesting"?

A) Picking data from a farm

B) Automatically scanning databases to collect metadata for an observability tool

C) Deleting metadata

D) Selling data

Answer: B

74. What is "Semantic Monitoring"?

A) Monitoring if the meaning of the data is correct (e.g., "Price cannot be negative")

B) Monitoring words

C) A security guard

D) A fast network

Answer: A

75. What is "Operational Metadata"?

A) Info about how the pipeline ran (start time, row count, status, compute cost)

B) Info about surgery

C) A list of employees

D) A security check

Answer: A

76. What is "Technical Metadata"?

A) Info about the structure (table names, column types, primary keys)

B) Info about machines

C) A list of passwords

D) A search engine

Answer: A

77. What is "Business Metadata"?

A) Definitions of columns in plain English, owners, and business importance

B) Info about money

C) A list of companies

D) A backup plan

Answer: A

78. What is "Data Lineage - Table Level"?

A) A high-level map showing how tables depend on each other

B) A table in a garden

C) A list of files

D) A security lock

Answer: A

79. What is "Observability Agent"?

A) A person who watches data

B) A small piece of code that runs inside your environment to collect health metrics

C) A security spy

D) A type of software

Answer: B

80. What is "Collector" in observability?

A) A person who collects stamps

B) A component that receives and aggregates metrics and logs from different systems

C) A waste bin

D) A type of disk

Answer: B

81. What is "Tracing" in data?

A) Drawing on paper

B) Tracking a single record or transaction as it moves through multiple systems

C) A security alarm

D) A type of code

Answer: B

82. What is "Historical Baseline"?

A) A history book

B) Using past data behavior to determine what is "normal" for current data

C) A line on a graph

D) A security lock

Answer: B

83. What is "Data Volume Drop"?

A) Losing your files

B) A sudden, unexpected decrease in the number of rows being ingested

C) A power outage

D) A deleted task

Answer: B

84. What is "Data Volume Spike"?

A) A sharp increase in data size

B) A sudden, unexpected increase in the number of rows (often indicating an error)

C) A fast network

D) A security breach

Answer: B

85. What is "Schema Evolution"?

A) Intentional and managed changes to a database structure over time

B) A random change

C) Deleting the schema

D) Renaming a table

Answer: A

86. What is "Data Quality - Functional"?

A) Does the data work?

B) Does the data meet the specific needs of the business process it supports?

C) A type of code

D) A security check

Answer: B

87. What is "Data Quality - Structural"?

A) Is the data stored correctly?

B) Does the data follow the technical rules and types of the database?

C) A type of math

D) A security lock

Answer: B

88. What is "Data Quality Score"?

A) A grade for data

B) A numeric value (e.g., 0-100) representing the overall health of a dataset

C) The price of data

D) A list of names

Answer: B

89. What is "Alert Routing"?

A) Sending the right alert to the right person or team (e.g., Marketing vs Tech)

B) Drawing a map for an alert

C) Deleting an alert

D) A security firewall

Answer: A

90. What is "Incidence Response Plan"?

A) A plan for an accident

B) A documented set of steps to follow when a critical data failure occurs

C) A backup plan

D) A security check

Answer: B

91. What is "Data Availability"?

A) Is the data online?

B) Ensuring that authorized users can access the data when they need it

C) A type of disk

D) A security password

Answer: B

92. What is "Granularity" in monitoring?

A) Monitoring at a very detailed level (e.g., hourly instead of daily)

B) Monitoring sugar

C) A type of math

D) A security lock

Answer: A

93. What is "Data Latency"?

A) How slow the data is

B) The time delay between when data is generated and when it is available in the warehouse

C) A network error

D) A deleted task

Answer: B

94. What is "User-Defined Check"?

A) A custom validation rule written by a user for their specific data

B) A check from a bank

C) A person's name

D) A security lock

Answer: A

95. What is "Out-of-the-box Check"?

A) A check found in a box

B) Pre-built common health checks provided by an observability tool (e.g., Null check)

C) A new computer

D) A type of code

Answer: B

96. What is "Data Lineage - Visual"?

A) A chart or diagram showing data flow that humans can easily read

B) A drawing on paper

C) A pie chart

D) A security lock

Answer: A

97. What is "Data Lineage - Technical"?

A) A list of SQL queries and code used to move and transform data

B) A technical book

C) A security password

D) A search engine

Answer: A

98. What is "Data Issue - Critical"?

A) An error that stops the whole business or makes important reports wrong

B) A small mistake

C) A deleted task

D) A security breach

Answer: A

99. What is "Data Issue - Minor"?

A) An error that is small and does not have a big impact on the business

B) A major problem

C) A deleted project

D) A security lock

Answer: A

100. What is "Data Literacy"?

A) Being able to read data

B) The ability for employees to understand, interpret, and use data correctly

C) A type of book

D) A security check

Answer: B

Phase 3: Master Level (101–150)

101. What is "Shift-Left" Data Quality?

A) Moving data to the left on a screen

B) Moving testing and quality checks as close to the source/creation as possible

C) Deleting data on the left

D) A type of code

Answer: B

102. What is "Data Mesh" Observability?

A) Decentralized observability where each domain team is responsible for its own health

B) Observability for a net

C) A type of database

D) A security guard

Answer: A

103. What is "Computational Governance"?

A) Using AI to manage data

B) Automating the enforcement of data policies and health checks using code

C) A very fast computer

D) A security lock

Answer: B

104. What is "Data Lineage - Programmatic"?

A) Extracting lineage by automatically parsing and analyzing SQL scripts and code

B) A TV program

C) A type of math

D) A security password

Answer: A

105. What is "Statistical Anomaly Detection - Seasonality"?

A) Accounting for normal changes over time (e.g., weekend sales spikes)

B) Monitoring the weather

C) A type of code

D) A security check

Answer: A

106. What is "Drift Detection - Statistical"?

A) Detecting when the fundamental statistical properties of a dataset change over time

B) Detecting if a ship is moving

C) A type of math

D) A security lock

Answer: A

107. What is "Data Lifecycle Management" (DLM)?

A) Managing data from creation to deletion

B) A school for data

C) A type of software

D) A security plan

Answer: A

108. What is "Self-Documenting Data"?

A) Data that writes its own metadata and descriptions through code and tags

B) Data that reads itself

C) A type of book

D) A security lock

Answer: A

109. What is "Data Quality - Predictive"?

A) Using AI to guess if a future data point will be an error

B) Seeing the future

C) A security check

D) A type of code

Answer: A

110. What is "Data Catalog - Automated"?

A) A catalog that updates itself by scanning systems daily for new data

B) A manual list

C) A security wall

D) A search engine

Answer: A

111. What is "Semantic Layer" Observability?

A) Monitoring the consistency of business definitions across different BI tools

B) Monitoring words

C) A type of code

D) A security lock

Answer: A

112. What is "Observability as Code"?

A) Defining all your monitors and alerts in configuration files (like YAML)

B) Writing code on paper

C) A manual process

D) A security rule

Answer: A

113. What is "Lineage - Logical vs Physical"?

A) Logical is the flow of meaning; Physical is the actual path through servers/code

B) They are the same

C) Logical is for text; Physical is for numbers

D) Logical is free; Physical is expensive

Answer: A

114. What is "Data Lineage - API Driven"?

A) Using APIs to push lineage info from pipelines directly into an observability tool

B) A search engine

C) A security firewall

D) A list of files

Answer: A

115. What is "Data Observability - Multi-cloud"?

A) Using observability tools that work across AWS, GCP, and Azure

B) Many clouds in the sky

C) A type of network

D) A security guard

Answer: A

116. What is "Root Cause Analysis - Automated"?

A) Using AI to suggest the most likely reason why a data alert was triggered

B) A manual check

C) A security alarm

D) A type of code

Answer: A

117. What is "Data Change Management"?

A) Managing how changes to data systems are planned and tested to avoid breaks

B) Renaming files

C) Deleting data

D) A security lock

Answer: A

118. What is "Data Quality - Cross-Dataset"?

A) Checking if values in Table A match the expected values in Table B

B) Checking one table

C) A type of math

D) A security firewall

Answer: A

119. What is "Schema - Versioned"?

A) Keeping a history of all schema changes so you can roll back if needed

B) A new schema

C) A deleted schema

D) A security lock

Answer: A

120. What is "Data Quality - Zero Trust"?

A) Never assuming data is correct; always verifying at every step

B) Not trusting the cloud

C) No quality checks

D) Deleting all data

Answer: A

121. What is "Data Observability - Real-time"?

A) Monitoring data health as it streams, with millisecond alerts

B) Monitoring daily

C) A slow query

D) A security guard

Answer: A

122. What is "Data Reliability Engineer" (DRE)?

A) A role focused specifically on the health and uptime of data platforms

B) A data scientist

C) A security guard

D) A job title

Answer: A

123. What is "SLI" (Service Level Indicator) for data?

A) A specific metric used to measure data health (e.g., "% of valid emails")

B) A type of database

C) A security password

D) A job title

Answer: A

124. What is "Error Budget" for data?

A) The amount of data downtime allowed before stopping new feature work

B) A list of prices

C) A backup plan

D) A security check

Answer: A

125. What is "Data Health Scorecard"?

A) A high-level report showing the health of the entire data org to leaders

B) A sports card

C) A list of names

D) A security lock

Answer: A

126. What is "Metadata Catalog - Graph based"?

A) Using a graph database to show complex connections between data assets

B) A drawing on paper

C) A pie chart

D) A security lock

Answer: A

127. What is "Data Quality - Sentiment Analysis"?

A) Monitoring if text data (like customer reviews) has the correct tone

B) Monitoring how people feel

C) A security guard

D) A type of code

Answer: A

128. What is "Data Lineage - Inference"?

A) Guessing lineage by analyzing which tables are frequently queried together

B) A scientific theory

C) A type of math

D) A security lock

Answer: A

129. What is "Data Observability - Open Standards"?

A) Using shared formats (like OpenLineage) so different tools can talk to each other

B) Free tools for everyone

C) A type of network

D) A security guard

Answer: A

130. What is "OpenLineage"?

A) An open-source standard for collecting data lineage across the ecosystem

B) A type of code

C) A security firewall

D) A list of files

Answer: A

131. What is "Data Observability - Cost Monitoring"?

A) Tracking how much money queries and storage are costing in real-time

B) Buying data

C) A list of prices

D) A security check

Answer: A

132. What is "FinOps" for Data?

A) Managing the financial efficiency of cloud data operations

B) A new type of bank

C) A security firewall

D) A backup plan

Answer: A

133. What is "Data Observability - Usage Metrics"?

A) Tracking which tables are most used and which are unused ghost data

B) Counting users

C) A security guard

D) A search engine

Answer: A

134. What is "Ghost Data"?

A) Scary data

B) Data that is stored and paid for but never used in any report or query

C) Deleted data

D) A security breach

Answer: B

135. What is "Data Observability - Privacy Compliance"?

A) Automatically alerting if PII data is found in an unauthorized location

B) Keeping data secret

C) A security firewall

D) A backup plan

Answer: A

136. What is "Sensitive Data Discovery"?

A) Using AI to find credit card numbers or names in large datasets

B) Finding hidden data

C) A search engine

D) A security lock

Answer: A

137. What is "Data Quality - Geometric"?

A) Monitoring geographic data for errors (e.g., coordinates in the ocean)

B) Monitoring shapes

C) A type of math

D) A security guard

Answer: A

138. What is "Data Observability - Alert Suppression"?

A) Temporarily silencing alerts (e.g., during planned maintenance)

B) Deleting alerts

C) A security firewall

D) A backup plan

Answer: A

139. What is "Data Observability - Collaborative Incident Response"?

A) Using tools where multiple people can work on a data fix together

B) Working alone

C) A security guard

D) A type of code

Answer: A

140. What is "Data Catalog - Crowdsourced"?

A) Allowing users to add their own notes and ratings to datasets

B) A public list

C) A security wall

D) A search engine

Answer: A

141. What is "Data Observability - Impact Analysis - Financial"?

A) Estimating the dollar cost of a specific data error to the business

B) Buying data

C) A list of prices

D) A security check

Answer: A

142. What is "Data Health - Historical Playback"?

A) Watching old videos of data

B) Being able to see the state of your data health at any point in the past

C) A backup plan

D) A security lock

Answer: B

143. What is "Observability - Integration with Orchestration"?

A) Having the observability tool talk to the orchestrator to stop bad jobs

B) Working alone

C) A security guard

D) A type of code

Answer: A

144. What is "Data Quality - Semantic Consistency"?

A) Ensuring that "Revenue" is calculated the exact same way in every system

B) Using the same words

C) A security firewall

D) A search engine

Answer: A

145. What is "Data Lineage - Automated Discovery"?

A) Tools that build a lineage map by watching data move without manual input

B) Drawing on paper

C) A pie chart

D) A security lock

Answer: A

146. What is "Data Observability - Trend Analysis"?

A) Monitoring how data quality metrics are improving or worsening over months

B) Watching the news

C) A type of math

D) A security guard

Answer: A

147. What is "Data Quality - Completeness - Columnar"?

A) Ensuring that a specific column is 100% populated in every row

B) Finishing a project

C) A type of code

D) A security lock

Answer: A

148. What is "Data Observability - Resource Efficiency"?

A) Monitoring if queries can be written more efficiently to save time and money

B) Working hard

C) A security firewall

D) A backup plan

Answer: A

149. What is "Data Trust Score"?

A) A metric based on user feedback and technical health checks

B) A grade for a bank

C) A security password

D) A job title

Answer: A

150. Why is Data Observability the "Final Frontier" of data engineering?

A) Because it is in space

B) Because once pipelines are built, the biggest challenge is keeping the data correct

C) Because it is the most expensive

D) Because it is from Google

Answer: B

Subtopic 12: Scalable Pipeline Deployment

Phase 1: Easy Level (1–50)

1. What is "Deployment" in data engineering?

A) Deleting old code

B) Moving code from a development environment to a live production environment

C) Buying a new server

D) Writing a SQL query

Answer: B

2. What does "CI" stand for?

A) Cloud Integration

B) Continuous Integration

C) Code Intelligence

D) Computer Interface

Answer: B

3. What does "CD" stand for?

A) Compact Disc

B) Continuous Deployment

C) Cloud Data

D) Code Design

Answer: B

4. What is the main goal of CI/CD?

A) To make code longer

B) To automate the testing and delivery of code updates safely and frequently

C) To save money on electricity

D) To replace developers with robots

Answer: B

5. What is "Production" (Prod)?

A) A movie set

B) The environment where real business data is processed and used by stakeholders

C) A testing area

D) A folder on a laptop

Answer: B

6. What is "Development" (Dev)?

A) A place to build and test new code before it goes live

B) A school for engineers

C) The final version of a product

D) A type of database

Answer: A

7. What is "Staging" (QA)?

A) A theater stage

B) An environment that mimics Production for final testing

C) A backup of the data

D) A list of users

Answer: B

8. What is "Git"?

A) A type of computer

B) A version control system to track changes in code

C) A programming language

D) A social media site

Answer: B

9. What is a "Repository" (Repo)?

A) A trash can

B) A storage location for your project's code and its history

C) A type of database

D) A security password

Answer: B

10. What is a "Commit"?

A) A promise to work hard

B) Saving a specific version of your changes to the Git repository

C) Deleting a file

D) Sending an email

Answer: B

11. What is a "Branch"?

A) A part of a tree

B) A parallel version of a repository, used to develop features without affecting the main code

C) A different office location

D) A type of cable

Answer: B

12. What is a "Merge"?

A) Combining two roads

B) Combining changes from one branch into another (usually back into 'main')

C) Deleting a branch

D) Renaming a file

Answer: B

13. What is a "Pull Request" (PR)?

A) Asking for a favor

B) A request to review and merge code changes into a branch

C) A way to download data

D) A type of security alert

Answer: B

14. What is "Automation" in deployment?

A) Using robots to type

B) Using scripts/tools to perform repetitive tasks (like testing) without human intervention

C) A type of car

D) A manual process

Answer: B

15. What is a "Build"?

A) Constructing a building

B) Compiling code and preparing it to be run

C) A type of database

D) A list of requirements

Answer: B

16. What is "Environment Variables"?

A) The weather outside the data center

B) Values (like API keys) that change depending on if the code is in Dev or Prod

C) A type of math

D) A security lock

Answer: B

17. What is "Docker"?

A) A person who works at a dock

B) A tool for "containerizing" applications so they run the same everywhere

C) A type of hard drive

D) A coding language

Answer: B

18. What is a "Container"?

A) A box for lunch

B) A lightweight, standalone package that includes everything needed to run a piece of software

C) A large database

D) A cloud region

Answer: B

19. What is "Kubernetes" (K8s)?

A) A Greek island

B) A system for automating the deployment and scaling of containers

C) A type of keyboard

D) A security protocol

Answer: B

20. What is "Scaling Up" (Vertical)?

A) Climbing a ladder

B) Adding more power (CPU/RAM) to a single machine

C) Adding more machines

D) Deleting data

Answer: B

21. What is "Scaling Out" (Horizontal)?

A) Spreading out on the floor

B) Adding more machines/nodes to a system to handle more work

C) Making a machine bigger

D) Renaming a project

Answer: B

22. What is "Infrastructure as Code" (IaC)?

A) Writing code to build houses

B) Managing servers and networks using configuration files (like Terraform) instead of manual setup

C) A type of database

D) A list of hardware

Answer: B

23. What is "Terraform"?

A) Changing the atmosphere of a planet

B) A popular IaC tool used to provision cloud resources

C) A type of soil

D) A data transformation tool

Answer: B

24. What is "YAML"?

A) A type of food

B) A human-readable data format often used for configuration files

C) A programming language like Java

D) A security password

Answer: B

25. What is "Cloud Native"?

A) Applications designed specifically to run in the cloud environment

B) People who live in the cloud

C) An offline computer

D) A physical server

Answer: A

26. What is a "Pipeline" in CI/CD?

A) A water pipe

B) The automated sequence of steps code goes through from commit to production

C) A list of employees

D) A type of database

Answer: B

27. What is "GitHub Actions"?

A) A tool to automate workflows (CI/CD) directly inside GitHub

B) A list of movies

C) A security alarm

D) A social network

Answer: A

28. What is "Unit Testing"?

A) Testing a single unit of code to make sure it works as expected

B) Testing the whole office

C) A physical exam

D) A type of math

Answer: A

29. What is "Rollback"?

A) Rolling a ball backwards

B) Returning to a previous, stable version of the code after a deployment fails

C) Deleting the database

D) Renaming a file

Answer: B

30. What is "Uptime"?

A) The time you wake up

B) The percentage of time a service is successfully running and available

C) A fast network

D) A security lock

Answer: B

31. What is "Latency"?

A) Being late to a meeting

B) The delay or time it takes for data to travel through a system

C) A network error

D) A type of code

Answer: B

32. What is "Throughput"?

A) The amount of data a pipeline can process in a given amount of time

B) Putting data through a hole

C) The size of the code

D) The number of users

Answer: A

33. What is "Provisioning"?

A) Providing food

B) Setting up the necessary infrastructure (servers, storage) for an application

C) Deleting a service

D) A type of code

Answer: B

34. What is a "Configuration File"?

A) A file that tells the software how to behave and what settings to use

B) A picture of a computer

C) A list of passwords

D) A search engine

Answer: A

35. What is "Secret Management"?

A) Keeping a secret from the boss

B) Securely storing and handling sensitive info like passwords and API keys

C) A type of encryption

D) A hidden folder

Answer: B

36. What is "Artifact"?

A) An old object from a museum

B) The output of a build process (like a ZIP file or Docker image) ready for deployment

C) A piece of data

D) A security breach

Answer: B

37. What is "Orchestration" in deployment?

A) Conducting an orchestra

B) Managing the lifecycle and coordination of multiple containers or services

C) Writing code

D) A manual task

Answer: B

38. What is "Auto-scaling"?

A) Automatically adjusting the number of active servers based on the current load

B) A type of scale for weight

C) A manual process

D) A security firewall

Answer: A

39. What is "High Availability" (HA)?

A) A very tall building

B) Systems designed to be operational 99.9% of the time with no single point of failure

C) A fast internet connection

D) A expensive service

Answer: B

40. What is "Fault Tolerance"?

A) Being nice to people who make mistakes

B) The ability of a system to continue functioning even if a component fails

C) A security error

D) A type of code

Answer: B

41. What is "Load Balancer"?

A) A person carrying heavy boxes

B) A device that distributes incoming network traffic across multiple servers

C) A type of battery

D) A data tool

Answer: B

42. What is "Scripting"?

A) Writing a movie

B) Writing small programs (Python, Bash) to automate tasks

C) A type of font

D) A security lock

Answer: B

43. What is "SSH"?

A) A way to tell someone to be quiet

B) A secure way to remotely log into and control a server

C) A type of internet

D) A security alarm

Answer: B

44. What is "Cloud Provider"?

A) A company like AWS, Google Cloud, or Azure that sells computing services

B) A weather company

C) A type of server

D) A person who works in the cloud

Answer: A

45. What is "Serverless"?

A) A system where you don't have to manage the physical or virtual servers yourself

B) A computer without a screen

C) An offline system

D) A free service

Answer: A

46. What is "Elasticity"?

A) How stretchy a rubber band is

B) The ability of a cloud system to grow and shrink resources instantly as needed

C) A type of code

D) A security guard

Answer: B

47. What is "Compute"?

A) The processing power (CPU and RAM) needed to run code

B) A mathematical calculation

C) A type of computer

D) A list of users

Answer: A

48. What is "Storage"?

A) A closet for boxes

B) Where data is kept permanently (disks, cloud buckets)

C) A type of RAM

D) A search engine

Answer: B

49. What is "Networking"?

A) Meeting people at a party

B) How different computers and services communicate with each other

C) A type of cable

D) A security firewall

Answer: B

50. Why is scalable deployment important for Data Engineering?

A) It makes the code prettier

B) It ensures pipelines can handle growing amounts of data without breaking or becoming too expensive

C) It is a fun hobby

D) It replaces SQL

Answer: B

Phase 2: Medium Level (51–100)

51. What is "Blue/Green Deployment"?

A) Using colors to label servers

B) Running two identical production environments; one is live (Green), one is idle (Blue) for testing

C) A type of security guard

D) A marketing strategy

Answer: B

52. What is "Canary Deployment"?

A) Testing code on a small bird

B) Releasing a new version of code to a small percentage of users before a full rollout

C) A yellow-colored server

D) A security breach

Answer: B

53. What is "Terraform Plan"?

A) A command to see what changes Terraform will make to your infrastructure before applying them

B) A map of the cloud

C) A type of code

D) A security check

Answer: A

54. What is "Terraform State"?

A) The country where the server is located

B) A file that keeps track of the current resources Terraform has created

C) A status message

D) A security lock

Answer: B

55. What is "Immutable Infrastructure"?

A) Servers that can never be changed

B) Infrastructure that is replaced entirely when an update is needed, rather than being modified

C) A very old server

D) A free service

Answer: B

56. What is a "Docker Image"?

A) A photo of a container

B) A read-only template used to create Docker containers

C) A type of database

D) A security password

Answer: B

57. What is "Docker Hub"?

A) A central place to find and share Docker images

B) A type of network cable

C) A security firewall

D) A search engine

Answer: A

58. What is "Kubernetes Pod"?

A) A small group of dolphins

B) The smallest deployable unit in Kubernetes, which can hold one or more containers

C) A type of server

D) A security lock

Answer: B

59. What is "Kubernetes Node"?

A) A single physical or virtual machine in a Kubernetes cluster

B) A point on a graph

C) A type of code

D) A security guard

Answer: A

60. What is a "Kubernetes Cluster"?

A) A group of nodes (machines) working together to run containerized applications

B) A list of users

C) A security wall

D) A search engine

Answer: A

61. What is "Helm"?

A) A steering wheel for a ship

B) A package manager for Kubernetes used to define and deploy complex applications

C) A type of code

D) A security lock

Answer: B

62. What is "Service Mesh" (e.g., Istio)?

A) A net for servers

B) A dedicated infrastructure layer for managing service-to-service communication

C) A type of database

D) A security firewall

Answer: B

63. What is "Configuration Management" (e.g., Ansible, Chef)?

A) Managing your computer's screen brightness

B) Tools used to automate the setup and maintenance of software on existing servers

C) A type of code

D) A manual process

Answer: B

64. What is "Environment Parity"?

A) Ensuring Dev, Staging, and Prod are as identical as possible to prevent "works on my machine" bugs

B) Making the office look the same

C) A type of math

D) A security lock

Answer: A

65. What is "Linting"?

A) Cleaning a clothes dryer

B) Automatically checking code for programmatic and stylistic errors

C) A type of test

D) A security check

Answer: B

66. What is "Static Code Analysis"?

A) Checking code without actually running it to find bugs or security risks

B) A slow way to read code

C) A type of database

D) A security alarm

Answer: A

67. What is "Dynamic Analysis"?

A) Testing code while it is running

B) A fast way to write code

C) A security guard

D) A type of math

Answer: A

68. What is "Integration Testing" in pipelines?

A) Testing if a new code change works correctly with existing databases and services

B) Combining two companies

C) A social meeting

D) A security firewall

Answer: A

69. What is "Secrets Vault" (e.g., HashiCorp Vault)?

A) A physical safe in the data center

B) A tool specifically designed to store and manage sensitive secrets securely

C) A type of database

D) A security lock

Answer: B

70. What is "Rolling Update"?

A) Updating while driving

B) Gradually replacing old versions of an app with new ones to ensure zero downtime

C) A type of math

D) A security breach

Answer: B

71. What is "Zero Downtime Deployment"?

A) Deploying code without the service ever going offline for users

B) A very fast deployment

C) A free service

D) A security lock

Answer: A

72. What is "Health Check" in deployment?

A) A doctor's visit for the engineer

B) An automated test to see if a container or service is running and ready for traffic

C) A security alarm

D) A type of code

Answer: B

73. What is "Liveness Probe"?

A) A tool to see if a container is still alive (if not, Kubernetes restarts it)

B) A medical test

C) A type of code

D) A security guard

Answer: A

74. What is "Readiness Probe"?

A) A tool to see if a container is ready to start accepting traffic/data

B) A school test

C) A type of database

D) A security lock

Answer: A

75. What is "Resource Limits" (CPU/Memory)?

A) Constraints put on a container to prevent it from using all the server's power

B) A list of requirements

C) A security firewall

D) A type of code

Answer: A

76. What is "Vertical Pod Autoscaler" (VPA)?

A) Automatically changing the CPU/Memory of existing pods

B) A tall robot

C) A type of math

D) A security guard

Answer: A

77. What is "Horizontal Pod Autoscaler" (HPA)?

A) Automatically adding more pods (instances) to handle more load

B) A wide robot

C) A type of code

D) A security lock

Answer: A

78. What is "Ingress" in Kubernetes?

A) A way to enter a building

B) An object that manages external access to services in a cluster (usually HTTP)

C) A type of database

D) A security wall

Answer: B

79. What is "Egress" in networking?

A) Data leaving a network

B) Data entering a network

C) A security breach

D) A type of code

Answer: A

80. What is "GitOps"?

A) A practice where Git is the "single source of truth" for infrastructure and deployments

B) A type of computer

C) A security alarm

D) A social network

Answer: A

81. What is "ArgoCD"?

A) A popular GitOps tool used to deploy applications to Kubernetes automatically

B) A type of CD player

C) A security firewall

D) A search engine

Answer: A

82. What is "Prometheus"?

A) A Greek god

B) A popular open-source monitoring and alerting toolkit

C) A type of database

D) A security lock

Answer: B

83. What is "Grafana"?

A) A tool used to create visual dashboards from monitoring data (like Prometheus)

B) A type of paint

C) A search engine

D) A security guard

Answer: A

84. What is "Log Aggregator" (e.g., ELK Stack)?

A) A tool that collects logs from many servers into one searchable place

B) A person who picks up wood

C) A type of code

D) A security alarm

Answer: A

85. What is "Container Registry"?

A) A list of all containers in the world

B) A secure place to store and manage Docker images (like GCR or ECR)

C) A type of database

D) A security lock

Answer: B

86. What is "Build Agent" / "Runner"?

A) The machine that actually executes the CI/CD pipeline steps

B) A person who builds houses

C) A type of code

D) A security guard

Answer: A

87. What is "Pipeline Orchestration"?

A) Managing complex workflows involving multiple CI/CD pipelines

B) Conducting a band

C) A manual task

D) A type of database

Answer: A

88. What is "Blue/Green Traffic Shifting"?

A) Moving users from the old version (Blue) to the new version (Green) using a load balancer

B) Painting the servers

C) A type of code

D) A security alarm

Answer: A

89. What is "Hotfix"?

A) A very fast fix for a critical bug in the Production environment

B) A warm meal

C) A security breach

D) A type of math

Answer: A

90. What is "Technical Debt"?

A) Money owed to a computer company

B) The future cost of choosing an easy, fast solution now instead of a better approach

C) A list of requirements

D) A security lock

Answer: B

91. What is "Idempotency" in deployment scripts?

A) The ability to run a script many times and get the same result without errors

B) A medical term

C) A type of code

D) A security guard

Answer: A

92. What is "Dry Run" in Terraform?

A) Using terraform plan to see what will happen without actually changing anything

B) Drying the computer

C) A type of math

D) A security lock

Answer: A

93. What is "Modular Infrastructure"?

A) Building infrastructure using small, reusable pieces of code (Terraform modules)

B) A type of house

C) A manual process

D) A security firewall

Answer: A

94. What is "StatefulSet" in Kubernetes?

A) A way to manage applications that need to remember data (like databases)

B) A list of states

C) A type of code

D) A security guard

Answer: A

95. What is "Deployment" vs "StatefulSet"?

A) Deployment is for apps that don't save data; StatefulSet is for apps that do

B) They are the same

C) Deployment is for text; StatefulSet is for numbers

D) Deployment is free; StatefulSet is expensive

Answer: A

96. What is "Sidecar Pattern"?

A) A motorcycle attachment

B) Running a secondary container alongside the main one to handle tasks like logging or security

C) A type of code

D) A security alarm

Answer: B

97. What is "Init Container"?

A) A container that runs and finishes before the main application container starts

B) A new container

C) A type of database

D) A security lock

Answer: A

98. What is "Namespace" in Kubernetes?

A) A name for a person

B) A way to divide cluster resources between different users or teams

C) A type of code

D) A security guard

Answer: B

99. What is "Taint and Toleration" in Kubernetes?

A) A way to ensure pods only run on specific, appropriate nodes

B) A type of food

C) A security breach

D) A manual task

Answer: A

100. What is "Pod Affinity"?

A) Ensuring that certain pods are placed on the same node (e.g., to reduce latency)

B) A list of pods

C) A type of code

D) A security lock

Answer: A

Phase 3: Master Level (101–150)

101. What is "Chaos Engineering"?

A) Working in a messy office

B) Intentionally breaking parts of a system to test its resilience and reliability

C) A type of math

D) A security breach

Answer: B

102. What is "Netflix Chaos Monkey"?

A) A famous pet

B) A tool that randomly shuts down production servers to ensure the system can handle failures

C) A type of code

D) A security alarm

Answer: B

103. What is "GitFlow" vs "Trunk-based Development"?

A) GitFlow uses many branches; Trunk-based focuses on small, frequent commits to 'main'

B) They are the same

C) Trunk-based is only for trees

D) GitFlow is for small teams only

Answer: A

104. What is "Feature Flags" (Toggles)?

A) Flags used in a race

B) A way to turn features on or off in production without deploying new code

C) A type of code

D) A security lock

Answer: B

105. What is "Dark Launching"?

A) Launching at night

B) Deploying code to production but keeping it hidden from users to test performance

C) A security breach

D) A type of math

Answer: B

106. What is "Service Level Objective" (SLO) vs "SLA"?

A) SLO is the internal goal; SLA is the external contract with users

B) They are the same

C) SLA is for text; SLO is for numbers

D) SLO is free; SLA is expensive

Answer: A

107. What is "Error Budget"?

A) The amount of downtime allowed before the team must stop new features and focus on stability

B) A list of prices

C) A type of code

D) A security guard

Answer: A

108. What is "Site Reliability Engineering" (SRE)?

A) A discipline that applies software engineering to system operations

B) Fixing a broken website

C) A type of database

D) A security alarm

Answer: A

109. What is "Golden Signals" in monitoring?

A) Latency, Traffic, Errors, and Saturation

B) A type of gold

C) A security breach

D) A list of requirements

Answer: A

110. What is "Observability" (Tracing, Metrics, Logs)?

A) Having deep insight into the internal state of a system based on its outputs

B) Watching the screen

C) A type of code

D) A security guard

Answer: A

111. What is "Distributed Tracing" (e.g., Jaeger)?

A) Tracking a request as it travels through dozens of different microservices

B) Drawing on a map

C) A type of math

D) A security lock

Answer: A

112. What is "Infrastructure Drift"?

A) When the actual cloud setup no longer matches the Terraform code (usually due to manual changes)

B) A continent moving

C) A security breach

D) A type of code

Answer: A

113. What is "Terraform Taint"?

A) Marking a resource to be deleted and recreated during the next apply

B) A dirty server

C) A type of math

D) A security lock

Answer: A

114. What is "Multi-Region Deployment"?

A) Deploying an app in multiple geographical areas to ensure global speed and reliability

B) A map of the world

C) A type of code

D) A security guard

Answer: A

115. What is "Active-Active" vs "Active-Passive"?

A) Active-Active has all regions serving traffic; Active-Passive has one serving and one waiting

B) They are the same

C) Active-Active is for sports

D) Active-Passive is for text

Answer: A

116. What is "Global Load Balancing"?

A) Distributing traffic to the server physically closest to the user

B) A person carrying many boxes

C) A type of math

D) A security lock

Answer: A

117. What is "Disaster Recovery" (DR) Plan?

A) A plan for a natural disaster

B) A documented process to restore data and services after a major system failure

C) A security breach

D) A backup plan

Answer: B

118. What is "RPO" (Recovery Point Objective)?

A) How much data (in time) a business is willing to lose in a crash

B) A type of database

C) A security guard

D) A list of requirements

Answer: A

119. What is "RTO" (Recovery Time Objective)?

A) How fast a business needs to be back online after a crash

B) A type of code

C) A security lock

D) A search engine

Answer: A

120. What is "Shadow Deployment"?

A) Deploying in the dark

B) Sending a copy of live production traffic to a new version of the app to see how it handles it

C) A security breach

D) A type of math

Answer: B

121. What is "Progressive Delivery"?

A) Delivering mail slowly

B) A set of practices (canary, feature flags) to release code with high control and safety

C) A type of code

D) A security guard

Answer: B

122. What is "Environment as a Service" (EaaS)?

A) Automatically spinning up a complete, temporary environment for every Pull Request

B) A type of bank

C) A security firewall

D) A search engine

Answer: A

123. What is "Serverless Containers" (e.g., Google Cloud Run, AWS Fargate)?

A) Running containers without having to manage the underlying Kubernetes or servers

B) A computer without a box

C) A type of code

D) A security lock

Answer: A

124. What is "Cold Start" in serverless?

A) Starting a computer in winter

B) The delay when a serverless function starts for the first time after being idle

C) A security breach

D) A type of math

Answer: B

125. What is "Warm Up" strategy?

A) Keeping serverless functions "active" to avoid cold start delays

B) A physical exercise

C) A type of code

D) A security guard

Answer: A

126. What is "Container Orchestration" vs "Choreography"?

A) Orchestration has a central controller; Choreography is decentralized communication

B) They are the same

C) Choreography is for dancing

D) Orchestration is for music

Answer: A

127. What is "Event-Driven Scaling"?

A) Scaling based on the number of messages in a queue (e.g., KEDA)

B) Scaling for a party

C) A type of code

D) A security lock

Answer: A

128. What is "Policy as Code" (e.g., OPA)?

A) Using code to define and enforce security and compliance rules across infrastructure

B) A law about code

C) A type of math

D) A security guard

Answer: A

129. What is "DevSecOps"?

A) Integrating security practices directly into the CI/CD pipeline from the start

B) A secret project

C) A type of code

D) A security lock

Answer: A

130. What is "Vulnerability Scanning"?

A) Automatically checking Docker images and code for known security holes

B) A doctor's visit

C) A type of database

D) A security alarm

Answer: A

131. What is "Secrets Rotation"?

A) Regularly changing passwords and API keys automatically to improve security

B) Spinning a safe

C) A type of code

D) A security lock

Answer: A

132. What is "Least Privilege" Principle?

A) Giving users/apps only the minimum access they need to do their job

B) No access for anyone

C) Full access for everyone

D) A security breach

Answer: A

133. What is "RBAC" (Role-Based Access Control)?

A) Assigning permissions based on a person's job role (e.g., Developer vs Viewer)

B) A type of car

C) A security wall

D) A list of users

Answer: A

134. What is "IAM" (Identity and Access Management)?

A) A system to manage who can access which cloud resources

B) A person's name

C) A type of code

D) A security lock

Answer: A

135. What is "External Secrets Operator"?

A) A tool that syncs secrets from a Vault (like AWS Secrets Manager) into Kubernetes

B) A person who works outside

C) A type of database

D) A security alarm

Answer: A

136. What is "Cost Anomaly Detection"?

A) Alerting if cloud spending suddenly spikes unexpectedly

B) Finding a cheap flight

C) A security breach

D) A type of math

Answer: A

137. What is "Spot Instances" / "Preemptible VMs"?

A) Cheap, spare cloud servers that can be shut down by the provider at any time

B) A very expensive server

C) A security lock

D) A type of code

Answer: A

138. What is "FinOps"?

A) Financial Operations; managing the cost and efficiency of cloud resources

B) A new type of bank

C) A security firewall

D) A search engine

Answer: A

139. What is "Vendor Lock-in"?

A) Being stuck with one cloud provider because it's too hard to move

B) A key for a server

C) A security breach

D) A type of math

Answer: A

140. What is "Multi-Cloud Strategy"?

A) Using multiple cloud providers (AWS and GCP) to avoid lock-in and improve reliability

B) Looking at the sky

C) A security guard

D) A type of code

Answer: A

141. What is "EKS" / "GKE" / "AKS"?

A) Managed Kubernetes services from AWS, Google, and Azure

B) A type of computer

C) A security alarm

D) A list of requirements

Answer: A

142. What is "Fargate" vs "EC2"?

A) Fargate is serverless; EC2 requires you to manage the servers

B) They are the same

C) Fargate is for gates

D) EC2 is for cars

Answer: A

143. What is "Terraform Cloud" / "Terraform Enterprise"?

A) Managed platforms to run Terraform as a team with security and collaboration tools

B) A type of soil

C) A search engine

D) A security lock

Answer: A

144. What is "GitOps - Push vs Pull"?

A) Push: CI tool sends changes to cluster; Pull: Cluster agent watches Git and pulls changes

B) They are the same

C) Pull is for sports

D) Push is for doors

Answer: A

145. What is "Container Security - Distroless Images"?

A) Using images that contain only the app, with no shell or package manager (very secure)

B) Images without a home

C) A security breach

D) A type of code

Answer: A

146. What is "Kubernetes Admission Controller"?

A) A tool that decides if a request to the Kubernetes API should be allowed or rejected

B) A person at a gate

C) A type of database

D) A security lock

Answer: A

147. What is "Network Policy" in Kubernetes?

A) Rules that control which pods can talk to each other

B) A law about the internet

C) A type of code

D) A security guard

Answer: A

148. What is "Horizontal Auto-scaling - Latency based"?

A) Adding more servers specifically because the response time for users has increased

B) Scaling based on weight

C) A security breach

D) A type of math

Answer: A

149. What is "Pre-deployment Hook"?

A) A script that runs just before code is deployed (e.g., to backup a database)

B) A tool for fishing

C) A type of code

D) A security lock

Answer: A

150. Why is Scalable Pipeline Deployment the "Final Step" of data engineering?

A) Because it is at the end of the book

B) Because it is how we turn a developer's idea into a reliable, automated reality for the business

C) Because it is the most expensive

D) Because it is from Google

Answer: B